1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

# SAFFORO_MLB,SCHEM

LAST_MODIFICATION=Fri Apr 20 15:11:56 2018

| REV | ECN | DESCRIPTION OF REVISION | CK APPD DATE |
|---|---|---|---|
| 5 | 0011084277 | ENGINEERING RELEASED | 2018-04-20 |



# Preliminary Test

Schematic / PCB #'s

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| [illegible] | 1 | PCB,MLB,SAFFORO | PCB | CRITICAL | [illegible] |
| [illegible] | 1 | STBF,MLB,SAFFORO | PCB | CRITICAL | [illegible] |
| | | | | | |

FAB#-EC315

DRAWING TITLE: SCHEM,MLB,SAFFORO

Apple Inc.

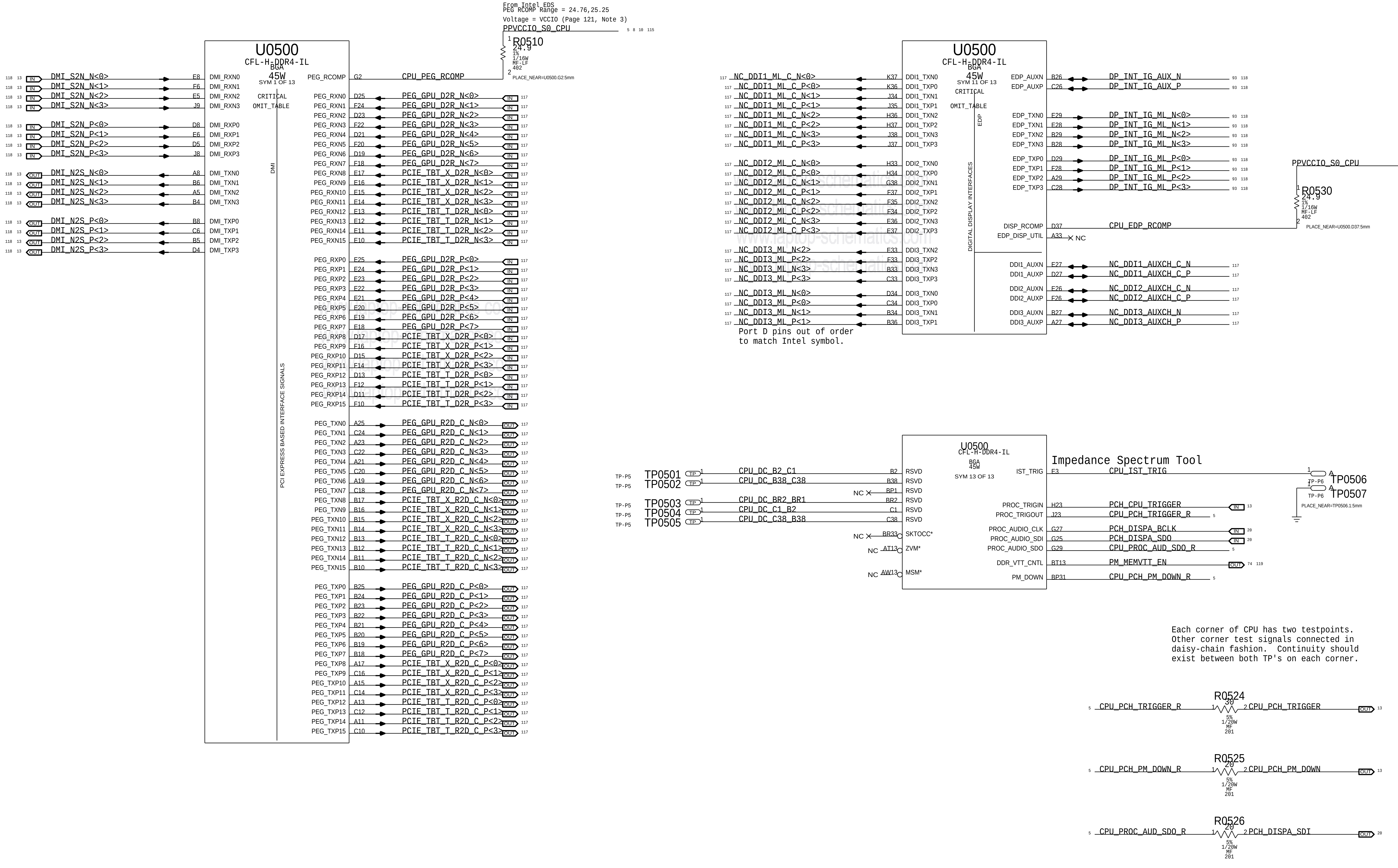
DRAWING NUMBER: 051-02888 SIZE: D

REVISION: 1.0.0

BRANCH: EVT

PAGE: 1 OF 200

SHEET: 1 OF 131



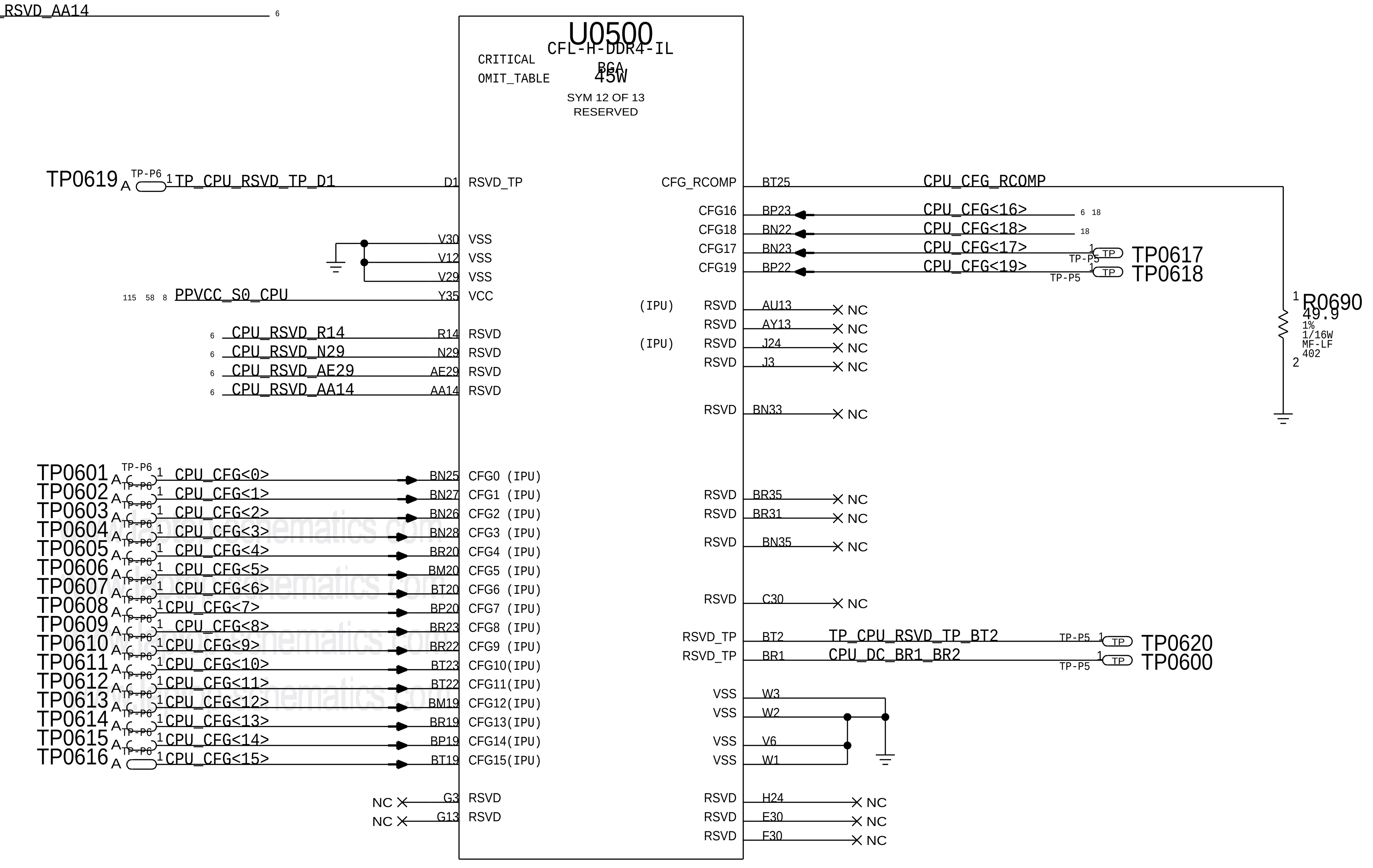

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE

# CPU DMI/PEG/FDI/RSVD

Apple Inc.

| DRAWING NUMBER | SIZE |
|---|---|
| 051-02888 | D |
| REVISION 1.0.0 | |
| BRANCH EVT | |
| PAGE 5 OF 200 | |
| SHEET 5 OF 131 | |


## CFL H

| Name | Balll# | Signal Name |
|---|---|---|
| BN25 | CFG[0] | NOA_N_0 |
| BN27 | CFG[1] | NOA_N_1 |
| BN26 | CFG[2] | NOA_N_2 |
| BN28 | CFG[3] | NOA_N_3 |
| BR20 | CFG[4] | NOA_N_4 |
| BM20 | CFG[5] | NOA_N_5 |
| BT20 | CFG[6] | NOA_N_6 |
| BP20 | CFG[7] | NOA_N_7 |
| BR23 | CFG[8] | NOA_N_8 |
| BR22 | CFG[9] | NOA_N_9 |
| BT23 | CFG[10] | NOA_N_10 |
| BT22 | CFG[11] | NOA_N_11 |
| BM19 | CFG[12] | NOA_N_12 |
| BR19 | CFG[13] | NOA_N_13 |
| BP19 | CFG[14] | NOA_N_14 |
| BT19 | CFG[15] | NOA_N_15 |
| | | |
| BN23 | CFG[17] | NOA_STB_P_0 |
| BP22 | CFG[19] | NOA_STB_P_1 |
| | | |
| D1 | RSVD_TP | PEG_VIEW0 |
| E1 | RSVD_TP | PEG_VIEW1 |
| E2 | RSVD_TP | PEG_VIEW3 |
| E3 | IST_TRIG | PEG_VIEW2 |
| BT2 | RSVD_TP | DDR_VIEW1 |
| BR1 | RSVD_TP | DDR_VIEW0 |

| BOM GROUP | BOM OPTIONS |
|---|---|
| CPUPEG:X8X8 | CPUCFG5_PD |
| CPUPEG:X8X4X4 | CPUCFG6_PD,CPUCFG5_PD |

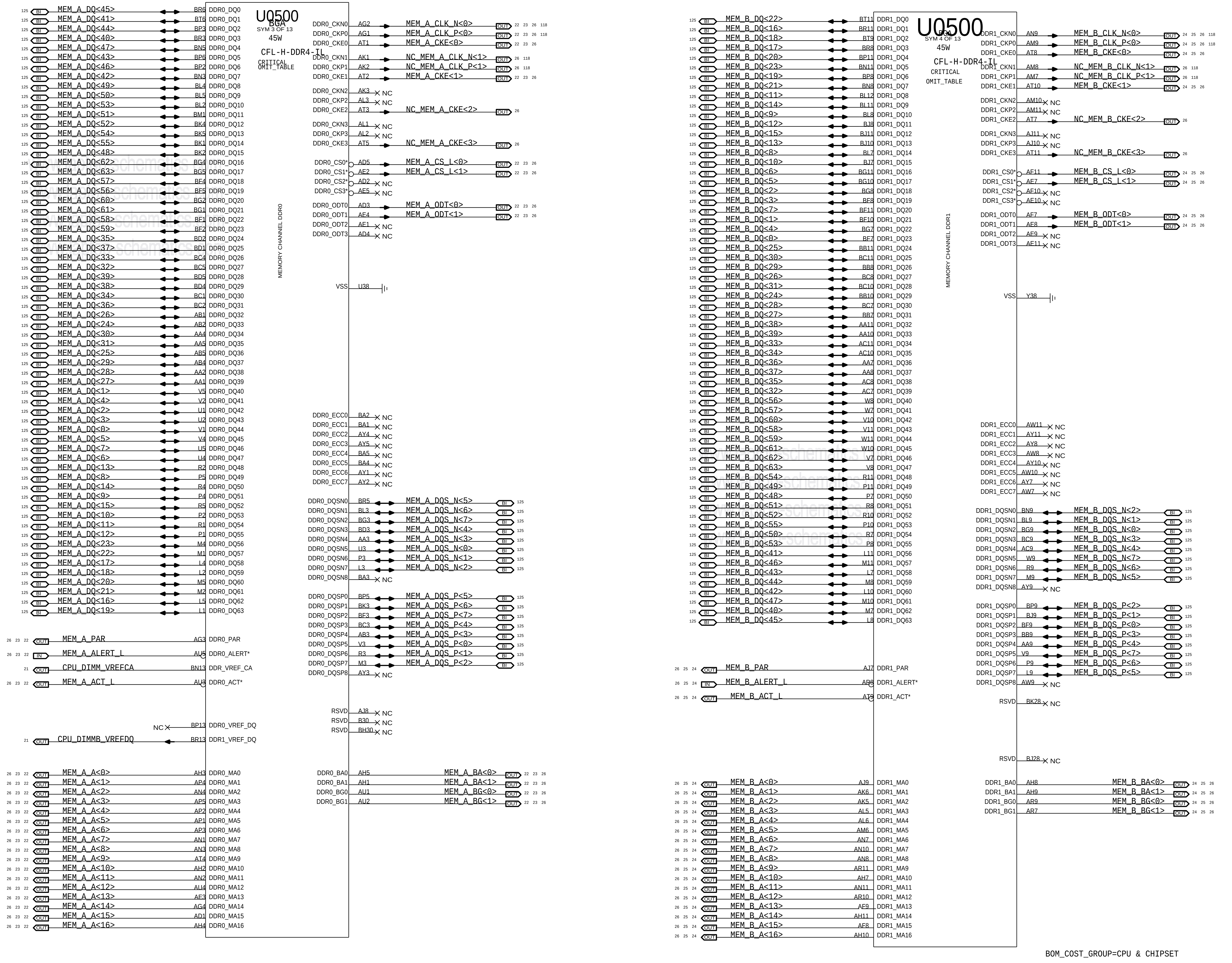
To use PEG X16 configuration, simply remove CPUPEG:X8X8 and CPUPEG:X8X4X4 from BOMs.

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
CPU Clock/Misc/JTAG/CFG

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D
REVISION 1.0.0
BRANCH EVT
PAGE 6 OF 200
SHEET 6 OF 131




PAGE TITLE

# CPU DDR4 Interfaces

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT

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SHEET 7 OF 131


PAGE TITLE

# CPU Power

Apple Inc.

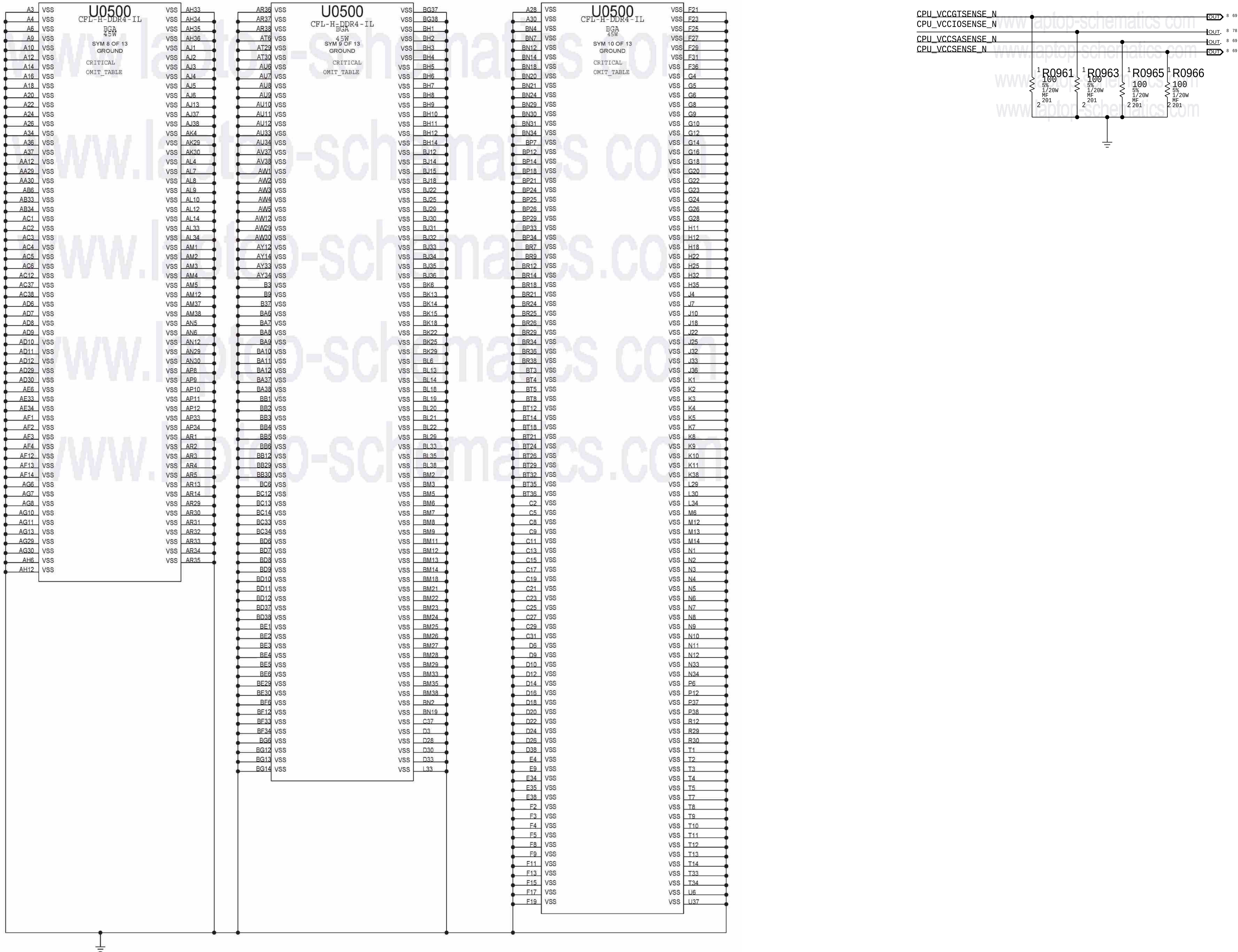
DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT

PAGE 8 OF 200

SHEET 8 OF 131






BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
# CPU Ground

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT

# Vcc CPU Core Decoupling from 20140905 BOM

## CPU VCORE Decoupling

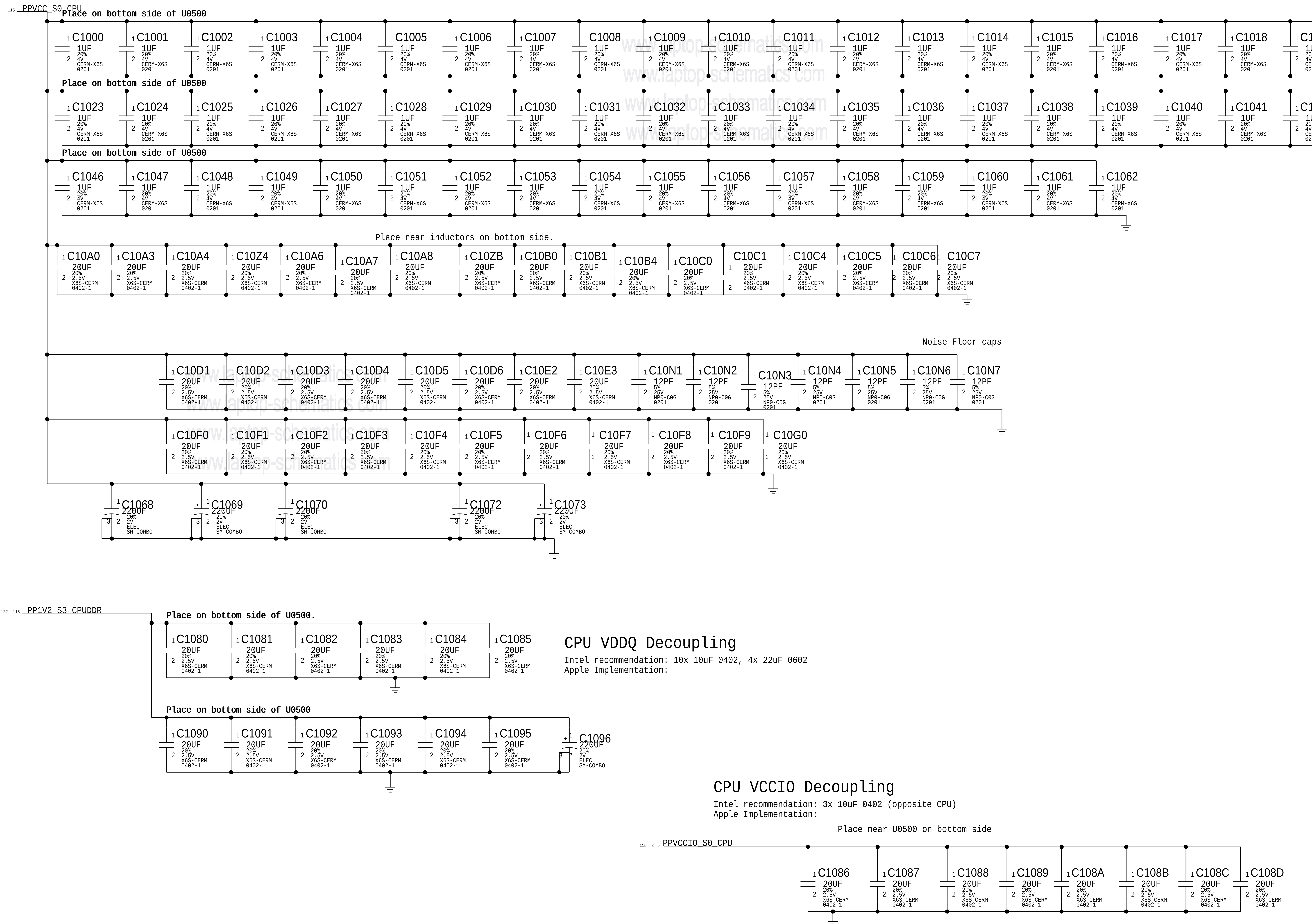
Intel recommendation: 5x 220uF ESR 5m ohms ESL 1.9nH each,4x 47uF 0805 8x22uF 0603, 28x 10uF 0402, 3x 10uF 0402, 69x 1uF 0201    Board Edge: 2x 220uF, 4x 47uF rest on the back side
Apple Implementation:

## CPU VDDQ Decoupling

Intel recommendation: 10x 10uF 0402, 4x 22uF 0602
Apple Implementation:

## CPU VCCIO Decoupling

Intel recommendation: 3x 10uF 0402 (opposite CPU)
Apple Implementation:




NOTE: Intel decoupling recommendations from CBR schematics for Skylake H doc#557227 and PDG section 48.1 (document# 546884)

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
CPU Decoupling 1

Apple Inc.

| | |
|---|---|
| DRAWING NUMBER | 051-02888 |
| SIZE | D |
| REVISION | 1.0.0 |
| BRANCH | EVT |
| PAGE | 10 OF 200 |
| SHEET | 10 OF 131 |

# Vcc GT Slice Core Decoupling from 20140905 BOM

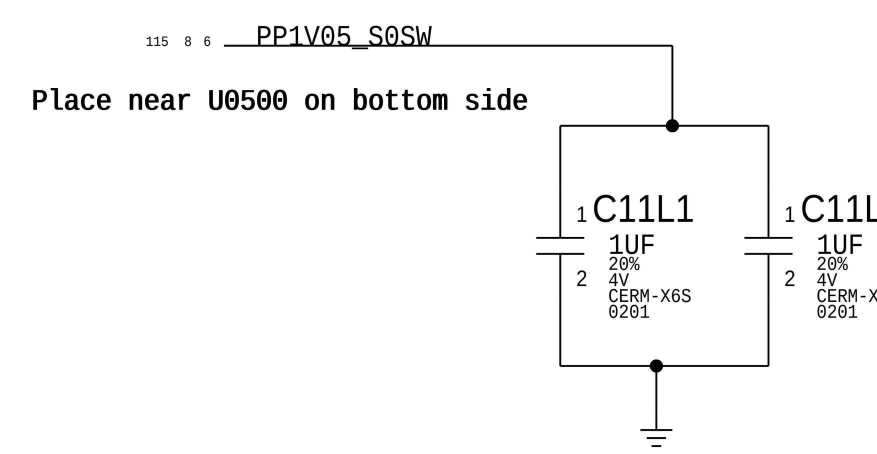
## CPU VGTSlice Decoupling

Intel recommendation: 7x 220uF, 6x 47uF 0805, 6x 22uF 0603, 35x 10uF 0402, 68 1uF 0201

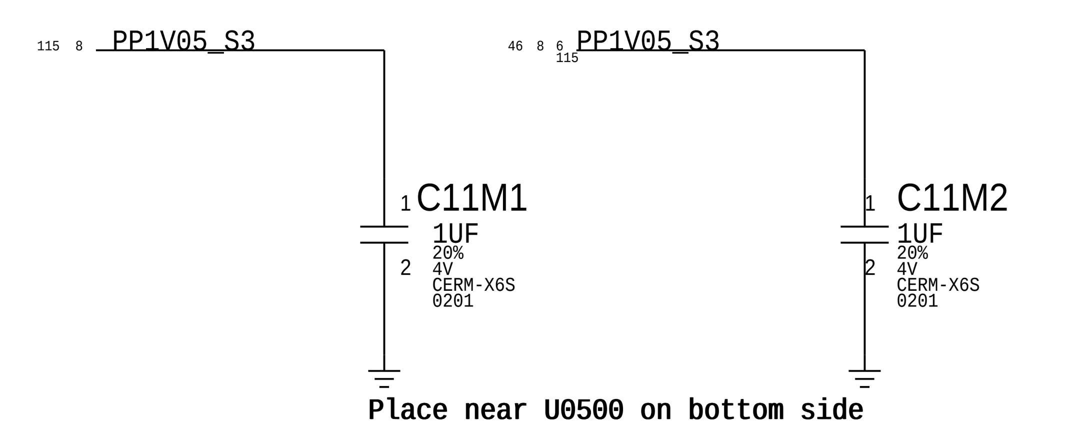
Apple Implementation:

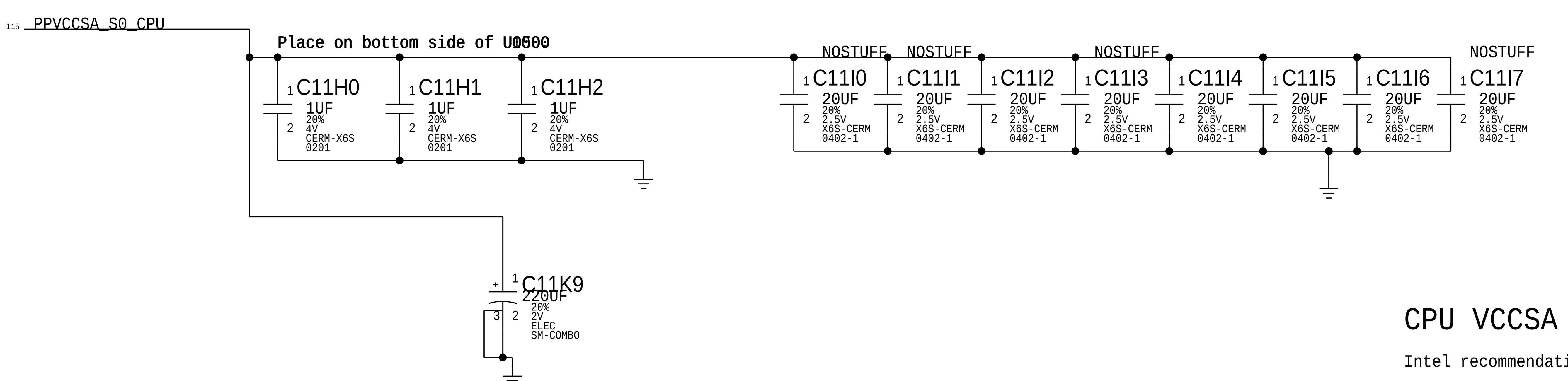
Board Edge: 4x220uF, 7x 47uF rest on back side


## CPU VCCSTG Decoupling




## CPU VCCPLL and VCCST Decoupling




## CPU VCCSA Decoupling

Intel recommendation: 2x 220uF, 1x 47uF 0805. 1x 22uF. 7x 10uF 0402, 3x 1uF 0201_
2x 220uF, 1x 22uF on board edge, everything else on back side

Apple Implementation:

NOTE: Intel decoupling recommendations from CBR schematics for Skylake H doc#557227 and PDG section 48.1 (document# 546884)

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE: CPU Decoupling 2

Apple Inc.

DRAWING NUMBER: 051-02888 SIZE: D

REVISION: 1.0.0

BRANCH: EVT


PAGE TITLE

# PCH RTC/CLK/ESPI/PM

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT

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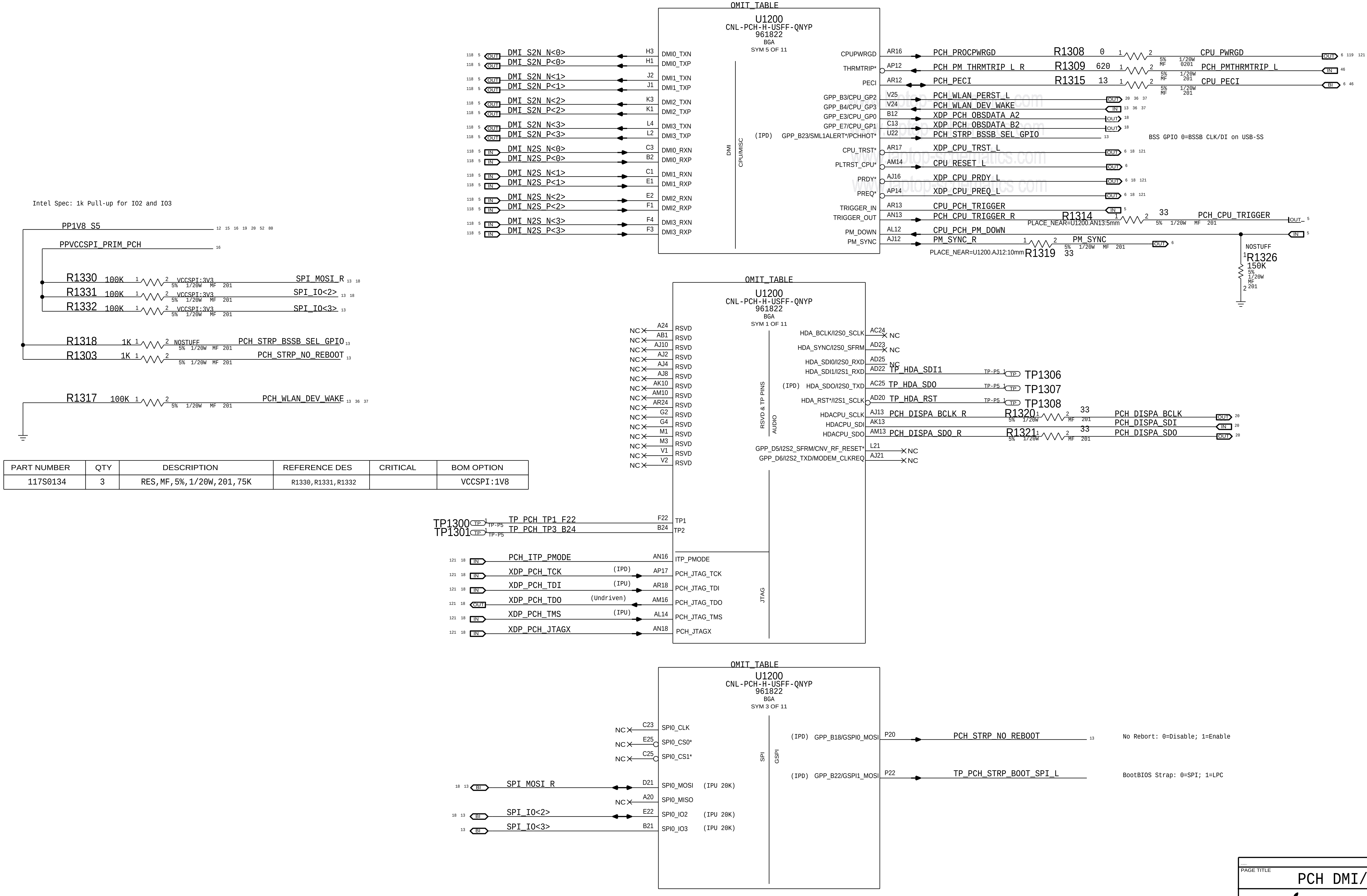
BOM_COST_GROUP=CPU & CHIPSET




| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 117S0134 | 3 | RES,MF,5%,1/20W,201,75K | R1330,R1331,R1332 | | VCCSPI:1V8 |

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE

# PCH DMI/JTAG/SPI/HDA

Apple Inc.

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BRANCH EVT

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PAGE TITLE

# PCH PCI-E/USB

Apple Inc.

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BRANCH EVT

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| SPEED | CONGIF3 |
|---|---|
| 2400MHZ | 0 |
| 2133MHZ | 1 |

| STORAGE | CONGIF2 |
|---|---|
| 16GB | 1 |
| 32GB | 0 |

| VENDOR | CONGIF1 | CONGIF0 |
|---|---|---|
| SAMSUNG | 1 | 0 |
| MICRON | 0 | 1 |
| HYNIX | 0 | 0 |


PAGE TITLE
# PCH GPIO/MISC/NCTF

Apple Inc.

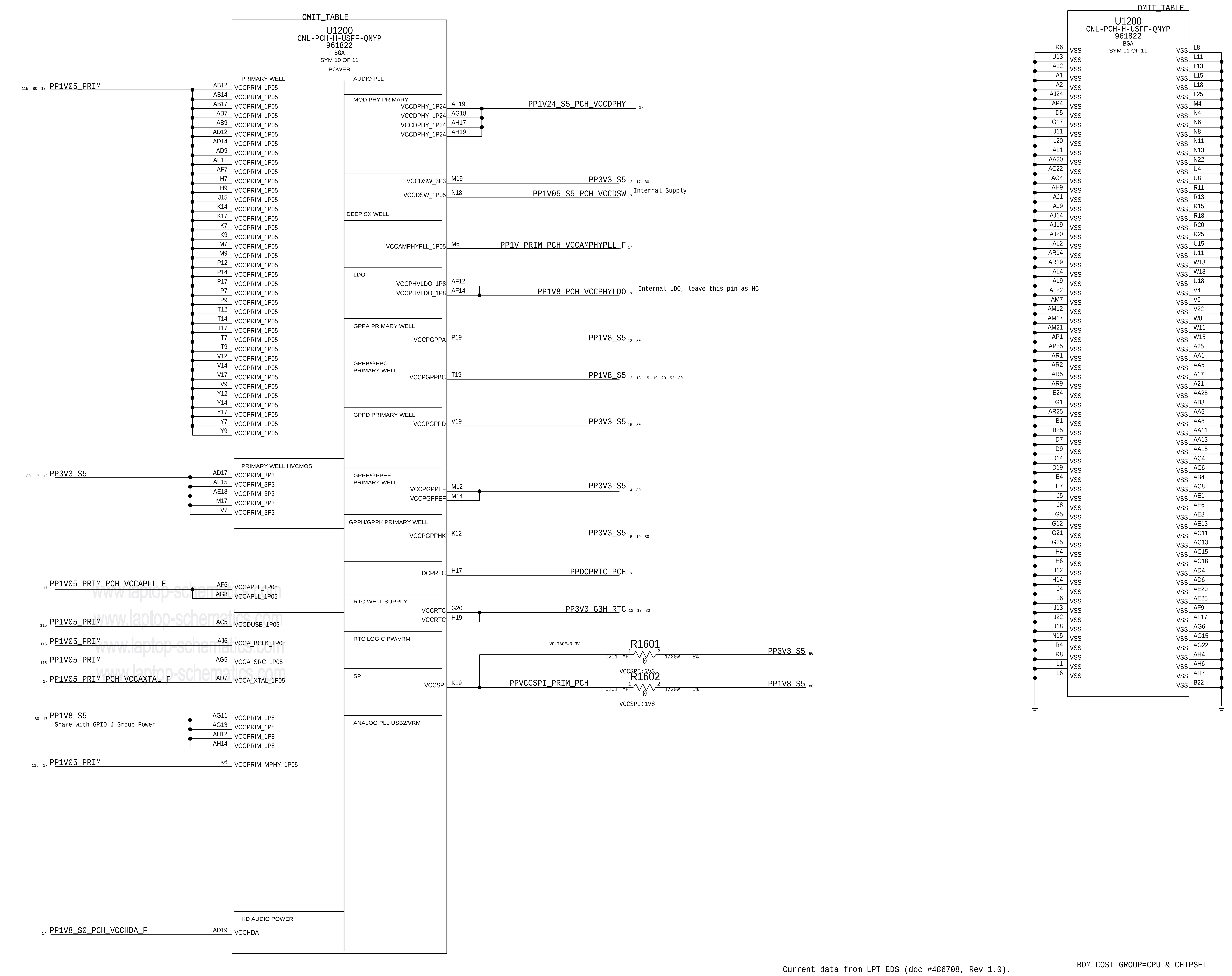
DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT

PAGE 15 OF 200

SHEET 15 OF 131




Current data from LPT EDS (doc #486708, Rev 1.0).

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
## PCH Power

Apple Inc.

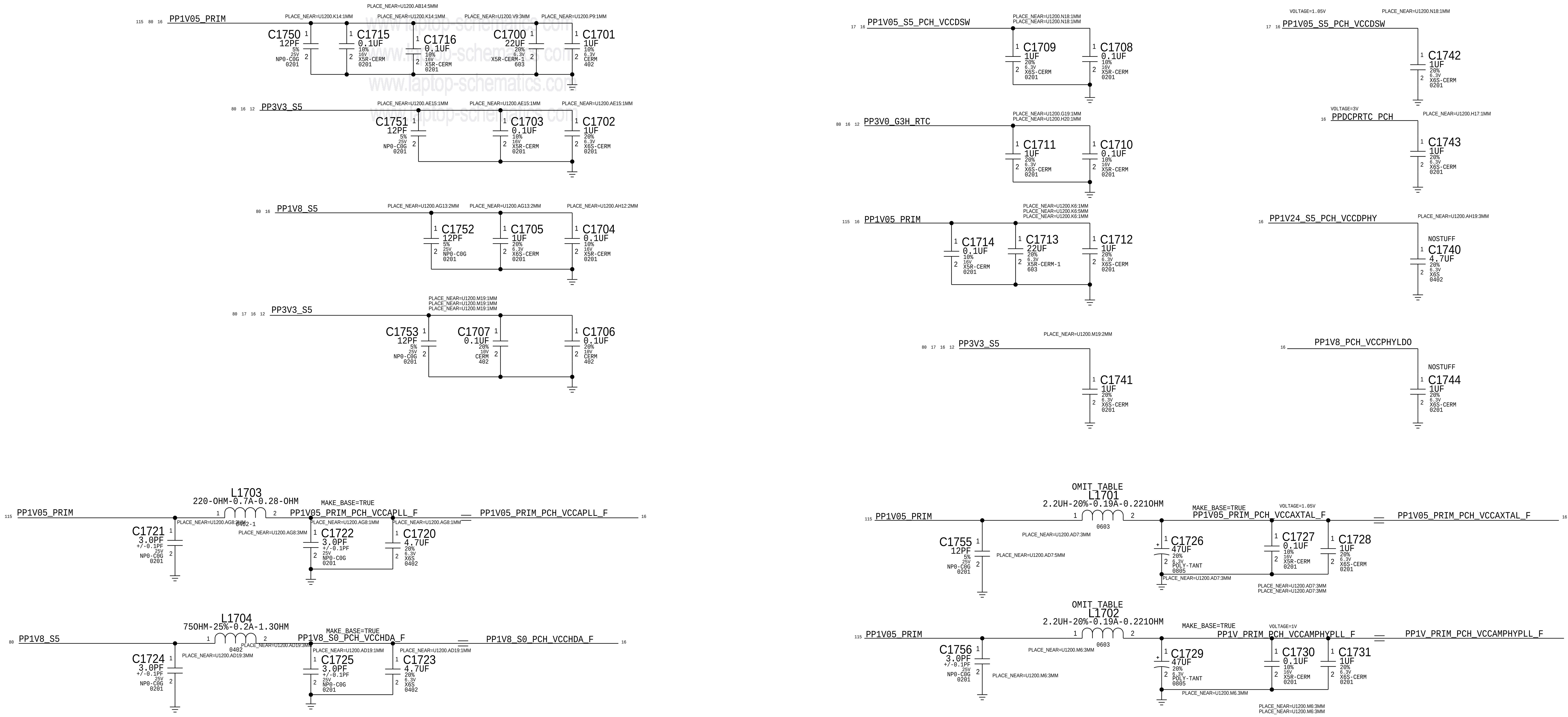
DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT

PAGE 16 OF 200

SHEET 16 OF 131




| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 113S0022 | 2 | RES,MF,1A MAX,0OHM,5%,0603 | L1701,L1702 | | |

Current data from LPT EDS (doc #486708, Rev 1.0).

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
# PCH Decoupling

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D
REVISION 1.0.0
BRANCH EVT


Unused GPIOs have TPs.

USB Overcurrents are aliased, do not cause USB OC# events during PCH debug.

JTAG_ISP (non-TMS) nets are aliased, do not attempt bit-banged JTAG during PCH debug.

DESIGN: X502/MLB

LAST CHANGE: Mon Jun 15 22:04:28 2015

PAGE TITLE

# CPU/PCH Merged XDP

Apple Inc.

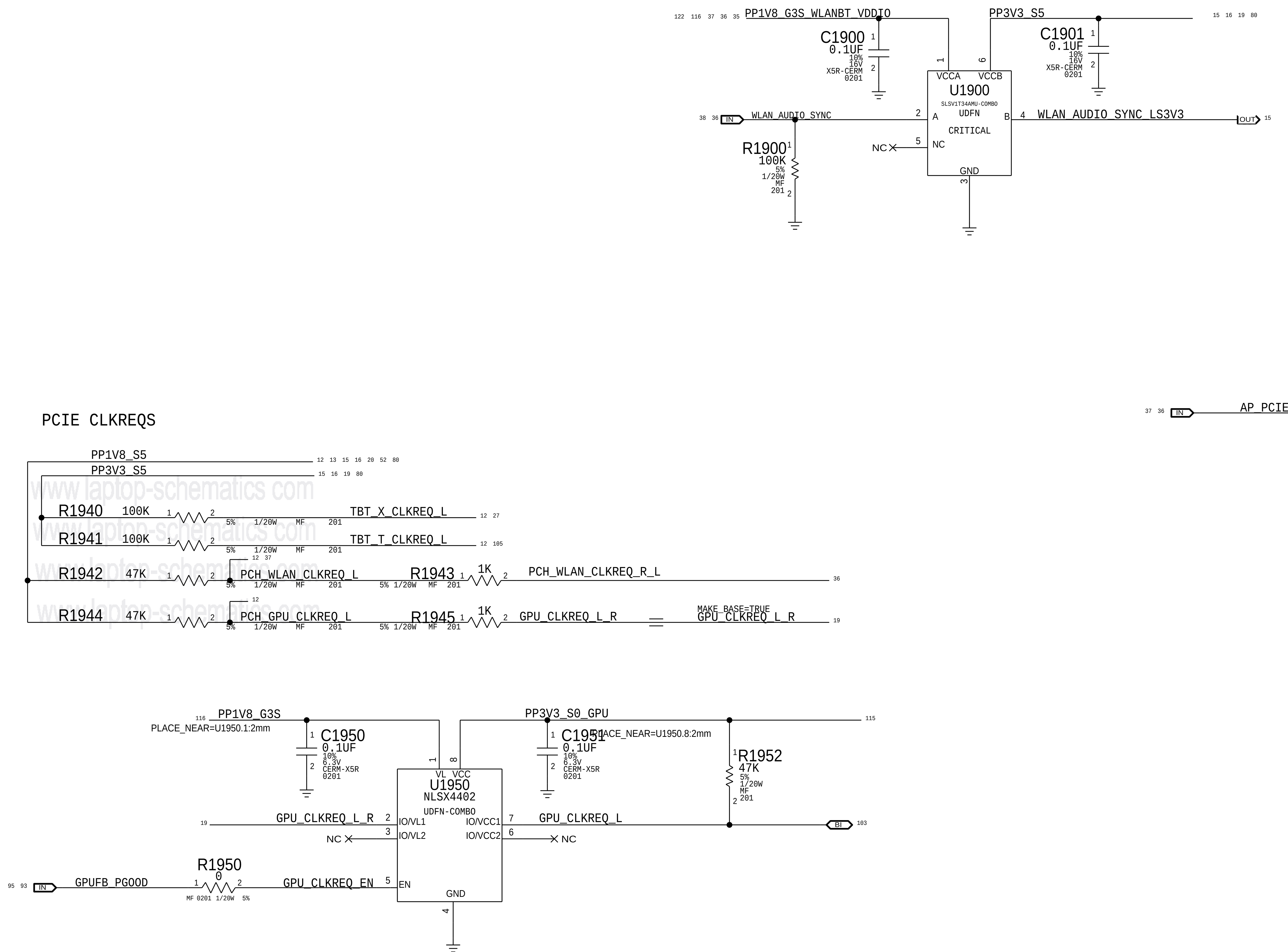
| DRAWING NUMBER | SIZE |
|---|---|
| 051-02888 | D |

REVISION 1.0.0

BRANCH EVT



BOM_COST_GROUP=DEBUG




FAB#:RC315

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE: Chipset Support 1

Apple Inc.

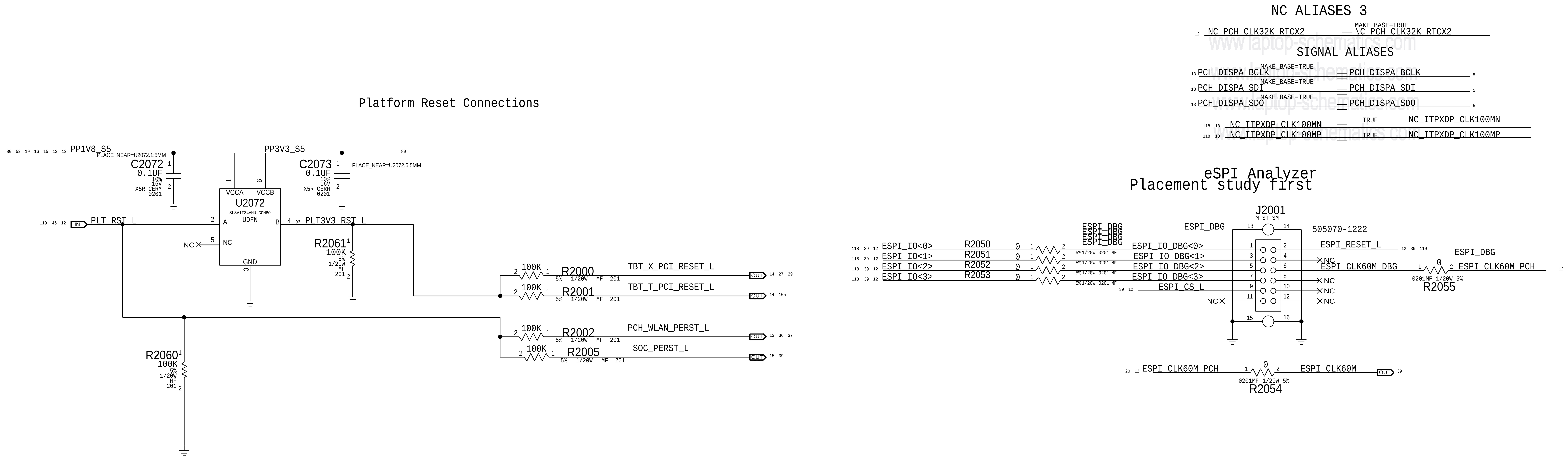
DRAWING NUMBER: 051-02888 SIZE: D

REVISION: 1.0.0

BRANCH: EVT

PAGE: 19 OF 200

SHEET: 19 OF 131




BOM_COST_GROUP=CPU & CHIPSET

| | |
|---|---|
| PAGE TITLE | Chipset Support 2 |
| Apple Inc. | DRAWING NUMBER 051-02888, SIZE D |
| | REVISION 1.0.0 |
| | BRANCH EVT |
| | PAGE 20 OF 200 |
| | SHEET 20 OF 131 |

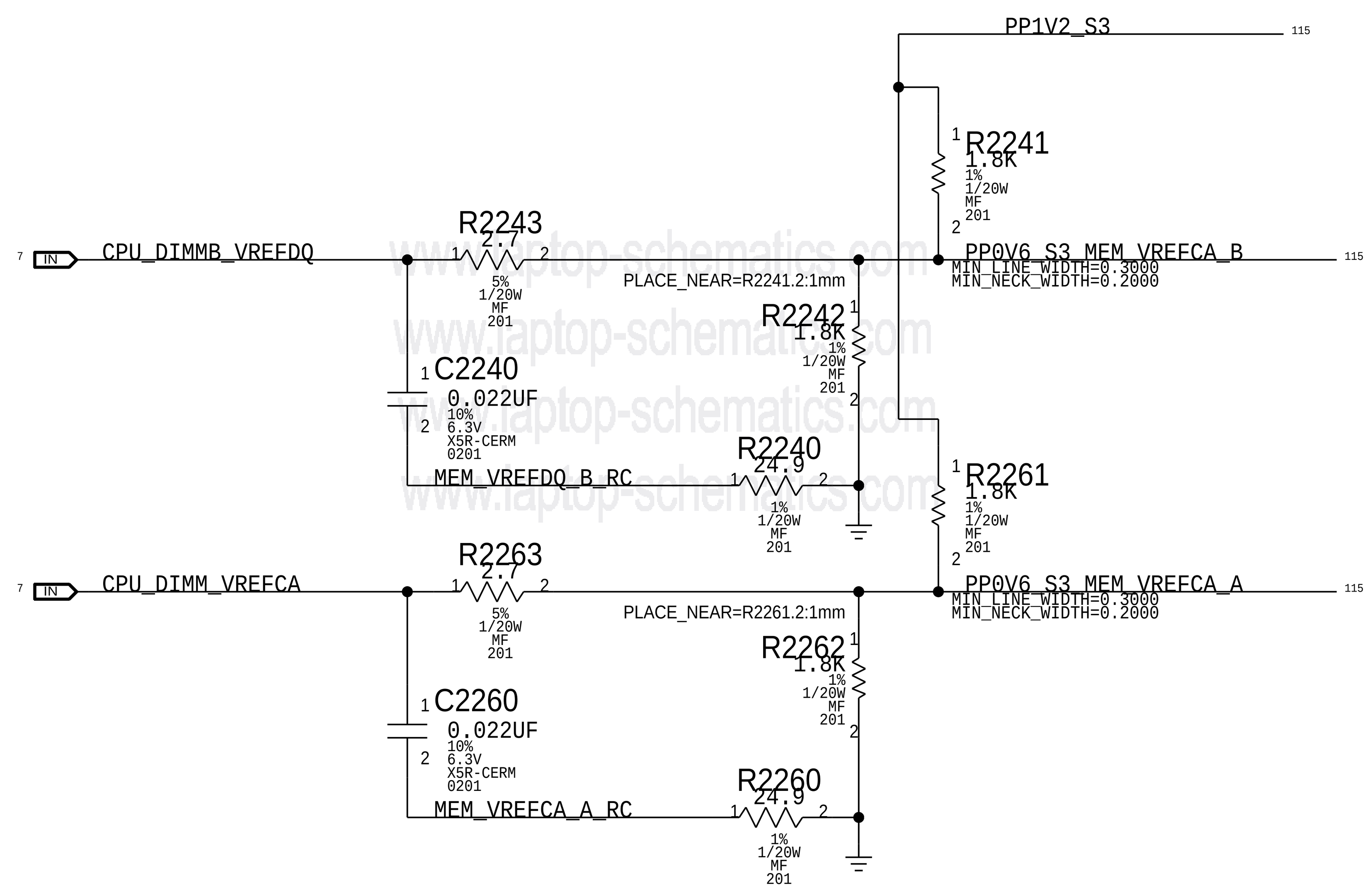
# CPU-Based Margining

## VRef Dividers

DDR4 VDDQ = 1.2V
-----------------
MIMINUM Step Size = 0.50% * VDDQ = 6.0mV per step
TYPICAL Step Size = 0.65% * VDDQ = 7.8mV per step
MAXIMUM Step Size = 0.80% * VDDQ = 9.6mV per step

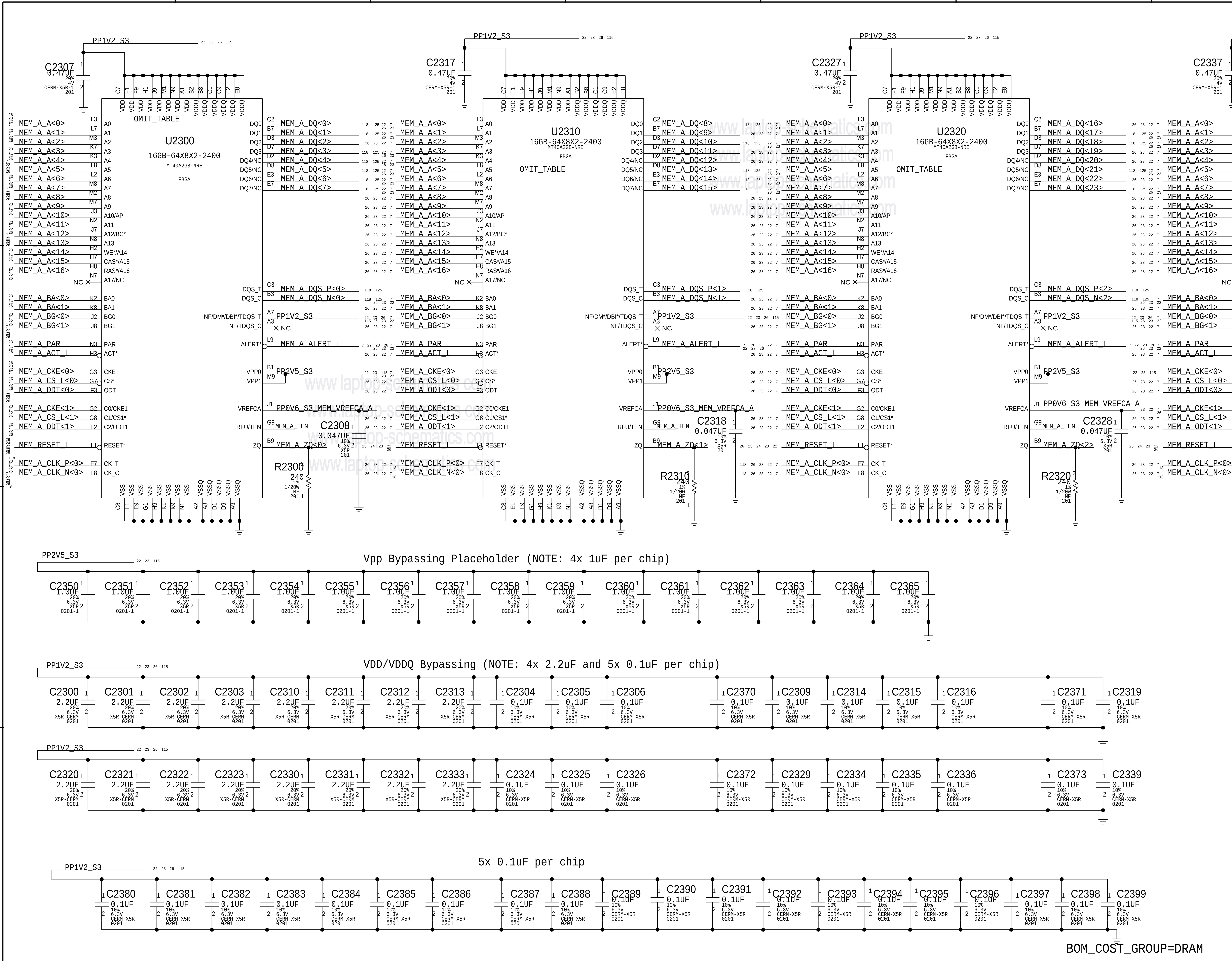
KBL_PLATFORM_GUIDE_Page.102_FOR_DDR4_X8_MEMORY_DOWN

DDR0_VREF_DQ = Not Used
DDR1_VREF_DQ = Reference For Channel B
DDR_VREF_CA = Reference For Channel A




BOM_COST_GROUP=DRAM

| | |
|---|---|
| PAGE TITLE | DDR4 VREF Margining |
| Apple Inc. | DRAWING NUMBER 051-02888 SIZE D |
| | REVISION 1.0.0 |
| | BRANCH EVT |
| | PAGE 22 OF 200 |
| | SHEET 21 OF 131 |




PAGE TITLE

# DDR4 SDRAM Channel A 1

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

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BRANCH EVT

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SHEET 22 OF 131


PAGE TITLE

# DDR4 SDRAM Channel A 2

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

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BRANCH EVT

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PAGE TITLE

# DDR4 SDRAM Channel B 1

Apple Inc.

| DRAWING NUMBER | SIZE |
|---|---|
| 051-02888 | D |

REVISION 1.0.0

BRANCH EVT

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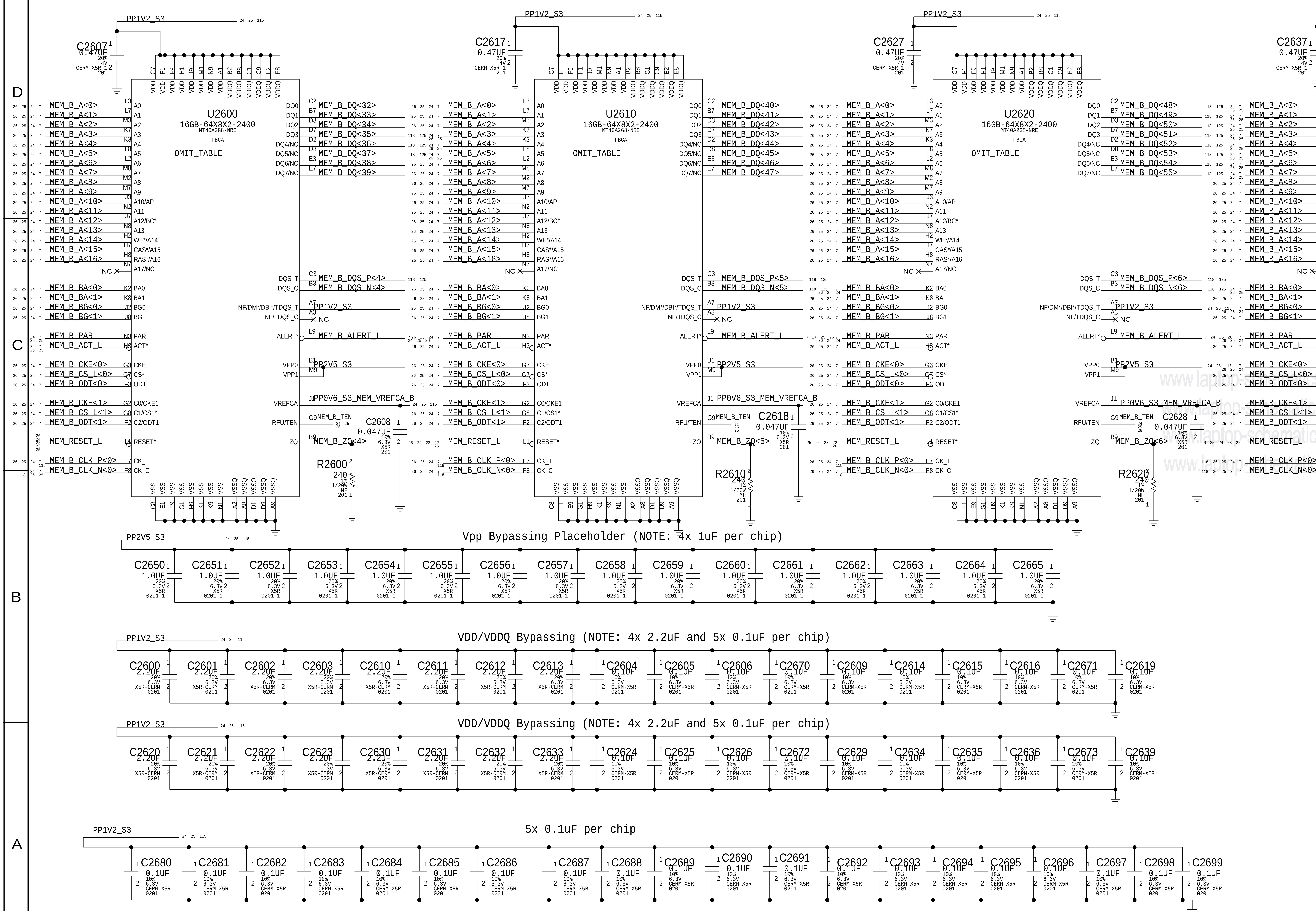


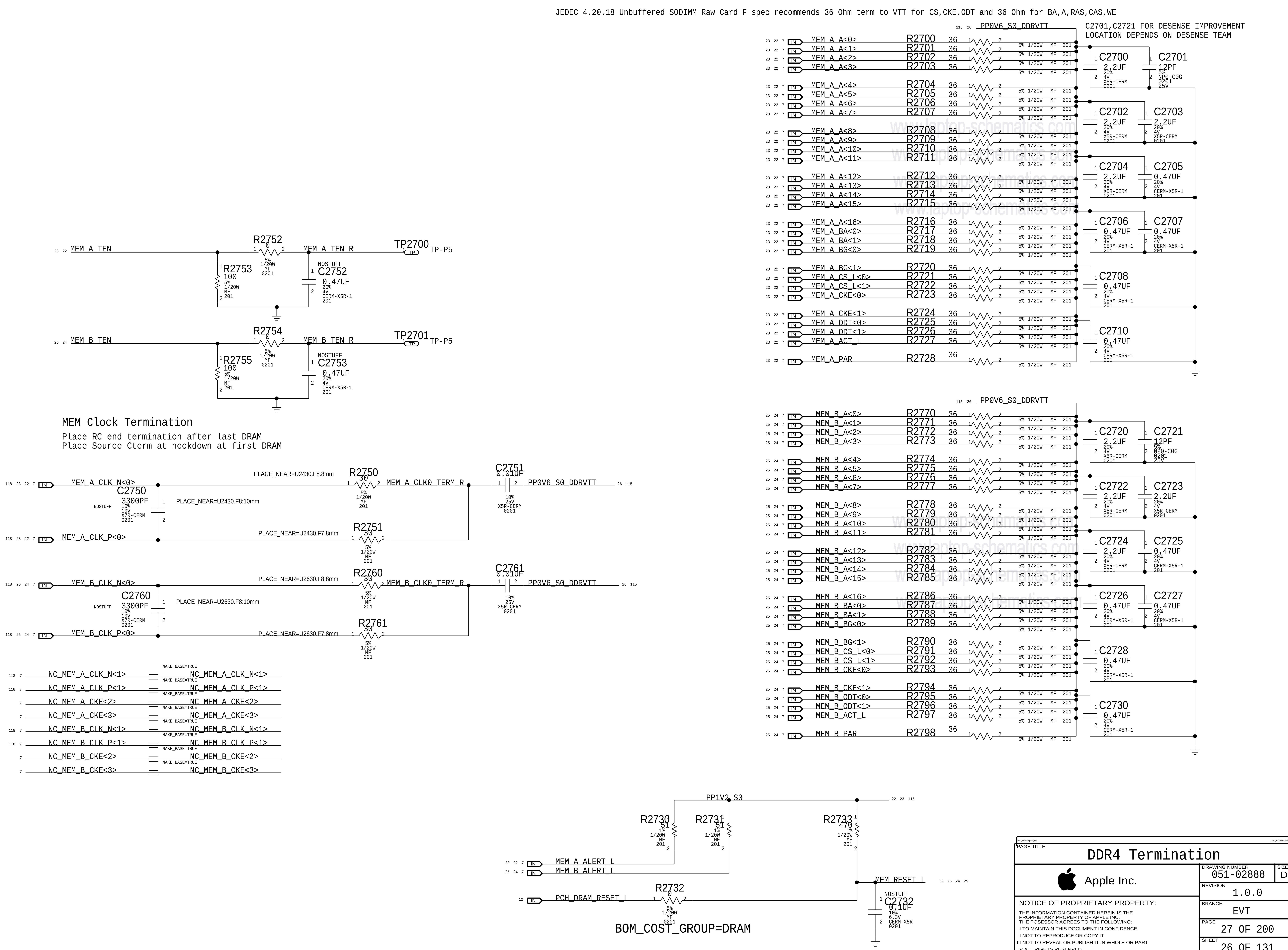

BOM_COST_GROUP=DRAM

PAGE TITLE

# DDR4 SDRAM Channel B 2

Apple Inc.

| | |
|---|---|
| DRAWING NUMBER | 051-02888 |
| SIZE | D |
| REVISION | 1.0.0 |
| BRANCH | EVT |
| PAGE | 26 OF 200 |
| SHEET | 25 OF 131 |




DDR4 Termination

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT

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SYNC_MASTER=J132 SYNC_DATE=05/11/2017

PAGE TITLE

# USB-C HIGH SPEED 1

Apple Inc.

| | |
|---|---|
| DRAWING NUMBER | 051-02888 |
| SIZE | D |
| REVISION | 1.0.0 |
| BRANCH | EVT |
| PAGE | 28 OF 200 |
| SHEET | 27 OF 131 |


SYNC_MASTER=ADITYA SYNC_DATE=03/30/2017

PAGE TITLE

USB-C HIGH SPEED 2

Apple Inc.

| DRAWING NUMBER | SIZE |
|---|---|
| 051-02888 | D |

REVISION: 1.0.0

BRANCH: EVT

PAGE: 29 OF 200

SHEET: 28 OF 131


PAGE TITLE

USB-C X Support

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT

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# PRIMARY ACE USB-C PORT CONTROLLER (UPC)


BOM_COST_GROUP=USB-C

SYNC_MASTER=ZIFENG SYNC_DATE=05/26/2017

PAGE TITLE

USB-C PORT CONTROLLER A

| DRAWING NUMBER | SIZE |
|---|---|
| 051-02888 | D |

Apple Inc.

REVISION 1.0.0

BRANCH EVT

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SHEET 30 OF 131

# SECONDARY ACE USB-C PORT CONTROLLER (UPC)


BOM_COST_GROUP=USB-C

SYNC_MASTER=ZIFENG SYNC_DATE=05/26/2017

PAGE TITLE: USB-C PORT CONTROLLER B

Apple Inc.

DRAWING NUMBER: 051-02888 SIZE: D

REVISION: 1.0.0

BRANCH: EVT

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LAST CHANGE: Wed Apr 1 22:57:37 2015

PAGE TITLE

# USB-C CONNECTOR A

Apple Inc.

| DRAWING NUMBER | SIZE |
| --- | --- |
| 051-02888 | D |

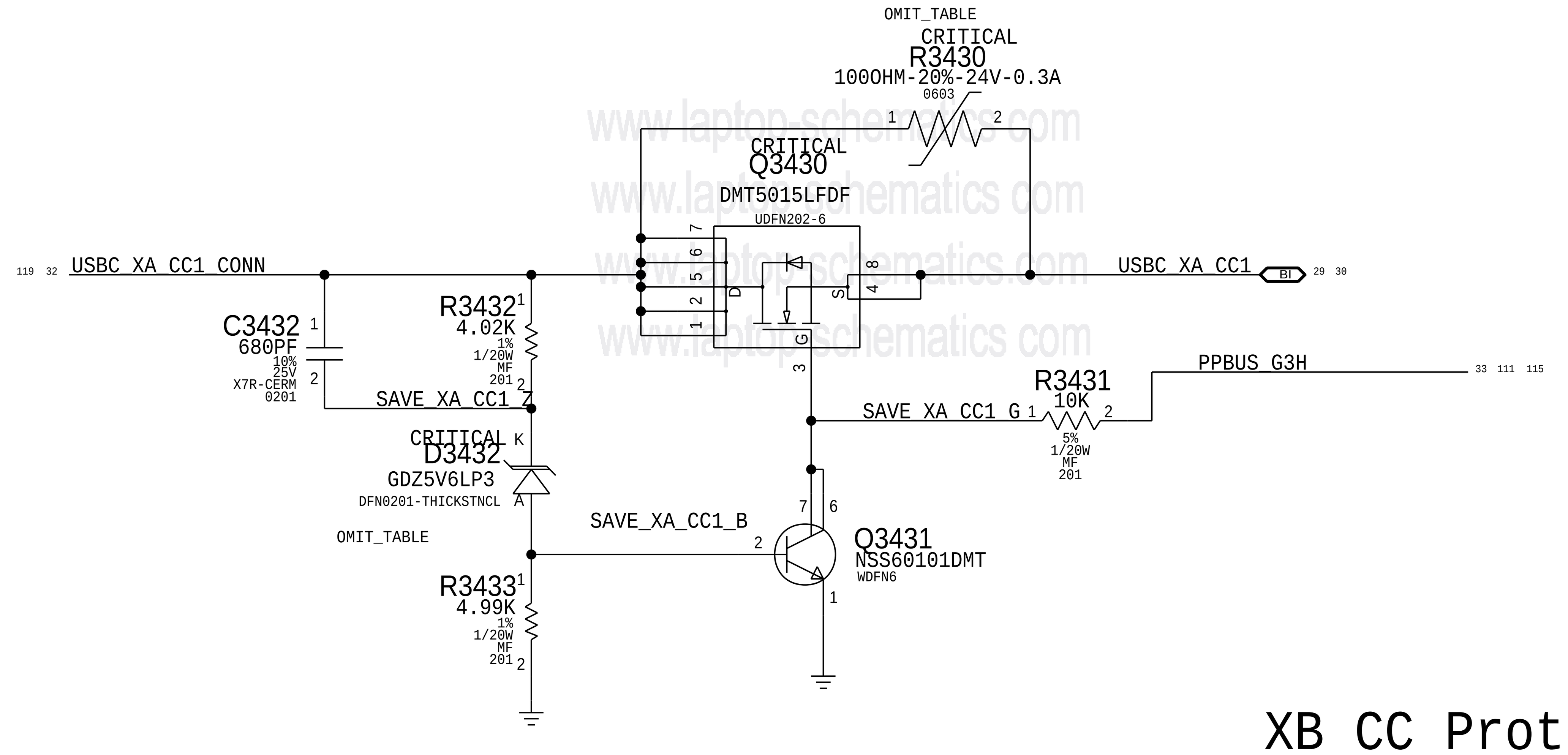
REVISION: 1.0.0

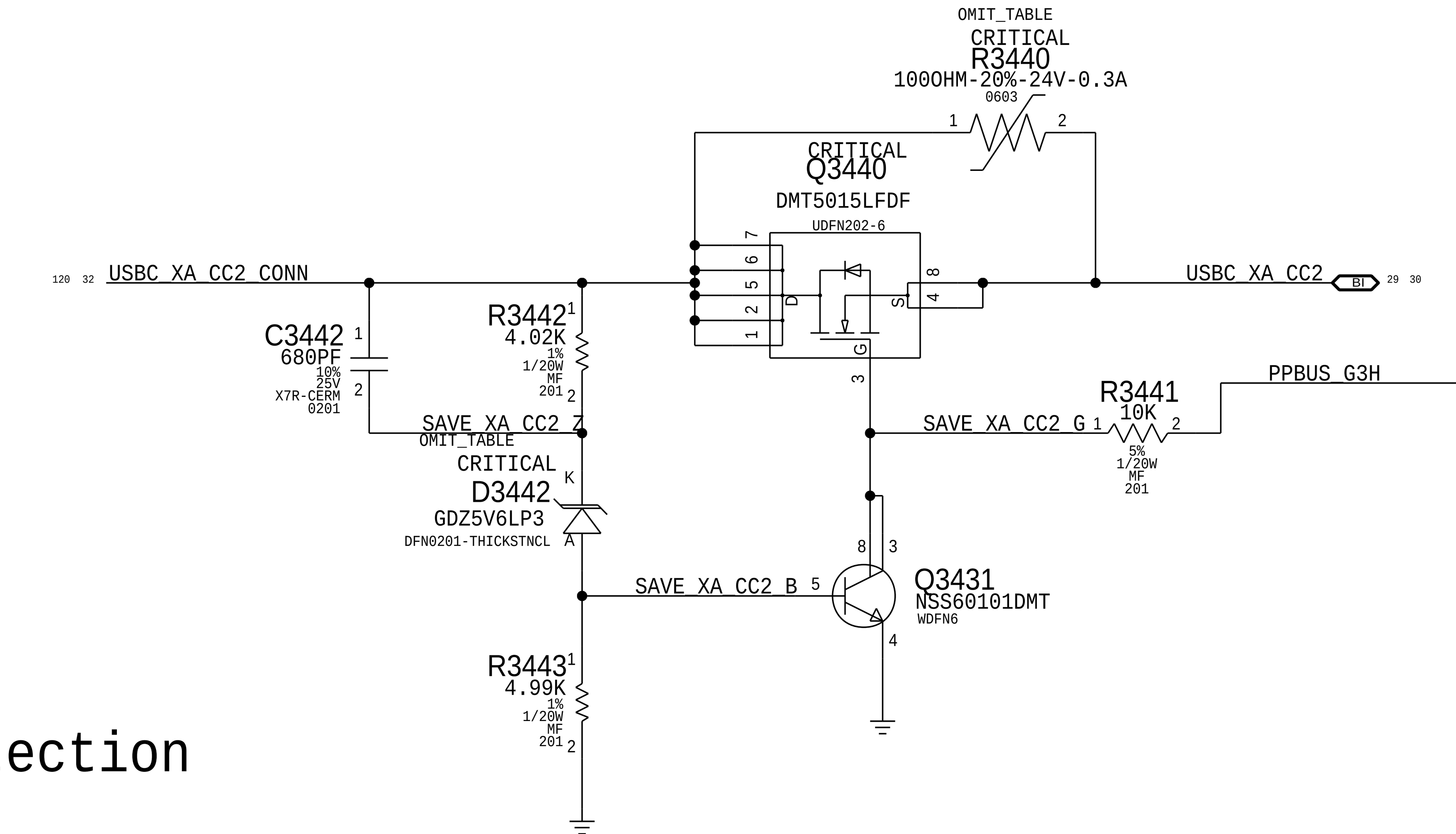
BRANCH: EVT

PAGE: 33 OF 200

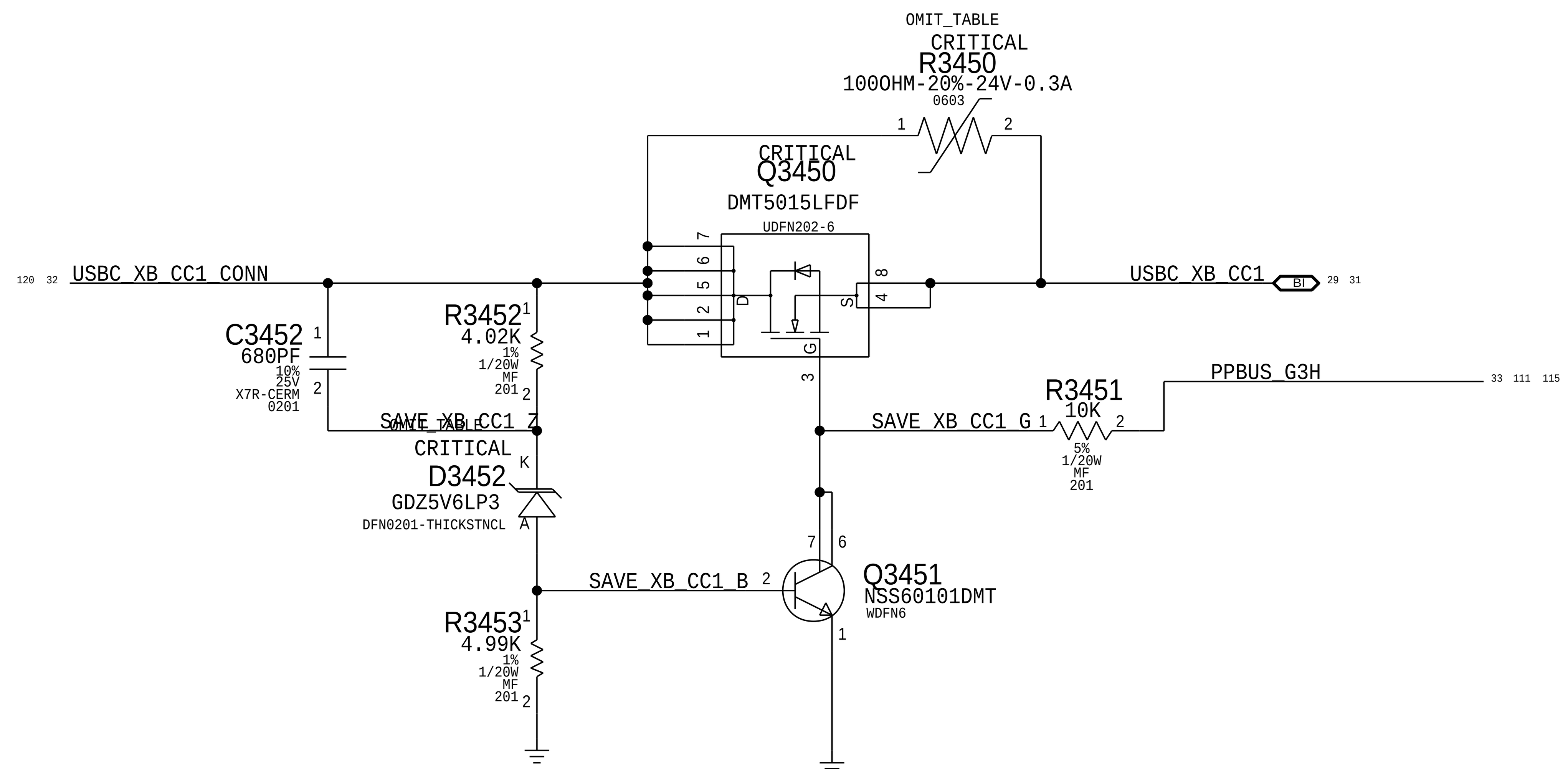
SHEET: 32 OF 131

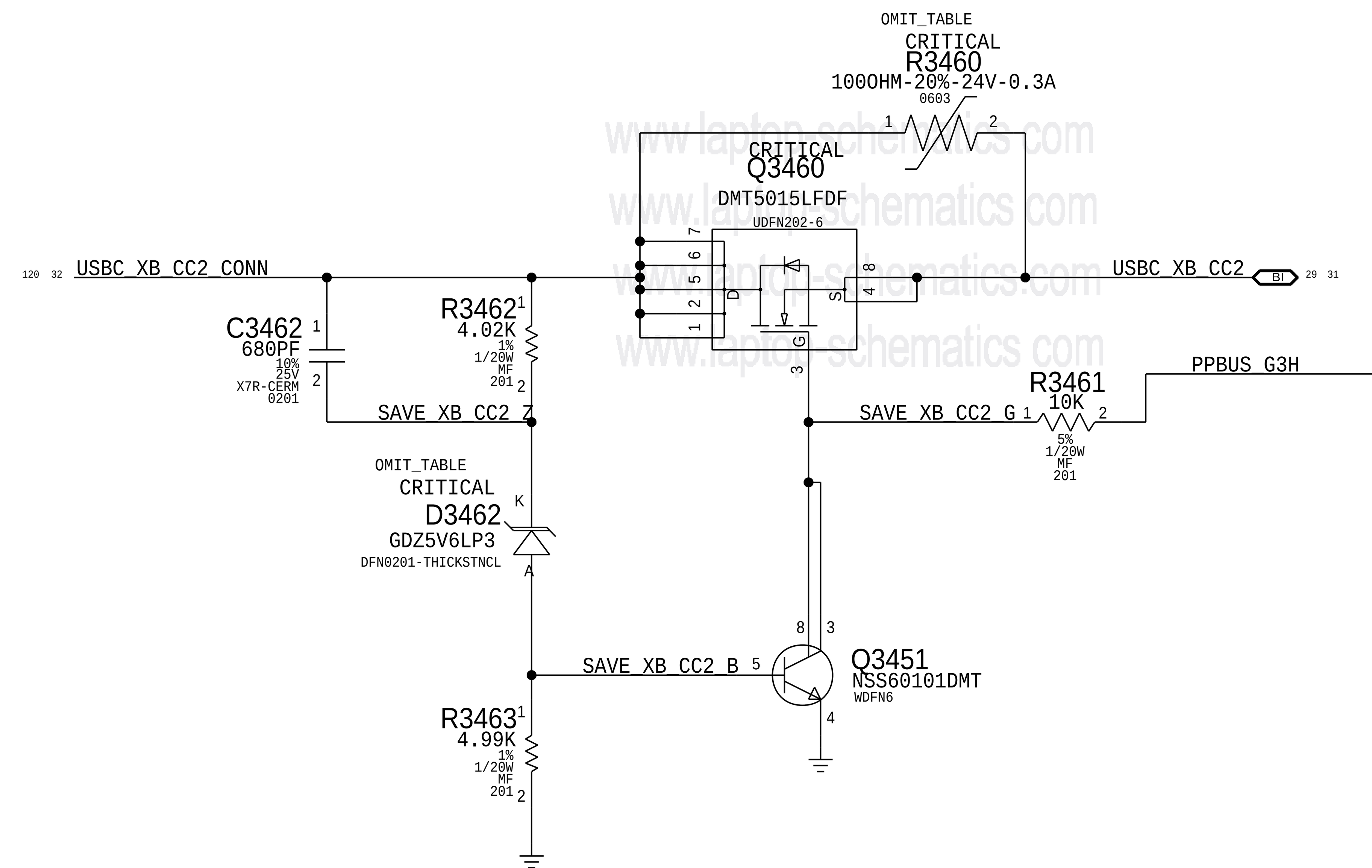
# XA CC Protection




# XB CC Protection




| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 371S00181 | 4 | DIODE,ZENER,5.6V,250MW,DFN0201 | D3432,D3442,D3452,D3462 | CRITICAL | |
| 740S00050 | 4 | FUSE,THRMSTR,PTC,470OHM,24V,60MA,0603 | R3430,R3440,R3450,R3460 | CRITICAL | |

BOM_COST_GROUP=USB-C

DESIGN: X1032/MLB_P4BP
LAST CHANGE: Fri Jan 6 16:01:21 2017

PAGE TITLE: USBC X Connector Support

Apple Inc.

DRAWING NUMBER: 051-02888 SIZE: D
REVISION: 1.0.0
BRANCH: EVT
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PAGE TITLE

# TBT 5V REGULATOR

Apple Inc.

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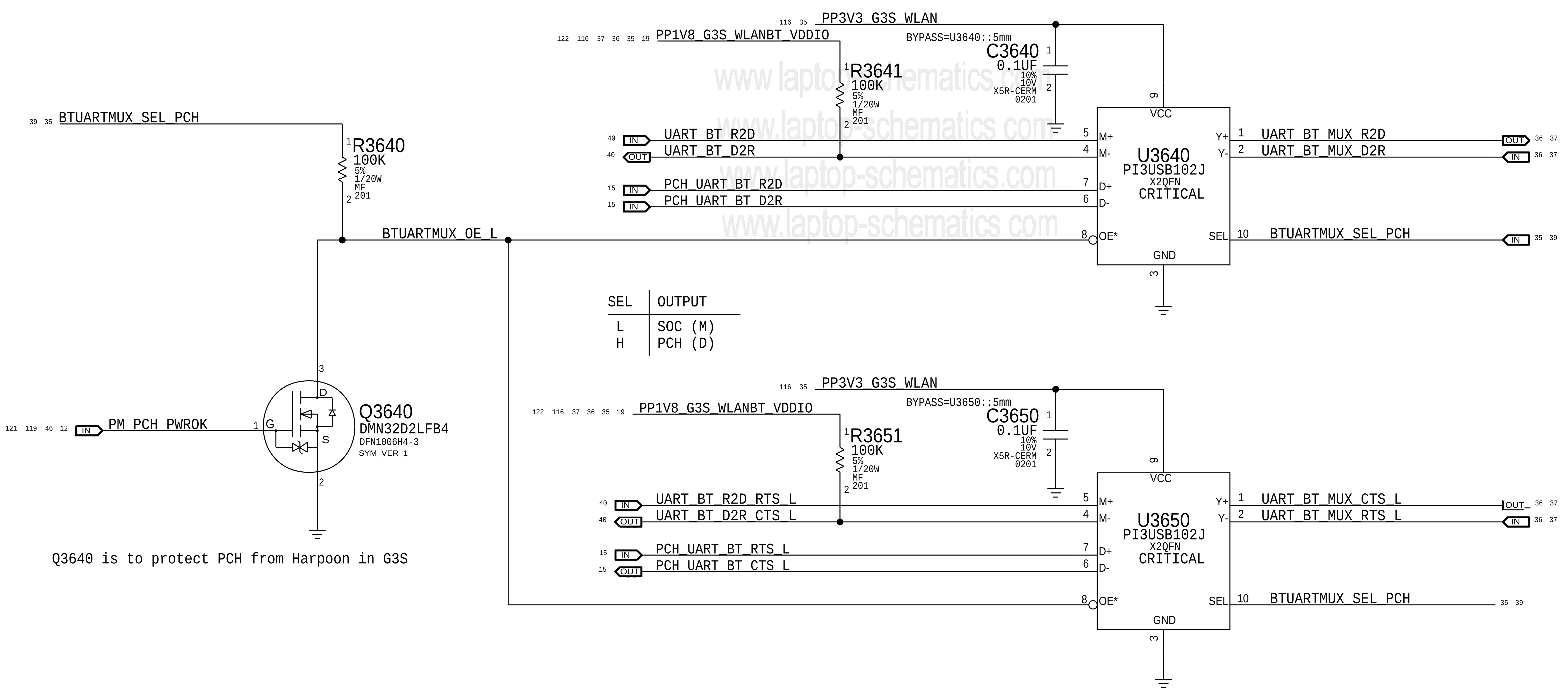
REVISION 1.0.0

BRANCH EVT



BOM_COST_GROUP=USB-C






BOM_COST_GROUP=WIRELESS

SYNC_MASTER=J132 SYNC_DATE=03/29/2017

PAGE TITLE

# WIFI/BT: Support

Apple Inc.

| | |
|---|---|
| DRAWING NUMBER | 051-02888 |
| SIZE | D |
| REVISION | 1.0.0 |
| BRANCH | EVT |
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| SHEET | 35 OF 131 |


PAGE TITLE

WIFI/BT: MODULE 1

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT

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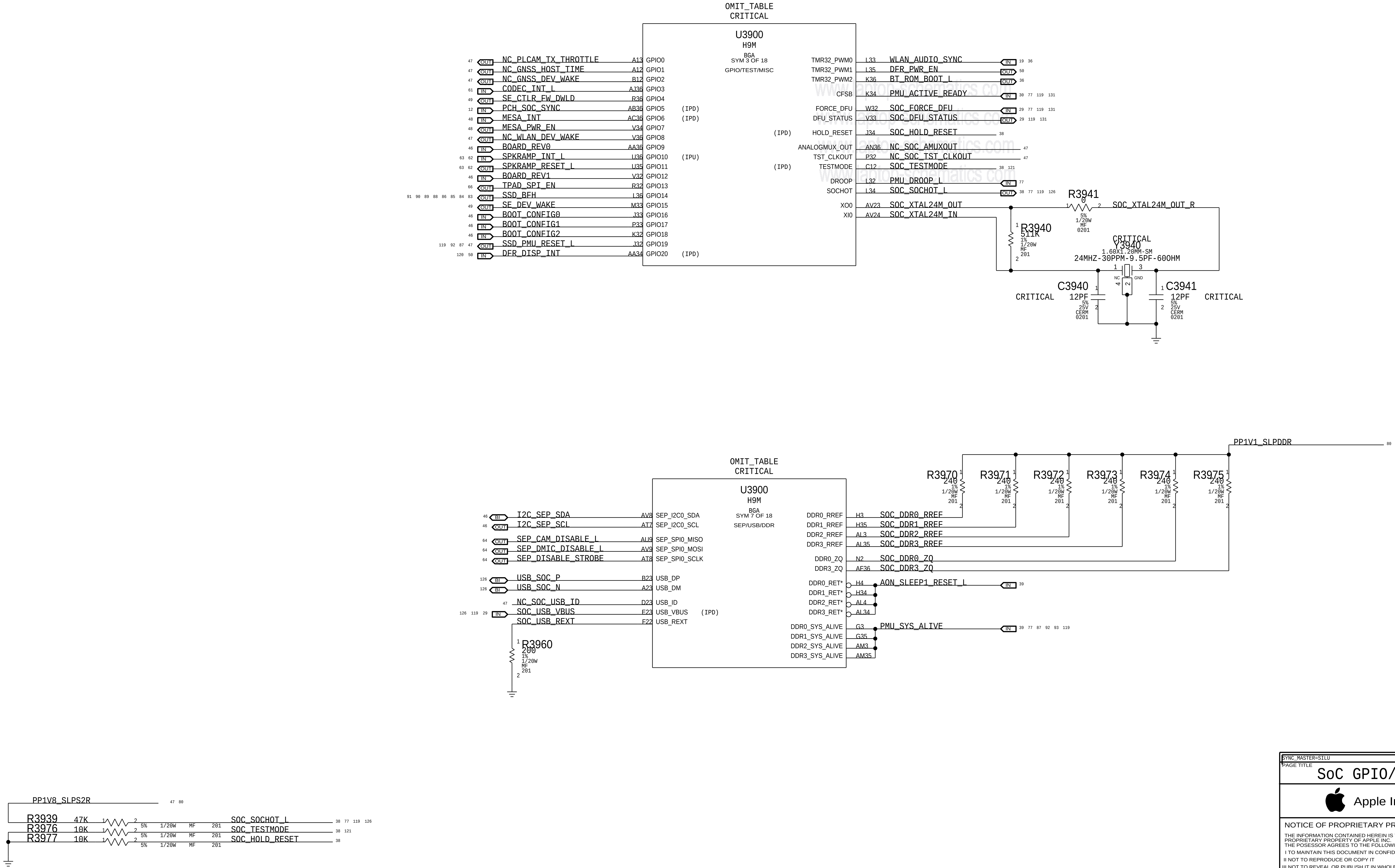
| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 155S00196 | 2 | FILTER,DIPLEXER,LB/HB,LGA6 | U3810,U3830 | CRITICAL | RF_TUNING |


BOM_COST_GROUP=WIRELESS

Note 1) IPU represents SW configured state, not HW default




BOM_COST_GROUP=SOC

SYNC_MASTER=SILU SYNC_DATE=05/26/2017

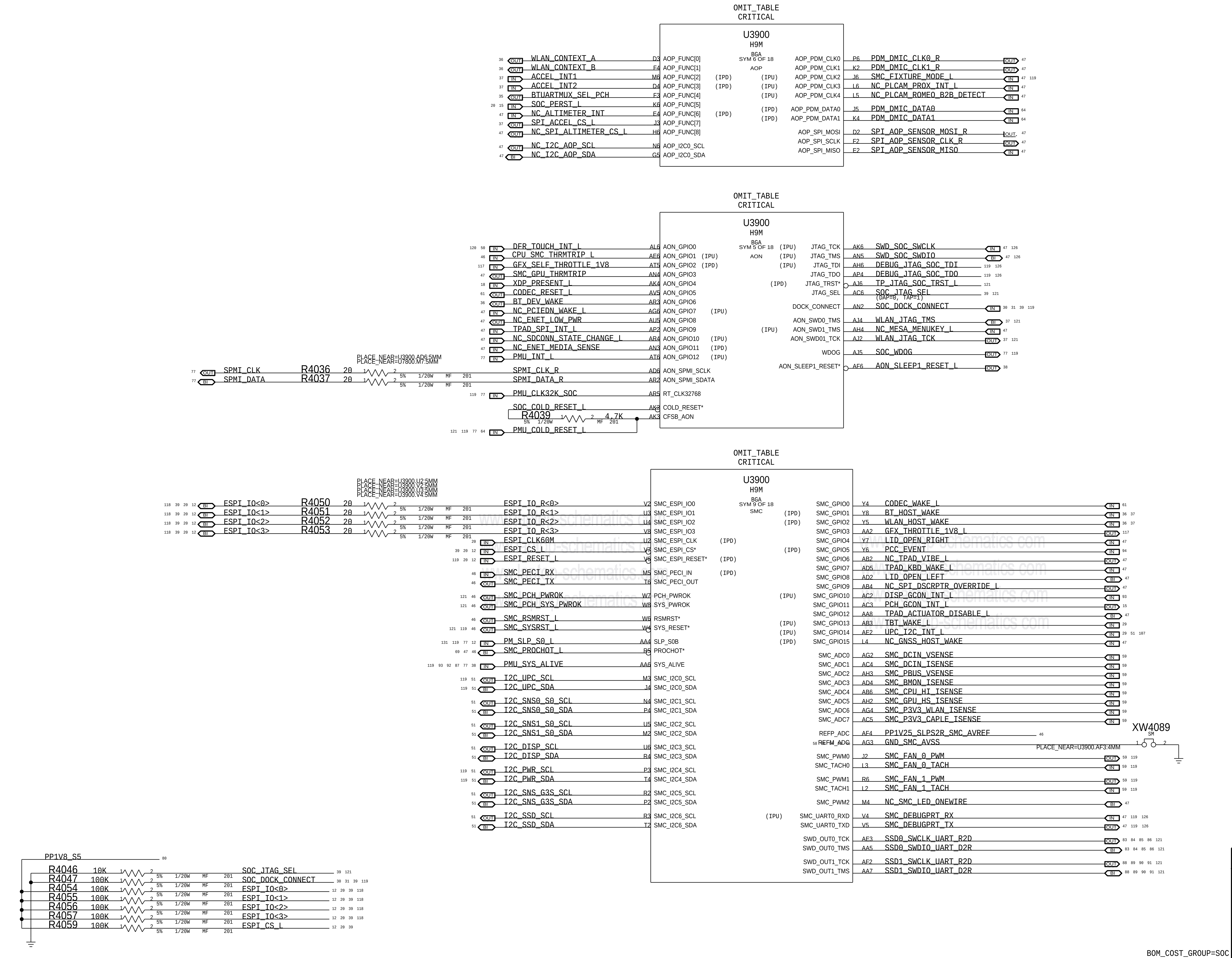
PAGE TITLE
SoC GPIO/SEP/USB/DDR/Test

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT




BOM_COST_GROUP=SOC

SYNC_MASTER=SILU SYNC_DATE=03/22/2017

PAGE TITLE
## SoC AOP/AON/SMC

Apple Inc.

| DRAWING NUMBER | SIZE |
|---|---|
| 051-02888 | D |

REVISION 1.0.0

BRANCH EVT

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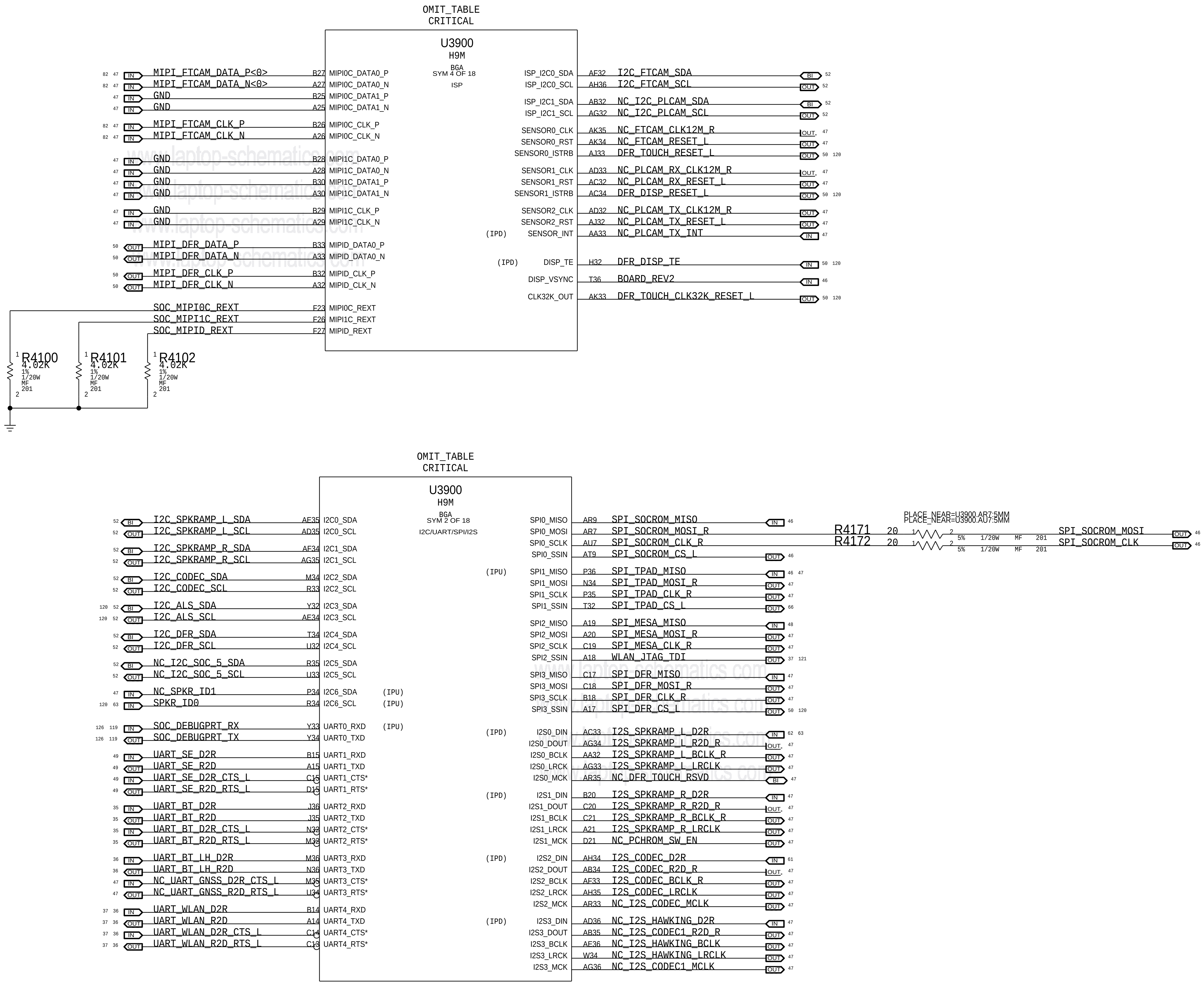



SYNC_MASTER=H9M and PMIC SYNC_DATE=03/02/2017

PAGE TITLE
# SoC ISP/I2C/UART/SPI/I2S

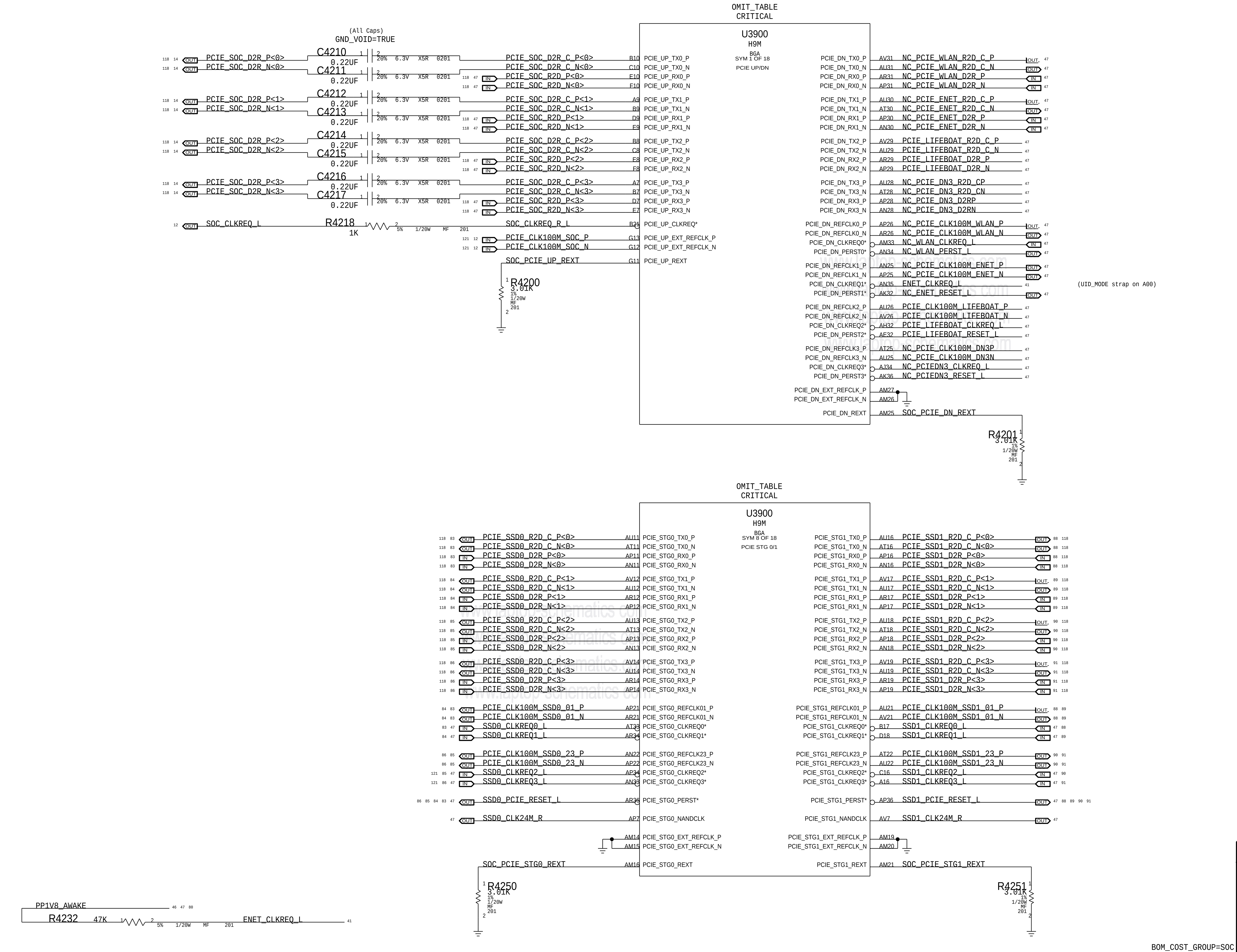
Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT

BOM_COST_GROUP=SOC




SYNC_MASTER=SILU SYNC_DATE=03/15/2017

PAGE TITLE

# SoC PCIe

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT

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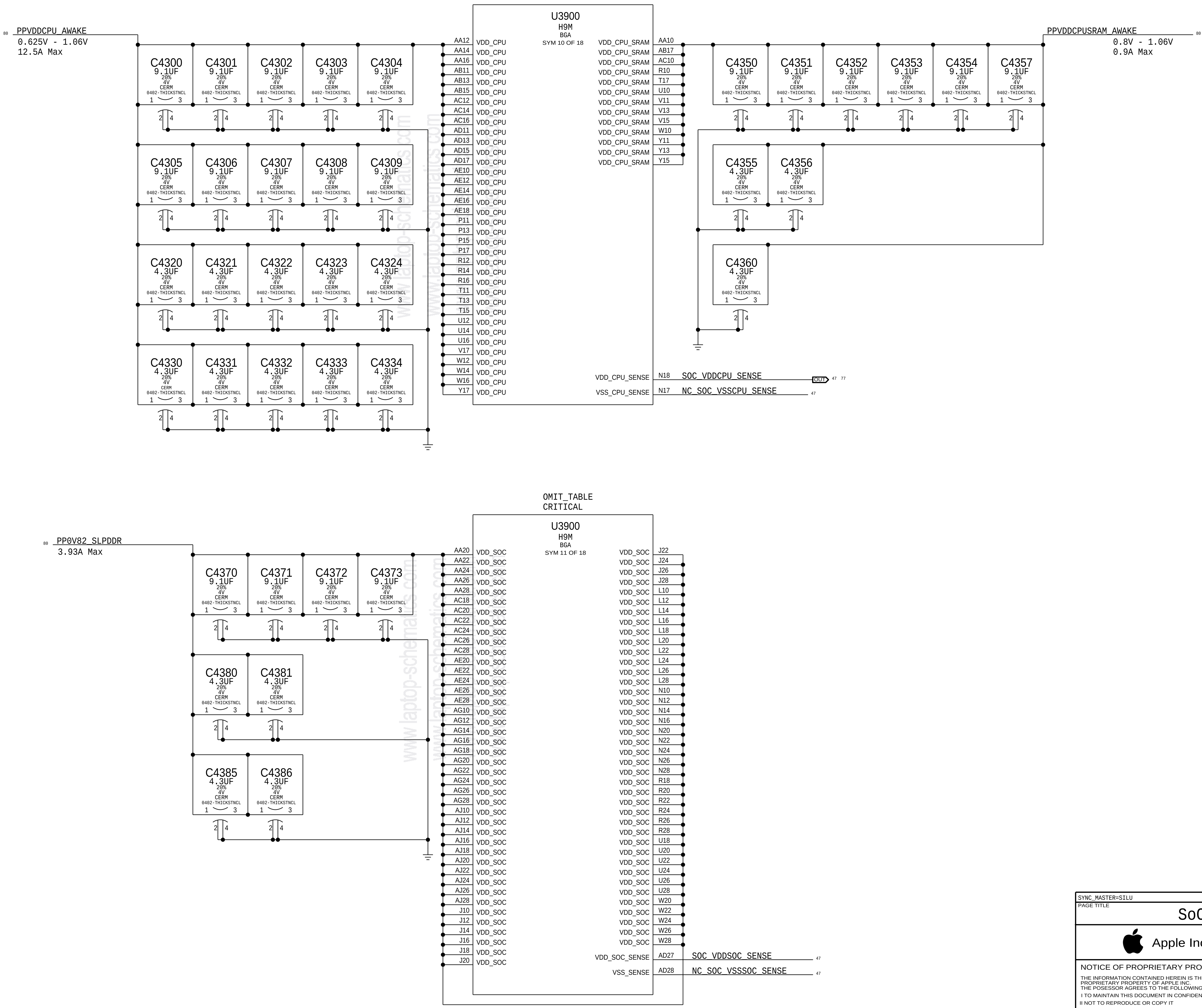
Current estimates @ 105C & 2GB from Gibraltar Power Specification Rev 0.5.3






BOM_COST_GROUP=SOC

SYNC_MASTER=SILU SYNC_DATE=04/04/2017

PAGE TITLE
SoC Power 1

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D
REVISION 1.0.0
BRANCH EVT
PAGE 43 OF 200
SHEET 42 OF 131

Current estimates @ 105C & 2GB from Gibraltar Power Specification Rev 0.5.3


BOM_COST_GROUP=SOC

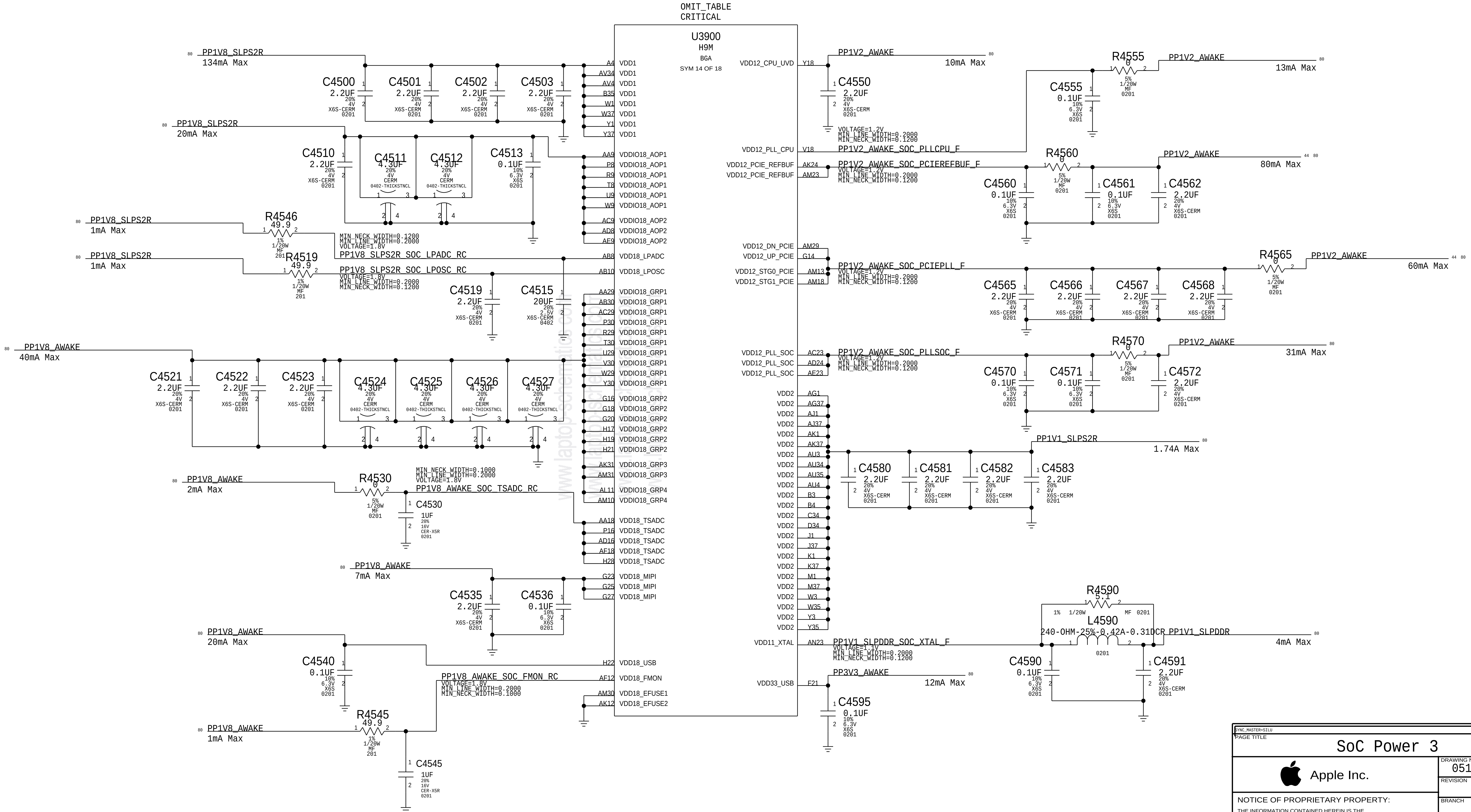
SYNC_MASTER=H9M and PMIC SYNC_DATE=03/02/2017

PAGE TITLE
SoC Power 2

Apple Inc.

| | |
|---|---|
| DRAWING NUMBER | 051-02888 |
| SIZE | D |
| REVISION | 1.0.0 |
| BRANCH | EVT |
| PAGE | 44 OF 200 |
| SHEET | 43 OF 131 |

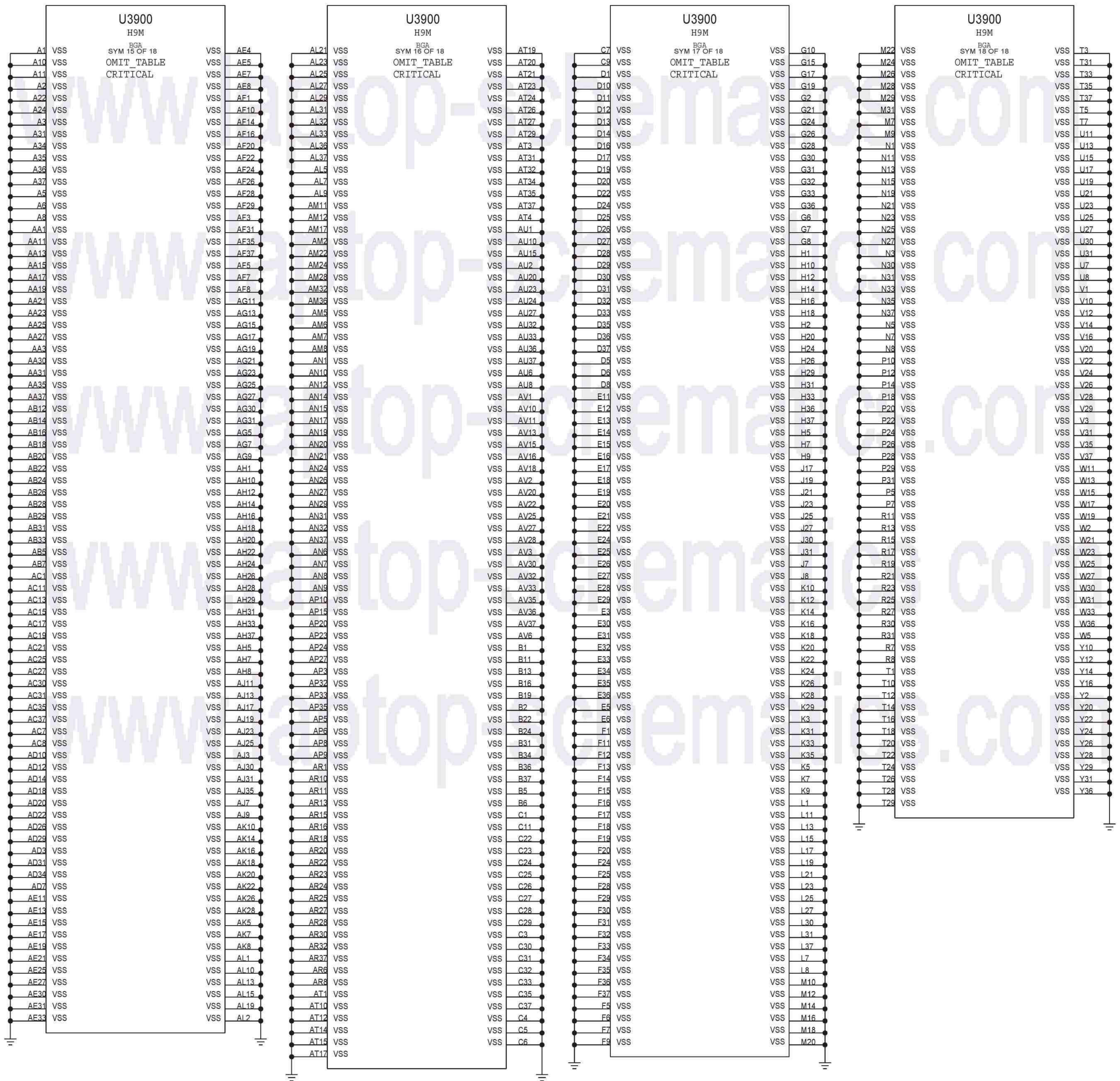
Current estimates @ 105C & 2GB from Gibraltar Power Specification Rev 0.5.3




BOM_COST_GROUP=SOC

| | |
|---|---|
| PAGE TITLE | SoC Power 3 |
| DRAWING NUMBER | 051-02888 |
| SIZE | D |
| REVISION | 1.0.0 |
| BRANCH | EVT |
| PAGE | 45 OF 200 |
| SHEET | 44 OF 131 |

Apple Inc.






BOM_COST_GROUP=SOC

SYNC_MASTER=H9M and PMIC SYNC_DATE=03/02/2017

PAGE TITLE

# SoC Ground

Apple Inc.

| DRAWING NUMBER | SIZE |
|---|---|
| 051-02888 | D |
| REVISION 1.0.0 | |
| BRANCH EVT | |
| PAGE 46 OF 200 | |
| SHEET 45 OF 131 | |


| BOOTCFG0 | Test Mode |
|---|---|
| 0 | Disabled |
| 1 | Enabled |

| BOOTCFG2 | BOOTCFG1 | Frequency |
|---|---|---|
| 0 | 0 | 40 MHz |
| 0 | 1 | 6 MHz |
| 1 | 0 | 24 MHz |
| 1 | 1 | Invalid |

| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 343S00234 | 1 | IC,SLG4AP41484,THERMTRIP,STQFN14 | U4740 | CRITICAL | |

SYNC_MASTER=SILU SYNC_DATE=08/09/2017

PAGE TITLE: SoC Shared Support

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT


| BOM GROUP | BOM OPTIONS |
|---|---|
| BOARD_REV:111 | BOARDREV2,BOARDREV1,BOARDREV0 |
| BOARD_REV:110 | BOARDREV2,BOARDREV1 |
| BOARD_REV:101 | BOARDREV2,BOARDREV0 |
| BOARD_REV:100 | BOARDREV2 |
| BOARD_REV:011 | BOARDREV1,BOARDREV0 |
| BOARD_REV:010 | BOARDREV1 |
| BOARD_REV:001 | BOARDREV0 |
| BOARD_REV:000 | |

Proto 0
Proto 1
Proto 2
Pre-EVT
EVT
DVT
PVT

BOM_COST_GROUP=SOC

PAGE TITLE

# SoC Project Support

Apple Inc.

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REVISION 1.0.0
BRANCH EVT
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BOM_COST_GROUP=T151

PAGE TITLE: MESA

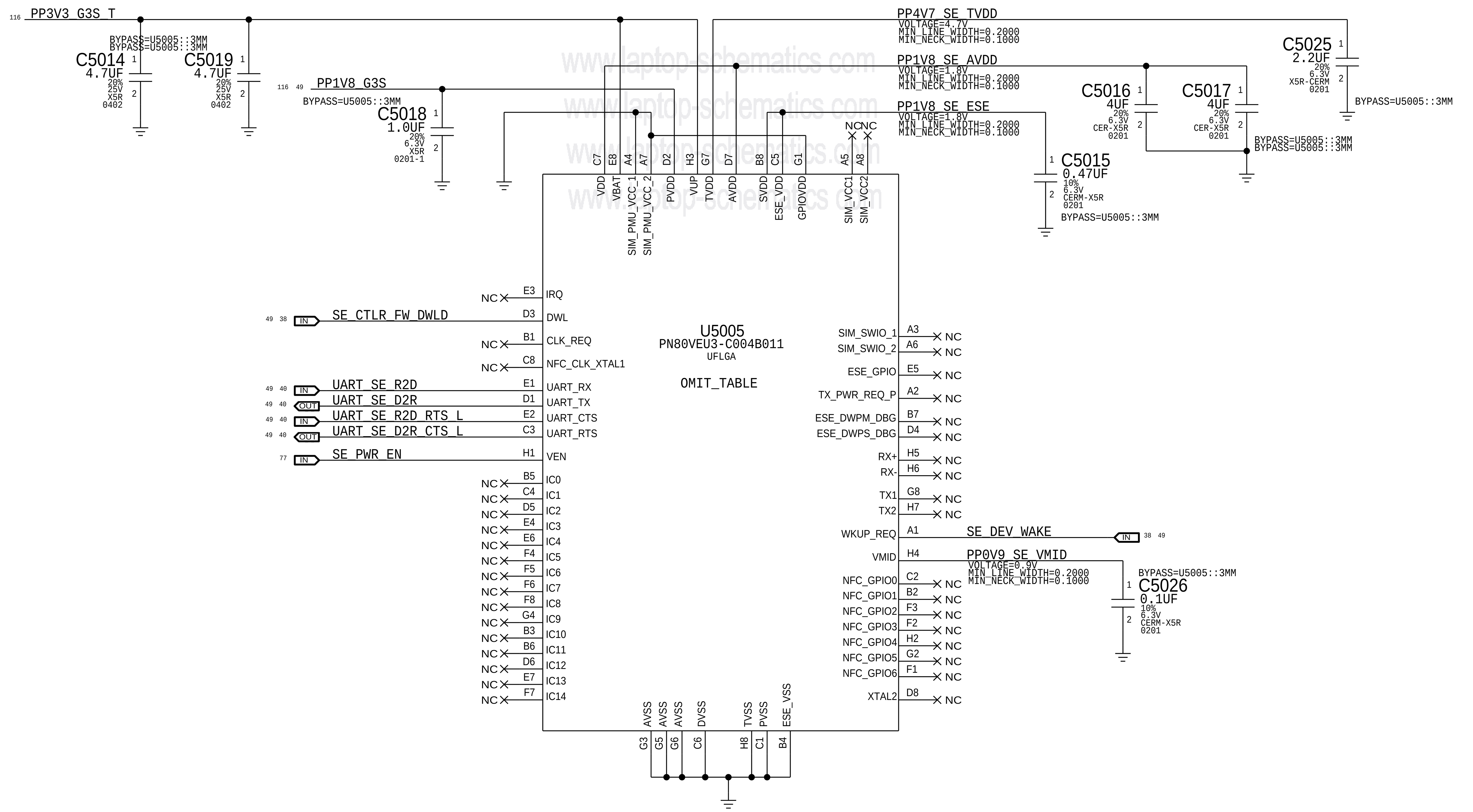
Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

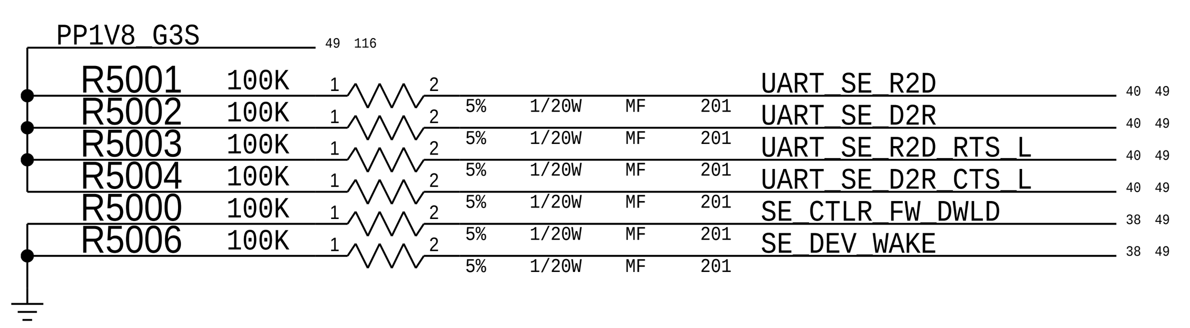


DRAWING NUMBER: 051-02888 SIZE: D
REVISION: 1.0.0
BRANCH: EVT

# STOCKHOLM 2017




| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 998-11132 | 1 | IC,RTM4,ES2.1,PN80V,DEV,UFLGA64 | U5005 | CRITICAL | SE:DEV_2017 |
| 338S00253 | 1 | IC,RTM4,ES2.1,PN80V,PRD,UFLGA64 | U5005 | CRITICAL | SE:PROD_2017 |




BOM_COST_GROUP=T151

SYNC_MASTER=SILU SYNC_DATE=05/05/2017

PAGE TITLE
Secure Element

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

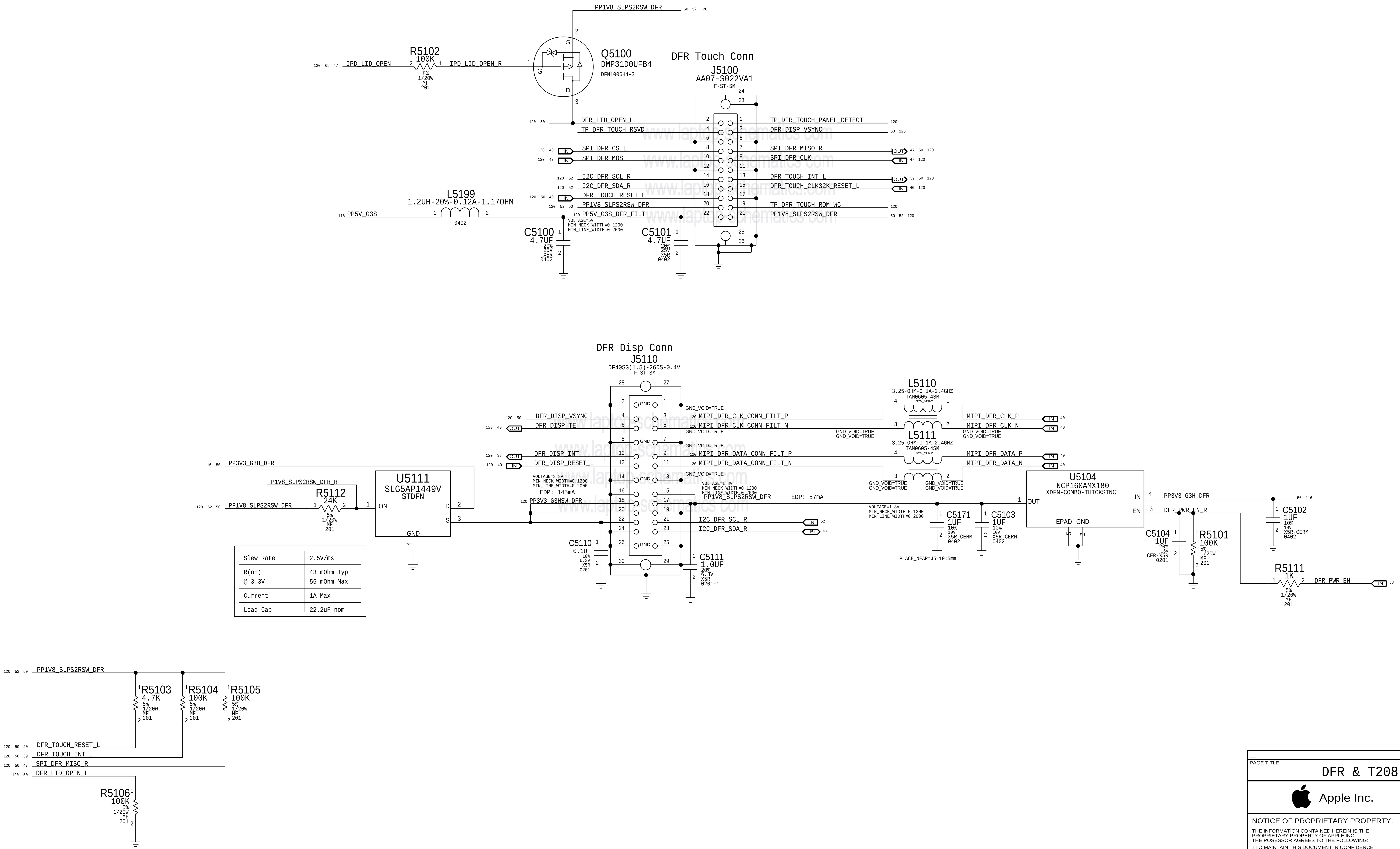
REVISION 1.0.0

BRANCH EVT

PAGE 50 OF 200

SHEET 49 OF 131

# T208 Support




| Slew Rate | 2.5V/ms |
|---|---|
| R(on) @ 3.3V | 43 mOhm Typ<br>55 mOhm Max |
| Current | 1A Max |
| Load Cap | 22.2uF nom |

BOM_COST_GROUP=T151

PAGE TITLE
DFR & T208 Support

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

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BRANCH EVT

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| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 117S0059 | 2 | RES,2.2K,1/20W,0201,SMD,MF | R5230,R5231 | | OCARINA_I2C:2.2K |

I2C Connections 1

Apple Inc.

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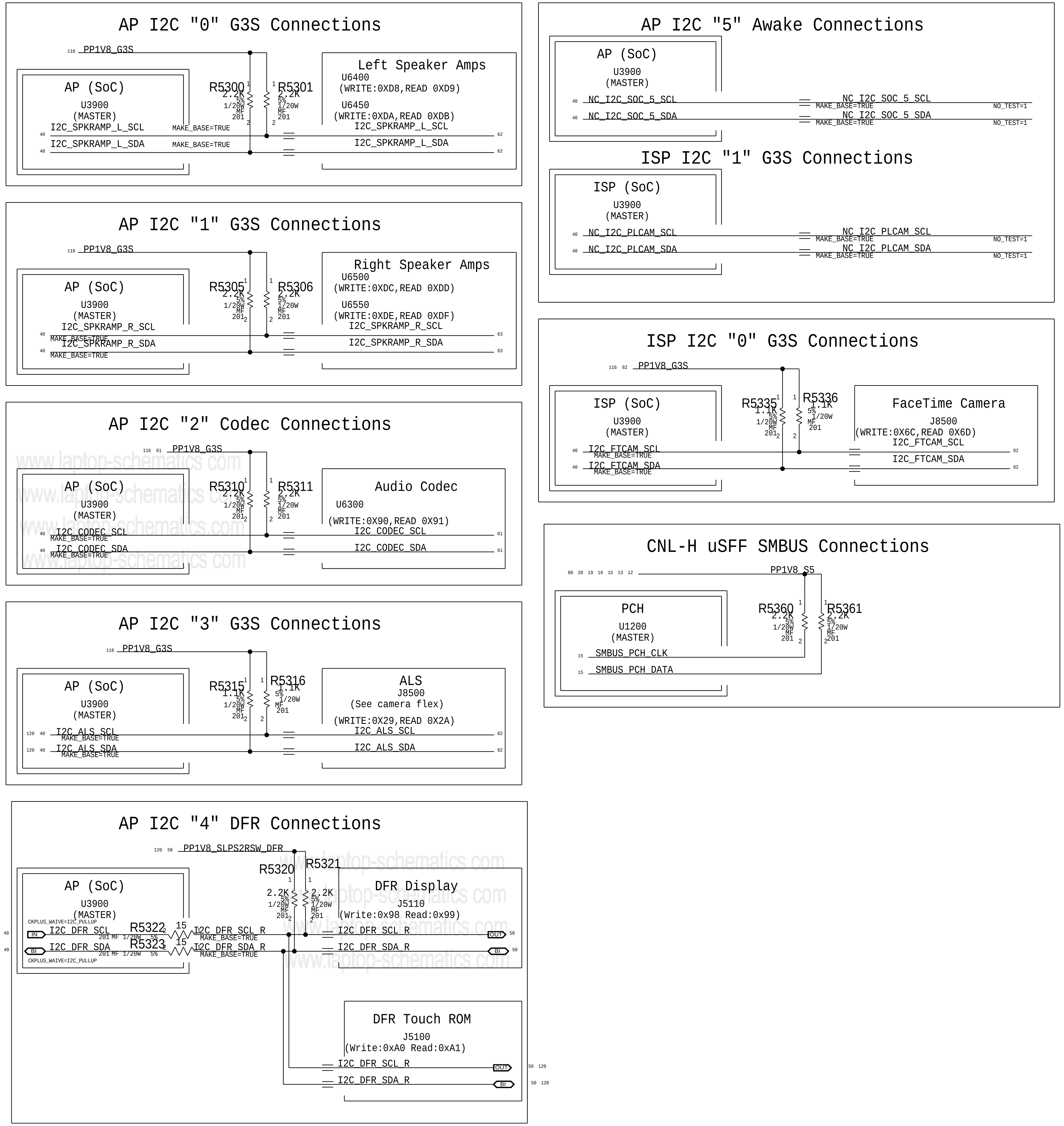
BRANCH EVT

PAGE 52 OF 200

SHEET 51 OF 131




## I2C Device Address

| Device | SMC IF | ADDR. (8b) |
|---|---|---|
| ACE XA | I2C0 | 0X70/1 |
| ACE XB | I2C0 | 0X7E/F |
| ACE TA | I2C0 | 0X40/1 |
| ACE TB | I2C0 | 0X4E/F |
| EADC1 | I2C1 | 0X10/1 |
| EADC2 | I2C1 | 0X12/3 |
| Temp. Sensor Left | I2C2 | 0X98/9 |
| Temp. Sensor Right | I2C2 | 0X96/7 |
| CPU,MEM,WLAN Thermal | I2C2 | 0X90/1 |
| GPU Analog Die Thermal | I2C2 | 0X92/3 |
| GPU Digital Die Thermal | I2C2 | 0X82/3 |
| TCON | I2C3 | 0X20-3F |
| GMUX IOEXP | I2C3 | 0X44/5 |
| Charger | I2C4 | 0X12/3 |
| Battery | I2C4 | 0X16/7 |
| Calpe | I2C4 | 0XE8/9 |
| Trackpad | I2C5 | 0X98/9 |
| SSD0 | I2C6 | 0XF2/3 |
| SSD1 | I2C6 | 0XF0/1 |
| | SoC IF | |
| Left Spkr Amp.(U6400) | I2C0 | 0XD8/9 |
| Left Spkr Amp.(U6450) | I2C0 | 0XDA/B |
| Right Spkr Amp.(U6500) | I2C1 | 0XDC/D |
| Right Spkr Amp.(U6550) | I2C1 | 0XDE/F |
| Audio Codec | I2C2 | 0X90/1 |
| ALS | I2C3 | 0X29/A |
| DFR Display | I2C4 | 0X98/9 |
| DFR Touch | I2C4 | 0XA0/1 |
| NC. | I2C5 | |
| NC. Spkr ID1 | I2C6_SDA | |
| Spkr ID0 | I2C6_SCL | |
| | ISP IF | |
| FT Camera | I2C0 | 0X6C/D |
| NC. | I2C1 | |
| | AOP IF | |
| NC. | I2C0 | |
| | PCH IF | |
| NC. | PULL-UP | |

BOM_COST_GROUP=SMC

PAGE TITLE
I2C Connections 2

Apple Inc.

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REVISION 1.0.0
BRANCH EVT
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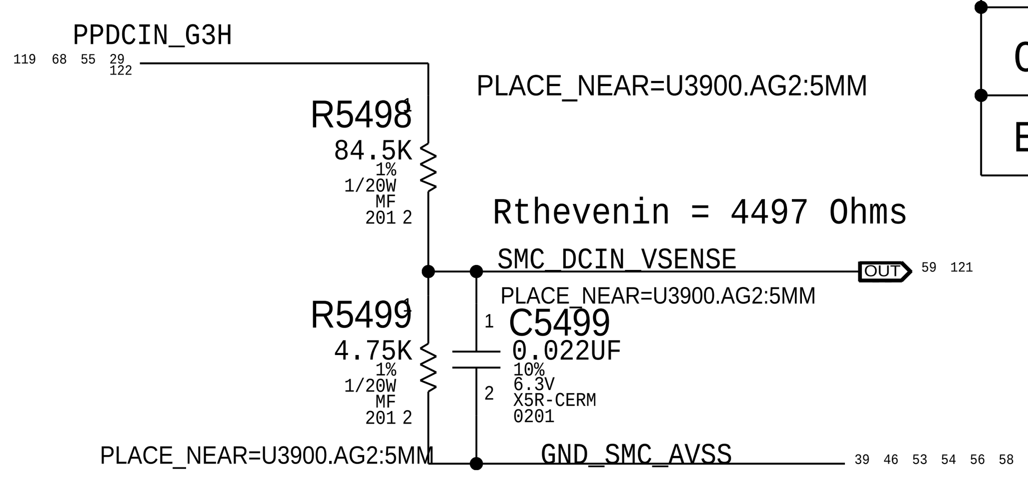
## DC-IN Voltage Sense (VD0R)

Gain: 0.05322x
Vnominal: 20 V, Range: 23.49 V
SMC ADC: 00
Enables DC-In VSense
divider when AC present.
2.2KHz

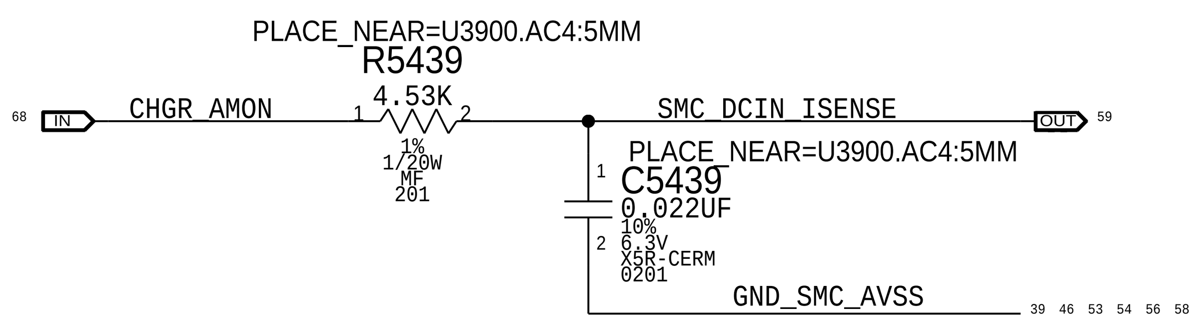



J680 SENSOR SETTINGS

| CHIP | Vref(V) | Vmax | SMC sample Freq. | ADR RC filter |
|---|---|---|---|---|
| H9M | 1.25 | 1.8 | 10khz | 0.1ms |
| CALPE | 1.5 | 5 | 100hz | 10ms |
| EADC | 2.5 | 5 | 1-2hz(10hz) | 100ms |

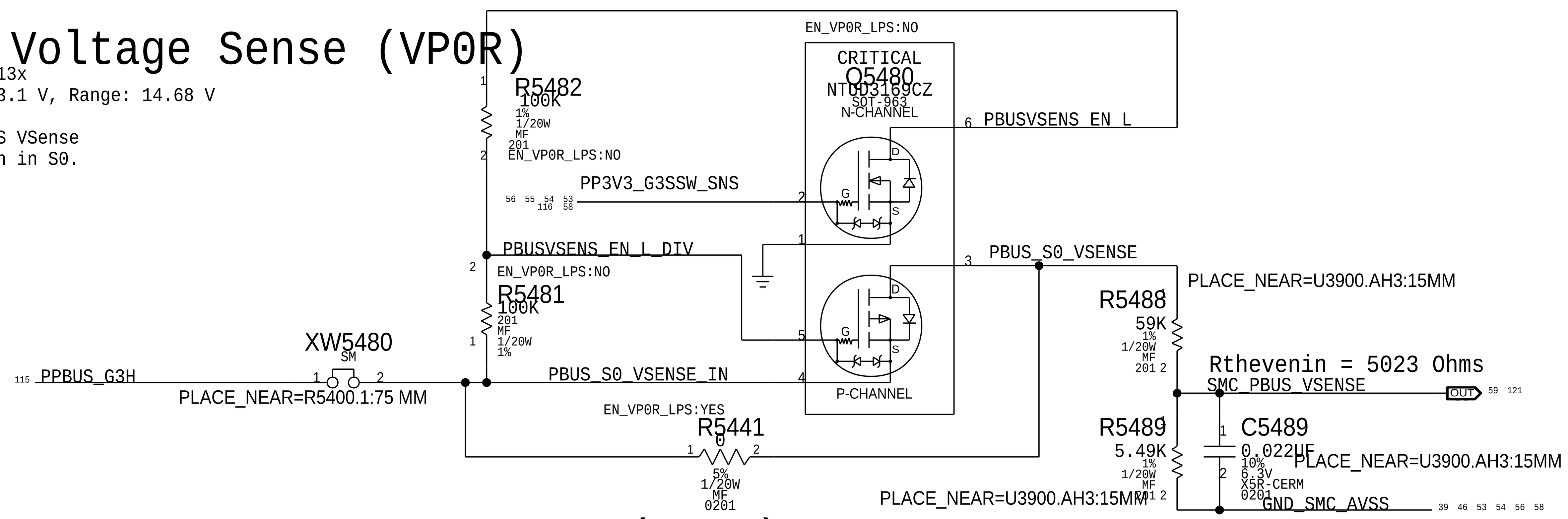
## DC-IN Current Sense (ID0R)

DISCharger Gain: 20x, EDP: 4.6 A
Rsense: 0.010 (R7020)
SMC ADC: 01



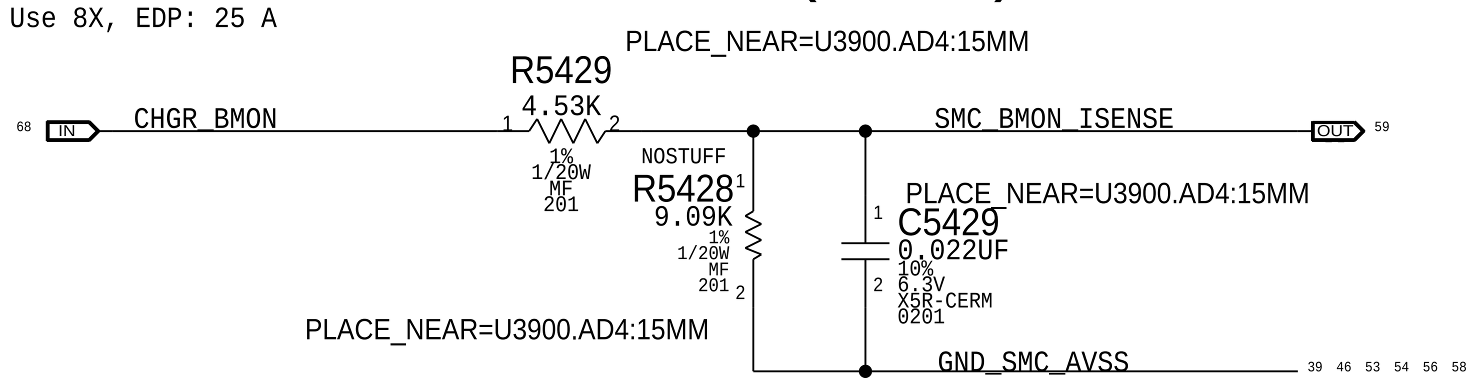

## PBUS Voltage Sense (VP0R)

Gain: 0.08513x
Vnominal: 13.1 V, Range: 14.68 V
SMC ADC: 02
Enables PBUS VSense
divider when in S0.



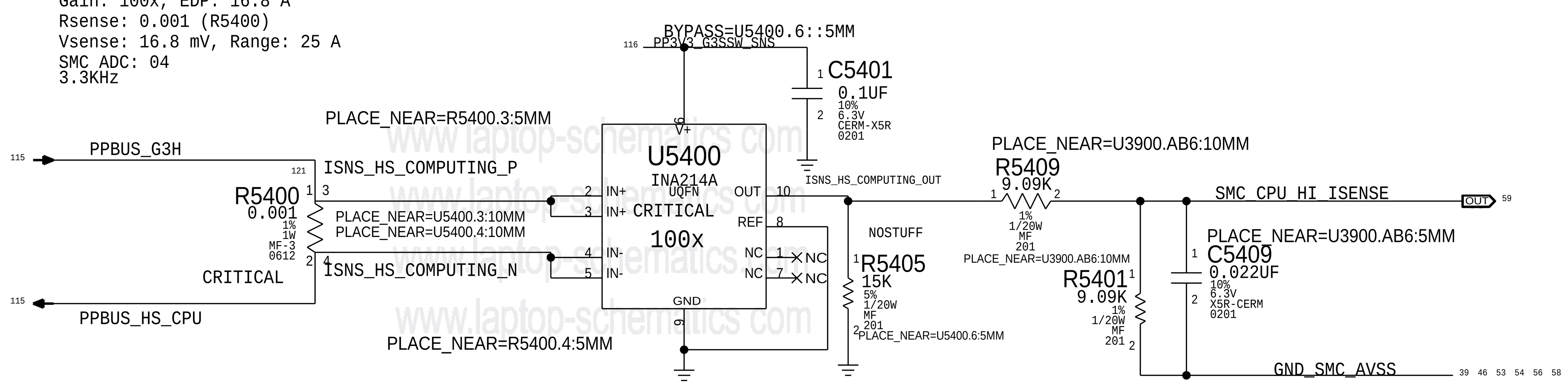

## Discharger BMON Current Sense (IPBR)

Charger Gain: 8X OR 64x, Use 8X, EDP: 25 A
Rsense: 0.005 (R7060)
SMC ADC: 03



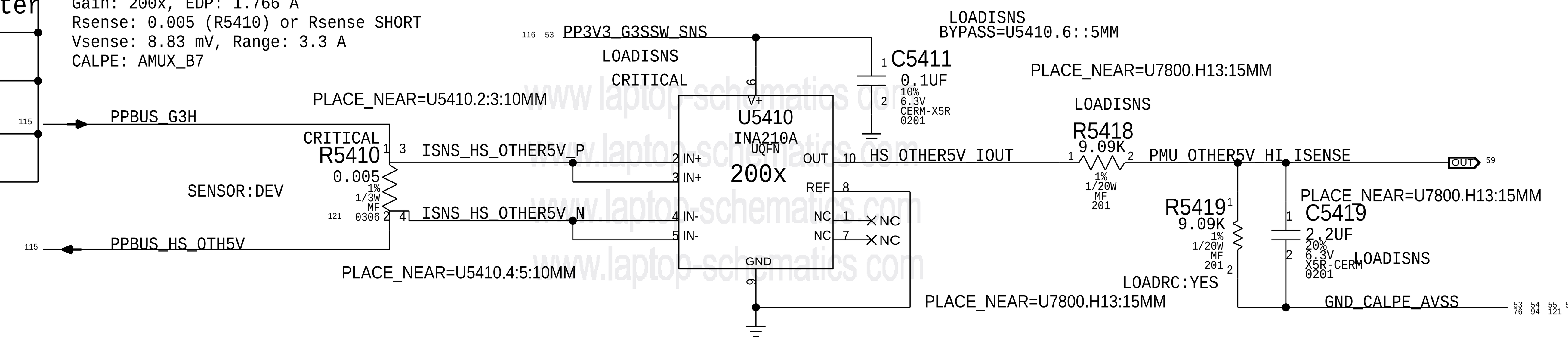

## CPU High Side Curent Sense (IC0R)

Gain: 100x, EDP: 16.8 A
Rsense: 0.001 (R5400)
Vsense: 16.8 mV, Range: 25 A
SMC ADC: 04
3.3KHz



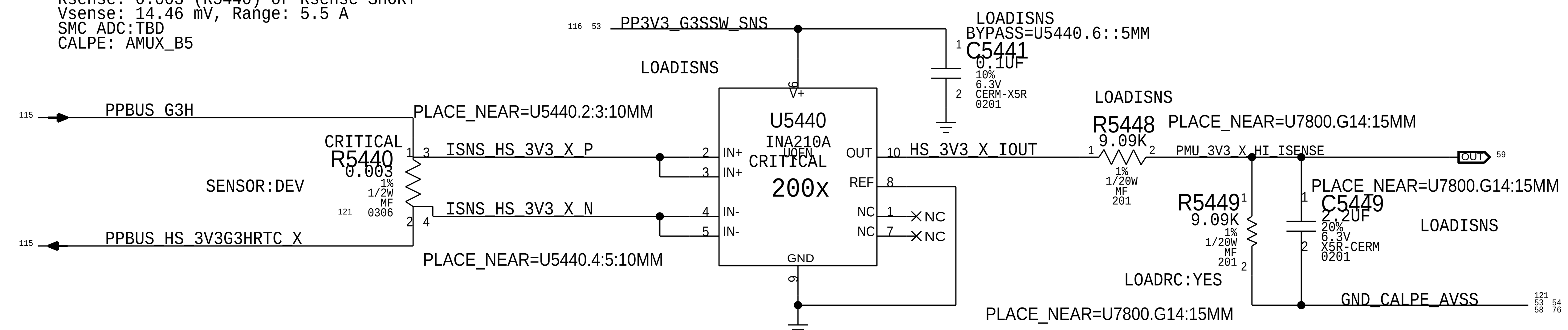

## OTHER 5V High Side Current Sense (IO5R)

Gain: 200x, EDP: 1.766 A
Rsense: 0.005 (R5410) or Rsense SHORT
Vsense: 8.83 mV, Range: 3.3 A
CALPE: AMUX_B7



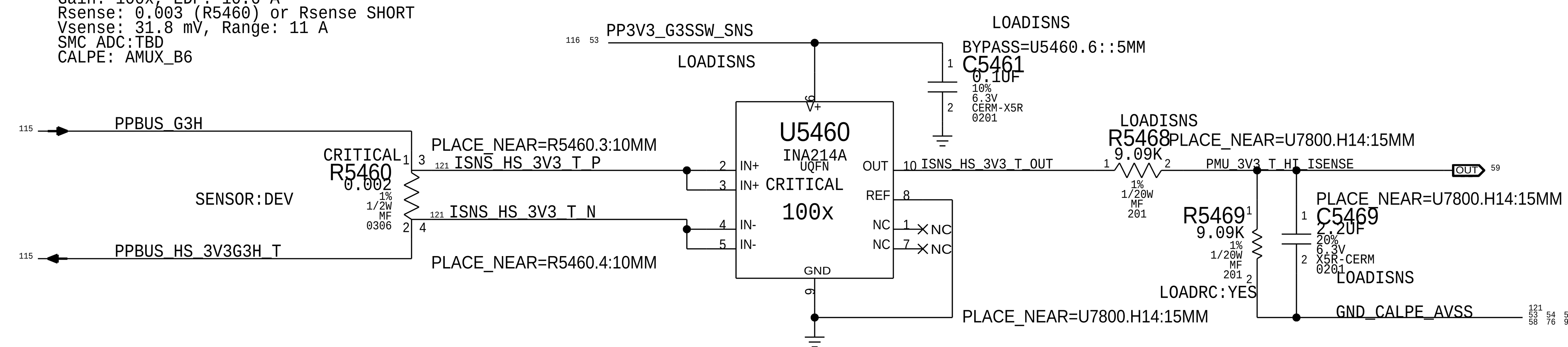

## LEFT SIDE 3.3V High Side Current Sense (IOLR)

Gain: 200x, EDP: 4.82 A
Rsense: 0.003 (R5440) or Rsense SHORT
Vsense: 14.46 mV, Range: 5.5 A
SMC ADC:TBD
CALPE: AMUX_B5



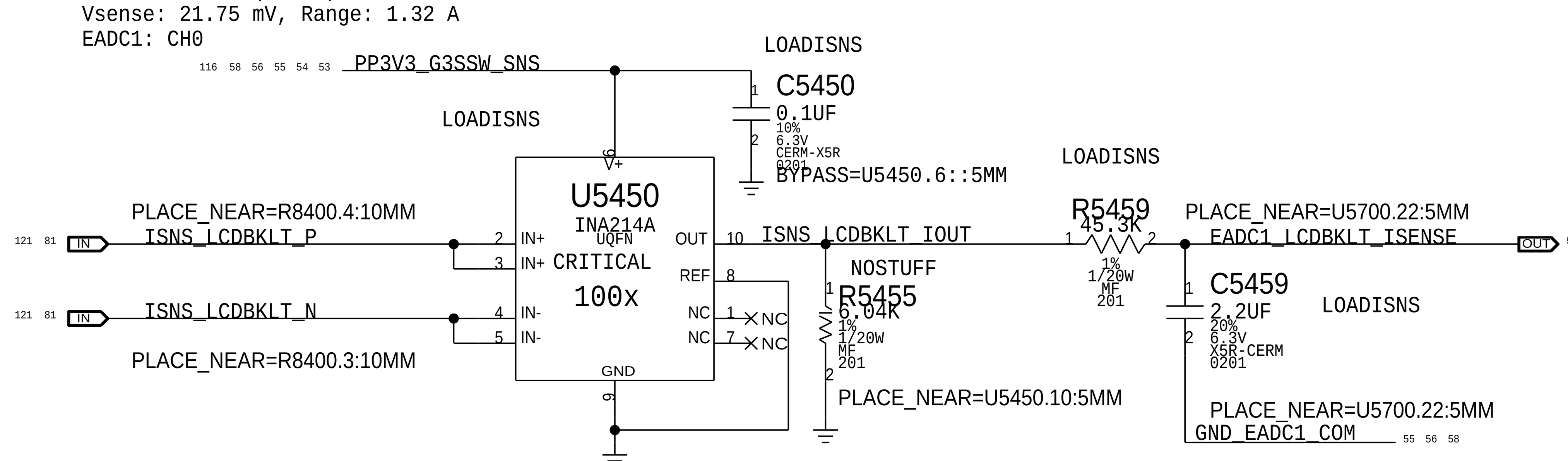

## RIGHT SIDE 3.3V High Side Current Sense (IORR)

Gain: 100x, EDP: 10.6 A
Rsense: 0.003 (R5460) or Rsense SHORT
Vsense: 31.8 mV, Range: 11 A
SMC ADC:TBD
CALPE: AMUX_B6




## LCD Backlight Current Sense (IBLR)

Gain: 100x. EDP: 0.87 A
Rsense: 0.025 (R8400)
Vsense: 21.75 mV, Range: 1.32 A
EADC1: CH0




| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 117S0008 | 3 | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | R5449,R5469,R5419 | | LOADRC:NO |

BOM_COST_GROUP=SENSORS

PAGE TITLE: Power Sensors High Side

Apple Inc.

DRAWING NUMBER: 051-02888 SIZE: D

REVISION: 1.0.0

BRANCH: EVT


| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 117S0008 | 5 | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | R5576,R5519,R5549,C5539,R5529 | | LOADRC:NO |

PAGE TITLE
Power Sensors Load Side

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D
REVISION 1.0.0
BRANCH EVT
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SHEET 54 OF 131


| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 117S0008 | 5 | RES,MTL FILM,100K,1/16W,0201,SMD,LF | R5602,R5612,R5622,R5632,R5679 | | LOADRC:NO |

PAGE TITLE: Power Sensors Extended 1

Apple Inc.

DRAWING NUMBER: 051-02888 SIZE: D
REVISION: 1.0.0
BRANCH: EVT


PAGE TITLE

Power Sensors Extended 2

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT

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PAGE TITLE
## Thermal Sensors

Apple Inc.

| | |
|---|---|
| DRAWING NUMBER | 051-02888 |
| SIZE | D |
| REVISION | 1.0.0 |
| BRANCH | EVT |
| PAGE | 58 OF 200 |
| SHEET | 57 OF 131 |

# GPU SENSORS


| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 117S0008 | 1 | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | R5969 | | LOADRC:NO |

BOM_COST_GROUP=SENSORS

PAGE TITLE
Power Sensor Extended 3

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D
REVISION 1.0.0
BRANCH EVT
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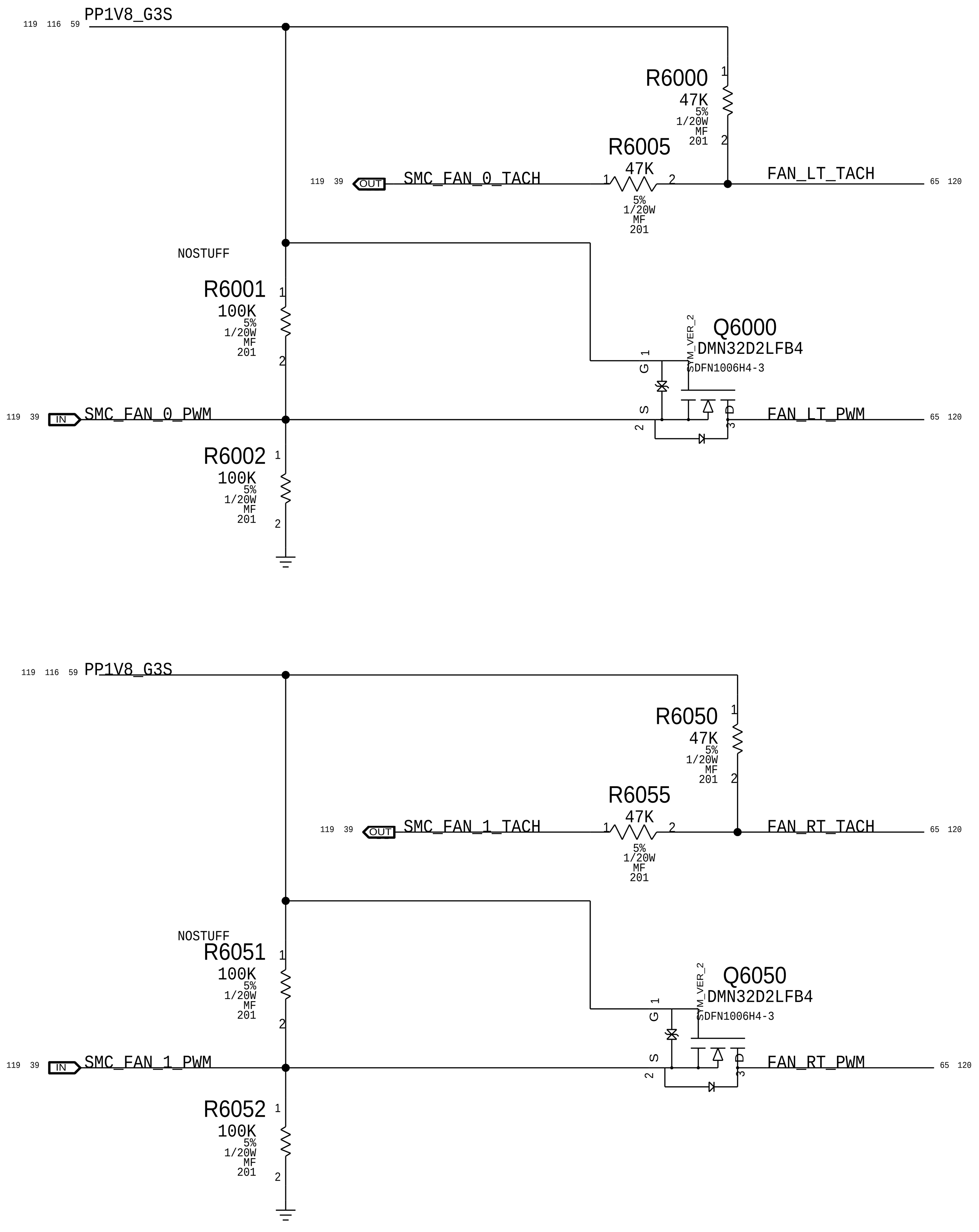
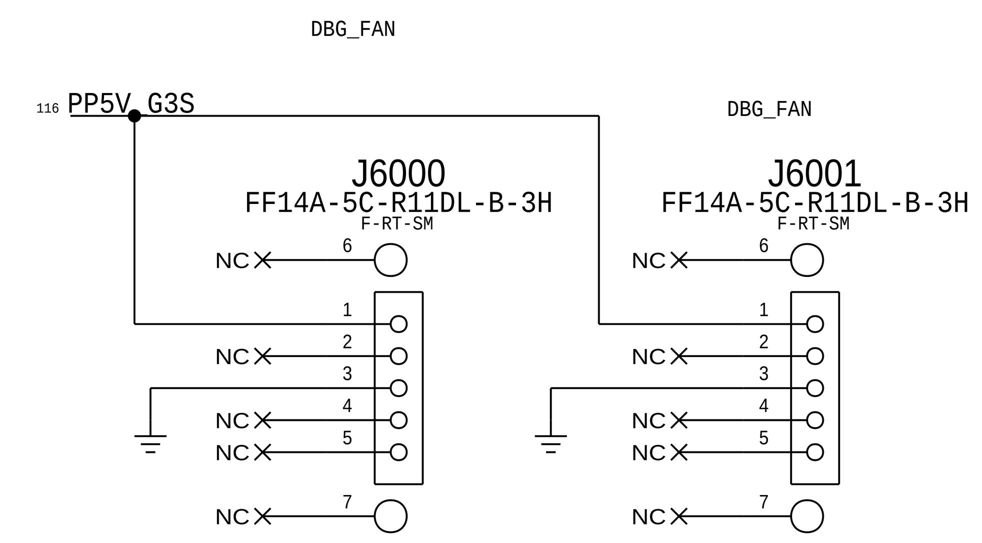
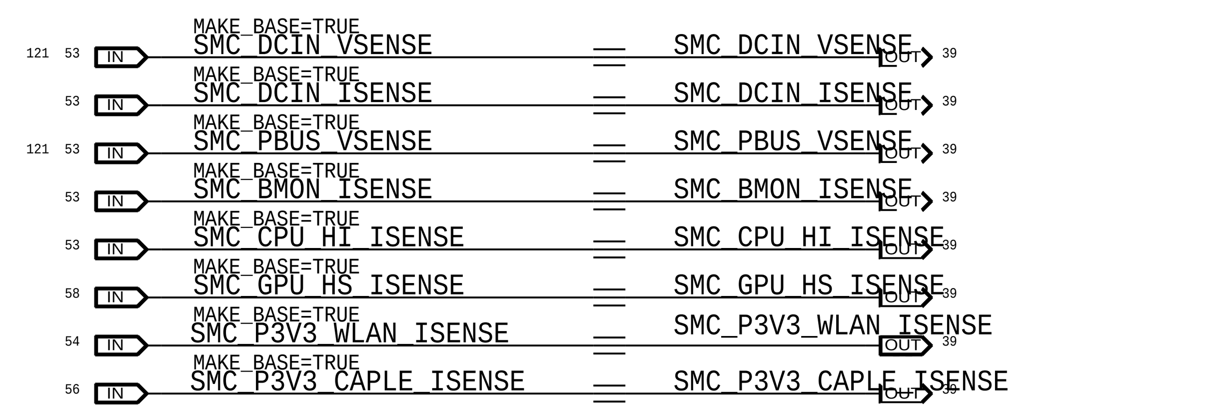
# FAN CONNECTOR

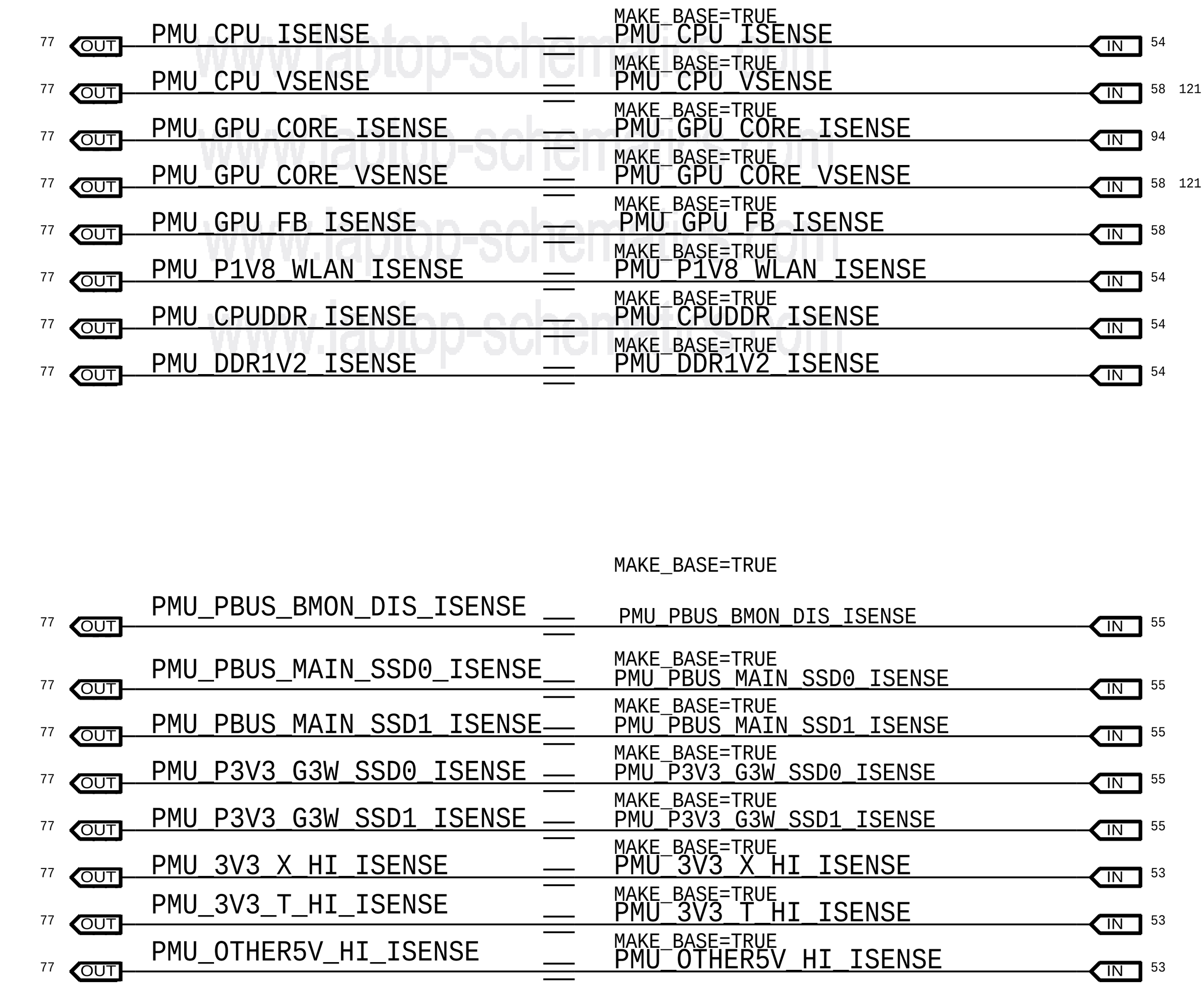



## H9M SMC ADC Assignments




## CALPE AMUX Assignments




BOM_COST_GROUP=FAN

PAGE TITLE
Fans/SMC/AMUX Support

Apple Inc.

DRAWING NUMBER 051-02888
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8 7 6 5 4 3 2 1

D C B A

BOM_COST_GROUP=AUDIO

DESIGN: X502/DEV_MLB_U
LAST CHANGE: Wed Feb 18 17:12:24 2015

PAGE TITLE
Audio Placeholder

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D
REVISION 1.0.0
BRANCH EVT
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| AUDIO JACK CODEC I2C ADDRESS | | |
|---|---|---|
| AD1 | AD0 | ADDRESS |
| GND | GND | 0x48 <-- |
| GND | 1.8V | 0x49 |
| 1.8V | GND | 0x4A |
| 1.8V | 1.8V | 0x4B |


DESIGN: X502/DEV_MLB_U
LAST CHANGE: Wed Feb 18 17:31:01 2015

PAGE TITLE
# Audio Jack Codec

Apple Inc.

| DRAWING NUMBER | SIZE |
|---|---|
| 051-02888 | D |

REVISION 1.0.0

BRANCH EVT

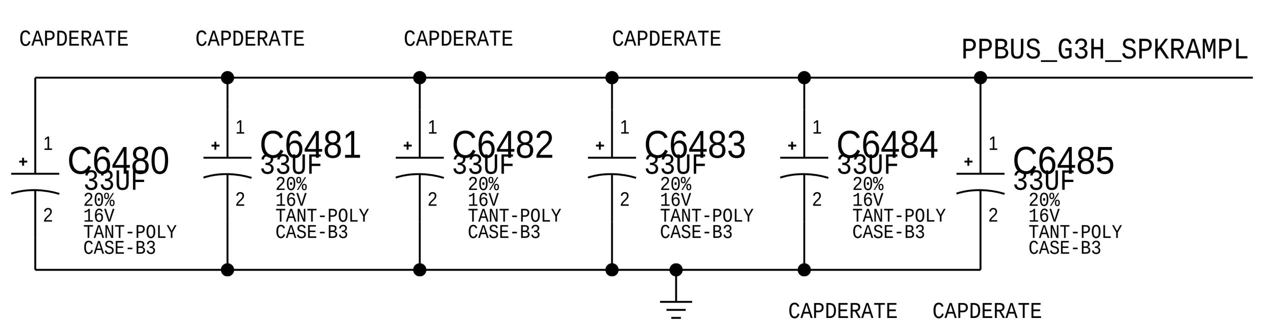
PAGE 63 OF 200

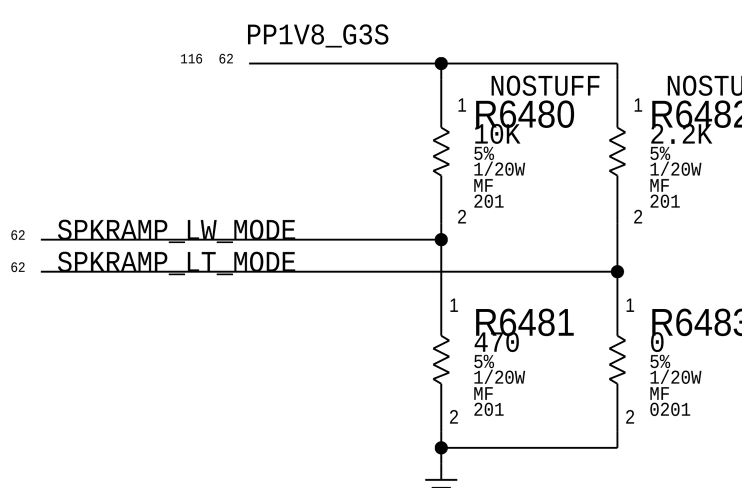
SHEET 61 OF 131

# 2X MONO SPEAKER LEFT AMPLIFIERS

APN: 353S01252
GAIN: 0DBFS = xxVRMS


LEFT BULK CAPACITANCE




I2C ADDRESS

| MODE PIN | 7-BIT | CHANNEL |
|---|---|---|
| GND | 0x31 | L TW |
| 470 to GND | 0x32 | L WF |
| 470 to IOVDD | 0x33 | R TW |
| 2k2 to GND | 0x34 | R WF |
| 2k2 to IOVDD | 0x35 | |
| 10k to GND | 0x36 | |
| 10k to IOVDD | 0x37 | |
| 47k to IOVDD | 0x38 | |

BOM_COST_GROUP=AUDIO

SYNC_MASTER=TROY SYNC_DATE=12/11/2017

PAGE TITLE
Audio Left Amplifiers

Apple Inc.

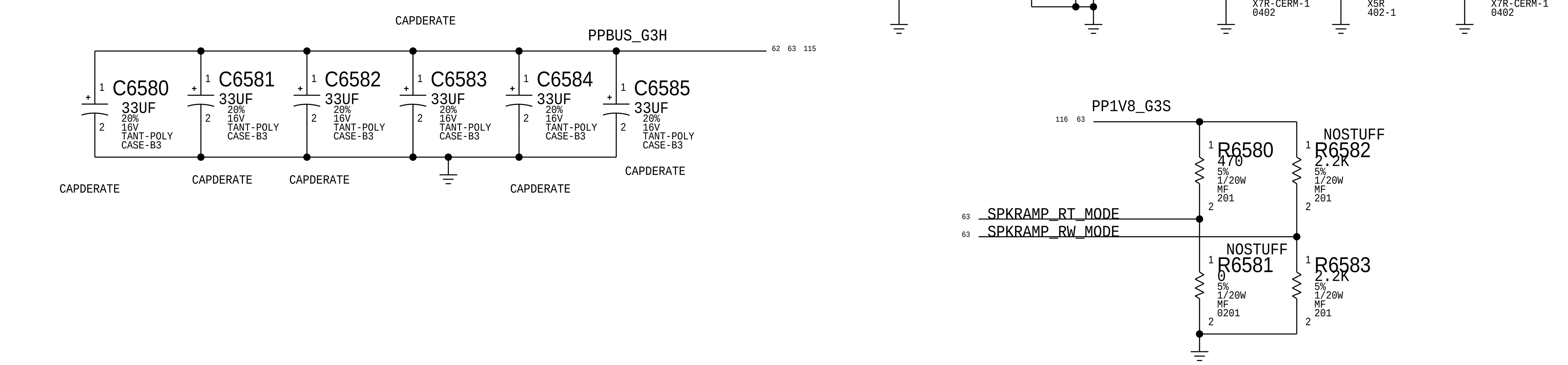
DRAWING NUMBER 051-02888 SIZE D

REVISION 1.0.0

BRANCH EVT

# 2X MONO SPEAKER RIGHT AMPLIFIERS

APN: 353S01252
GAIN: 0DBFS = xxVRMS


I2C ADDRESS

| MODE PIN | 7-BIT | CHANNEL |
|---|---|---|
| GND | 0x31 | L TW |
| 470 to GND | 0x32 | L WF |
| 470 to IOVDD | 0x33 | R TW |
| 2k2 to GND | 0x34 | R WF |
| 2k2 to IOVDD | 0x35 | |
| 10k to GND | 0x36 | |
| 10k to IOVDD | 0x37 | |
| 47k to IOVDD | 0x38 | |

BOM_COST_GROUP=AUDIO

PAGE TITLE
Audio Right Amplifiers

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D
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DESIGN: X502/DEV_MLB_U
LAST CHANGE: Wed Feb 18 17.12.24 2015

PAGE TITLE
# Audio Flex Connectors

Apple Inc.

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|---|---|
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| BRANCH EVT | |
| PAGE 66 OF 200 | |
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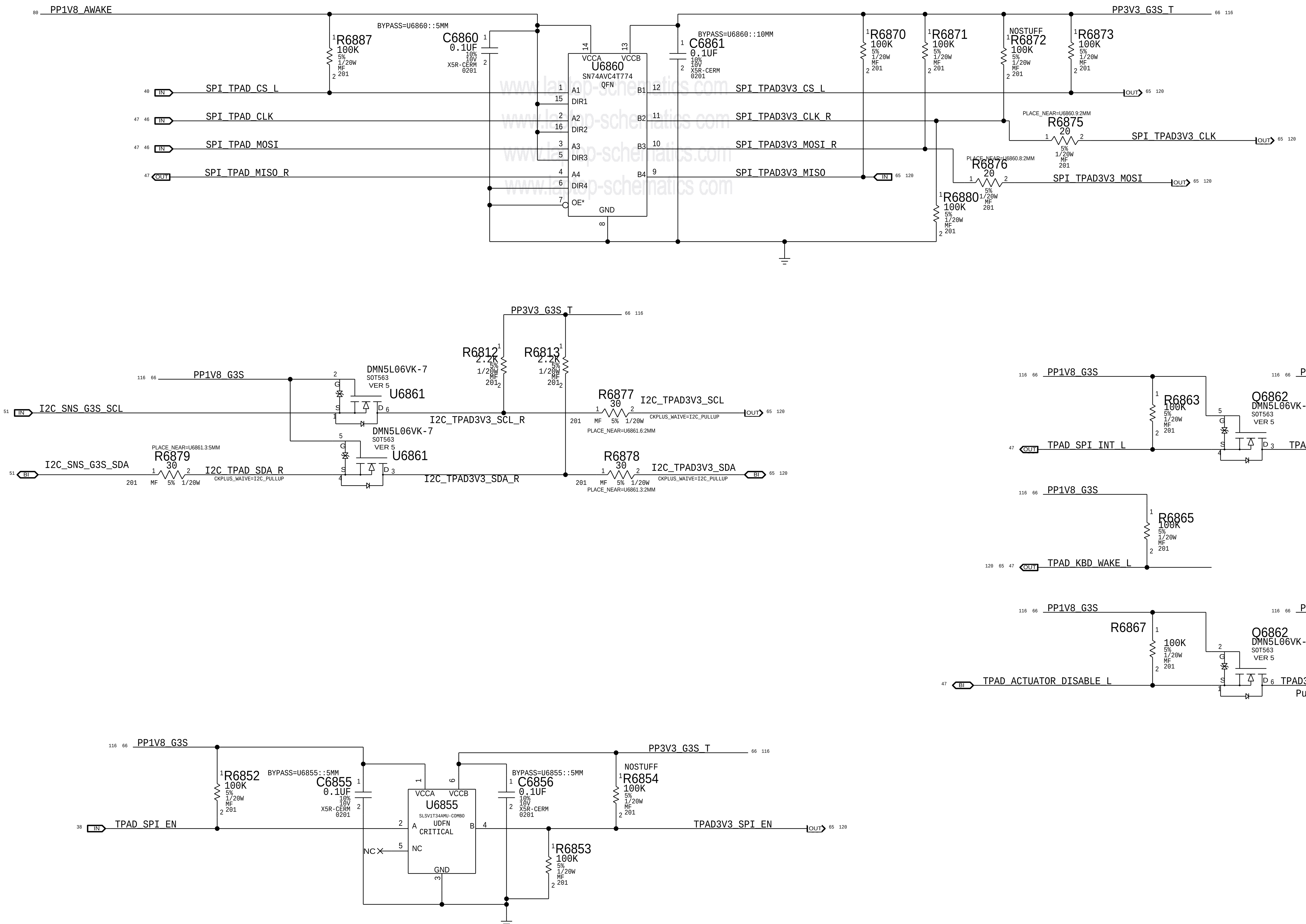
# KBD CONNECTOR

# TPAD CONNECTOR


BOM_COST_GROUP=KEYBOARD

| | |
|---|---|
| PAGE TITLE | Keyboard & Trackpad 1 |
| Apple Inc. | DRAWING NUMBER 051-02888 SIZE D |
| | REVISION 1.0.0 |
| | BRANCH EVT |
| | PAGE 67 OF 200 |
| | SHEET 65 OF 131 |

# Trackpad Level Shifting




BOM_COST_GROUP=KEYBOARD

SYNC_MASTER=j132 SYNC_DATE=03/23/2017

PAGE TITLE: Keyboard & Trackpad 2

Apple Inc.

DRAWING NUMBER: 051-02888 SIZE: D

REVISION: 1.0.0

BRANCH: EVT

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SHEET: 66 OF 131


BMU POWER FLEX HOTBAR'd TO THE MLB:

| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 632-00862 | 1 | PCBA,FLEX,BMU PWR,X363 | J6950 | CRITICAL | |

BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE: VR 3.3V G3H & Battery Conn

Apple Inc.

DRAWING NUMBER: 051-02888 SIZE: D

REVISION: 1.0.0

BRANCH: EVT

PAGE: 69 OF 200

SHEET: 67 OF 131


PAGE TITLE

# PBUS Supply & Battery Charger

DRAWING NUMBER 051-02888 SIZE D

Apple Inc.

REVISION 1.0.0

BRANCH EVT

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PAGE TITLE

IMVP IC

Apple Inc.

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|---|---|
| 051-02888 | D |
| REVISION 1.0.0 | |
| BRANCH EVT | |
| PAGE 71 OF 200 | |
| SHEET 69 OF 131 | |




| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 353S00519 | 3 | IC,SIC635,DRMOS,IMVP-8,40A,PQFN31,5X5 | U7210,U7220,U7230 | CRITICAL | |

PAGE TITLE
## IMVP VCC Block

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D
REVISION 1.0.0
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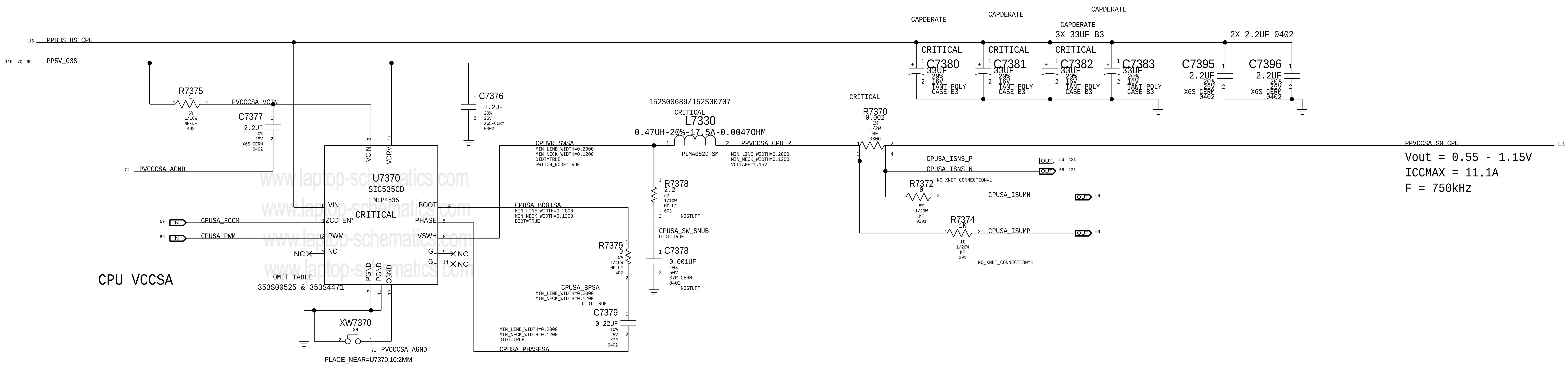


| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 353S00525 | 1 | IC,SIC635,DRMOS,IMVP-8,40A,PQFN31,5X5 | U7370 | CRITICAL | |

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE

IMVP SA Block

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D

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BRANCH EVT

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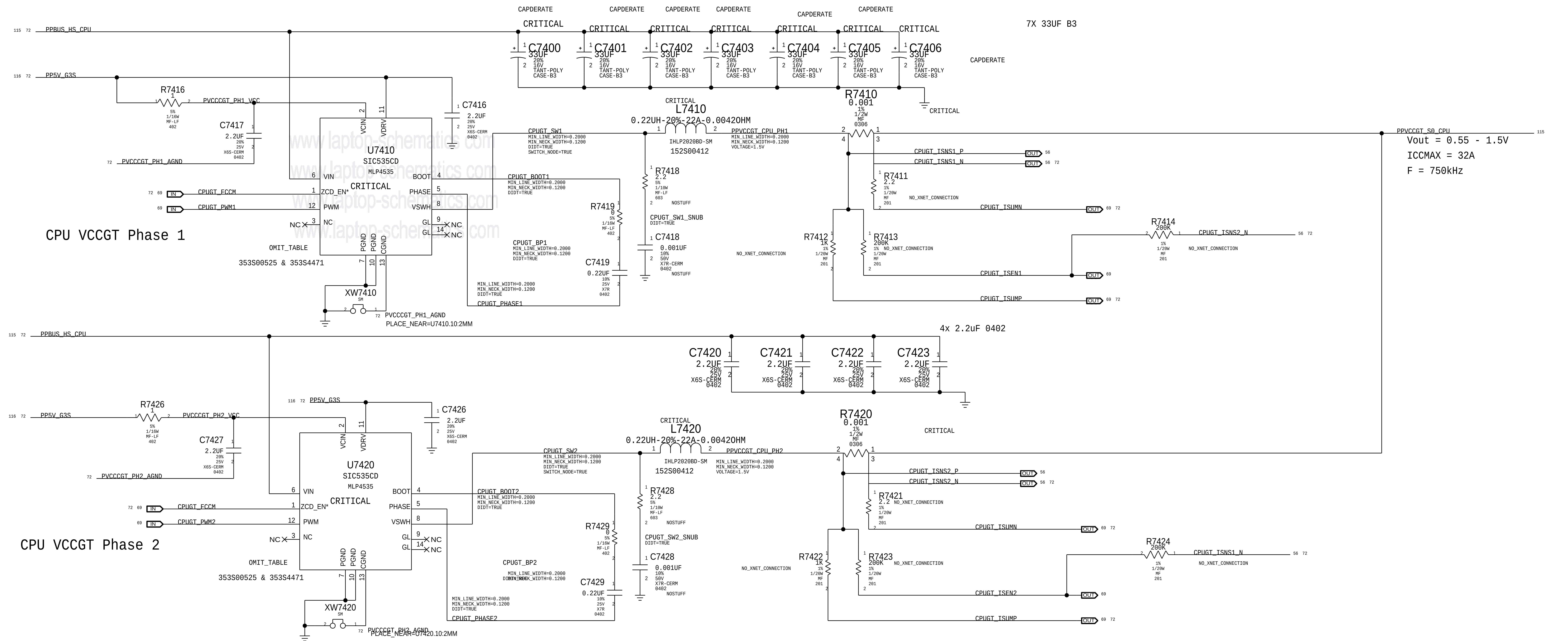


| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 353S00525 | 2 | IC,SIC635,DRMOS,IMVP-8,40A,PQFN31,5X5 | U7410,U7420 | CRITICAL | |

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
IMVP GT Block

Apple Inc.

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| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 152S00703 | 1 | IND,MLD,2.2UH,20%,43mOHM,5.5A,4.2x4x2MM | L7600 | CRITICAL | |

BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE
Power - 5V 3.3V Supply

Apple Inc.

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VR 2.5V & 1.1/1.2V/VTT

Apple Inc.

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Note : Design based on Calpe ERS - D2449-A0-110-00_0v3.pdf (Radar# 24696002)
System Block Diagram - T290 Power System Architecture . v9
Optimize componentS for individual projects based on EDP(A)


| | |
|---|---|
| PAGE TITLE | PMIC BUCKS AND SWs |
| DRAWING NUMBER | 051-02888 |
| SIZE | D |
| REVISION | 1.0.0 |
| BRANCH | EVT |
| PAGE | 78 OF 200 |
| SHEET | 75 OF 131 |

Apple Inc.


Part : SLG5AP1668V
R(ON) : 7.8 mohm (Typical) , 9.6 mohm (max)
Current: 5.3A Max

BOM_COST_GROUP=SOC

# PMIC LDOs

Apple Inc.

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# VDD_HI Threshold Select

| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 118S0481 | 1 | RES,MF,649K,1%,1/20W,0201 | R8050 | CRITICAL | PBUS:2S |
| 118S00077 | 1 | RES,MF,1.15M,1%,1/16W,0201 | R8050 | CRITICAL | PBUS:3S |
| 118S00077 | 1 | RES,MF,1.15M,1%,1/16W,0201 | R8050 | CRITICAL | PBUS:12V |

| PBUS | Rising Vth | Falling Vth |
|---|---|---|
| 2S | 6.059V | 5.636V |
| 3S | 9.076V | 8.443V |
| 12V | 9.076V | 8.443V |

BOM_COST_GROUP=SOC

PAGE TITLE: PMIC GPIOs & Control

Apple Inc.

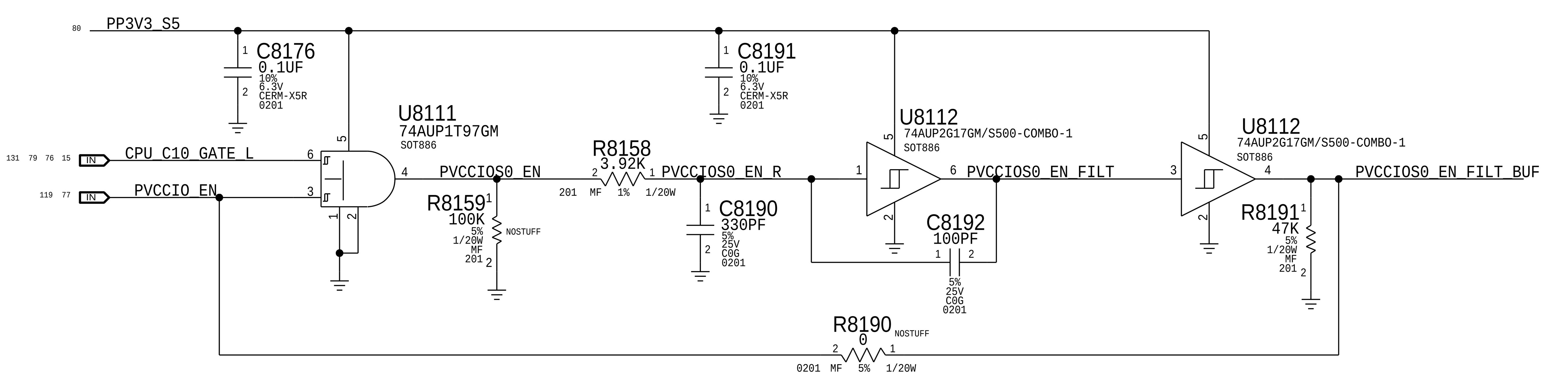
DRAWING NUMBER: 051-02888 | SIZE: D

REVISION: 1.0.0

BRANCH: EVT

# 0.95V VCCIO


# VCCIO ENABLE LOGIC




BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE
VR VCCIO

Apple Inc.

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# 3.3V G3 Standby Switch X

# 3.3V G3 Standby Switch T

# 1.8V G3 Standby Switch

# 1.1V S0SW VCCPLL_OC Switch

# 3.3V S0SW TBT X Switch

# 3.3V Sensor Switch

# 3.3V S0SW TBT T Switch


| Part | SLG5AP1445V |
|---|---|
| R(on) @ 3.6V | 7.8 mOhm Typ 8.5 mOhm Max |
| Current | 4A Max |

| Part | SLG5AP1445V |
|---|---|
| R(on) @ 3.6V | 7.8 mOhm Typ 8.5 mOhm Max |
| Current | 4A Max |

| Part | SLG5AP1445V |
|---|---|
| R(on) @ 3.6V | 7.8 mOhm Typ 8.5 mOhm Max |
| Current | 4A Max |

| Part | SLG5AP1635V |
|---|---|
| Type | Load Switch |
| R(on) @ 3.3V | 27.5 mOhm Typ 31 mOhm Max |
| Current | 2.5A Max |

| Part | SLG5AP1445V |
|---|---|
| Type | Load Switch |
| R(on) @ 4A | 7.8 mOhm Typ TBD mOhm Max |
| Current | 4A Max |

| Part | SLGAP1569V |
|---|---|
| Type | Load Switch |
| R(on) @ 3.6V | 34 mOhm Typ 46 mOhm Max |
| Current | 1A Max |

| Part | SLG5AP1445V |
|---|---|
| Type | Load Switch |
| R(on) @ 4A | 7.8 mOhm Typ TBD mOhm Max |
| Current | 4A Max |

BOM_COST_GROUP=PLATFORM POWER

SYNC_MASTER=SILU

PAGE TITLE: Power FETs

Apple Inc.

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PAGE TITLE

# SOC/PMIC Aliases

Apple Inc.

| | |
|---|---|
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| BRANCH | EVT |
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BOM_COST_GROUP=DISPLAY

PAGE TITLE

# LCD Backlight Driver

Apple Inc.

| | |
|---|---|
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# LCD PANEL INTERFACE (eDP) + Camera (MIPI)


PAGE TITLE

eDP Display Connector

Apple Inc.

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# S4E0


BOM_COST_GROUP=SDD

SYNC_MASTER=j137_gs5_redhead SYNC_DATE=01/26/2017

PAGE TITLE: SSD0 S4E 0

Apple Inc.

DRAWING NUMBER: 051-02888 SIZE: D

REVISION: 1.0.0

BRANCH: EVT

# S4E1


SYNC_MASTER=j137_gs5_redhead SYNC_DATE=01/26/2017

PAGE TITLE: SSD0 S4E 1

Apple Inc.

DRAWING NUMBER: 051-02888 SIZE: D

REVISION: 1.0.0

BRANCH: EVT

# S4E2


SYNC_MASTER=j137_gs5_redhead SYNC_DATE=01/26/2017

PAGE TITLE: SSD0 S4E 2

Apple Inc.

| | |
|---|---|
| DRAWING NUMBER | 051-02888 |
| SIZE | D |
| REVISION | 1.0.0 |
| BRANCH | EVT |
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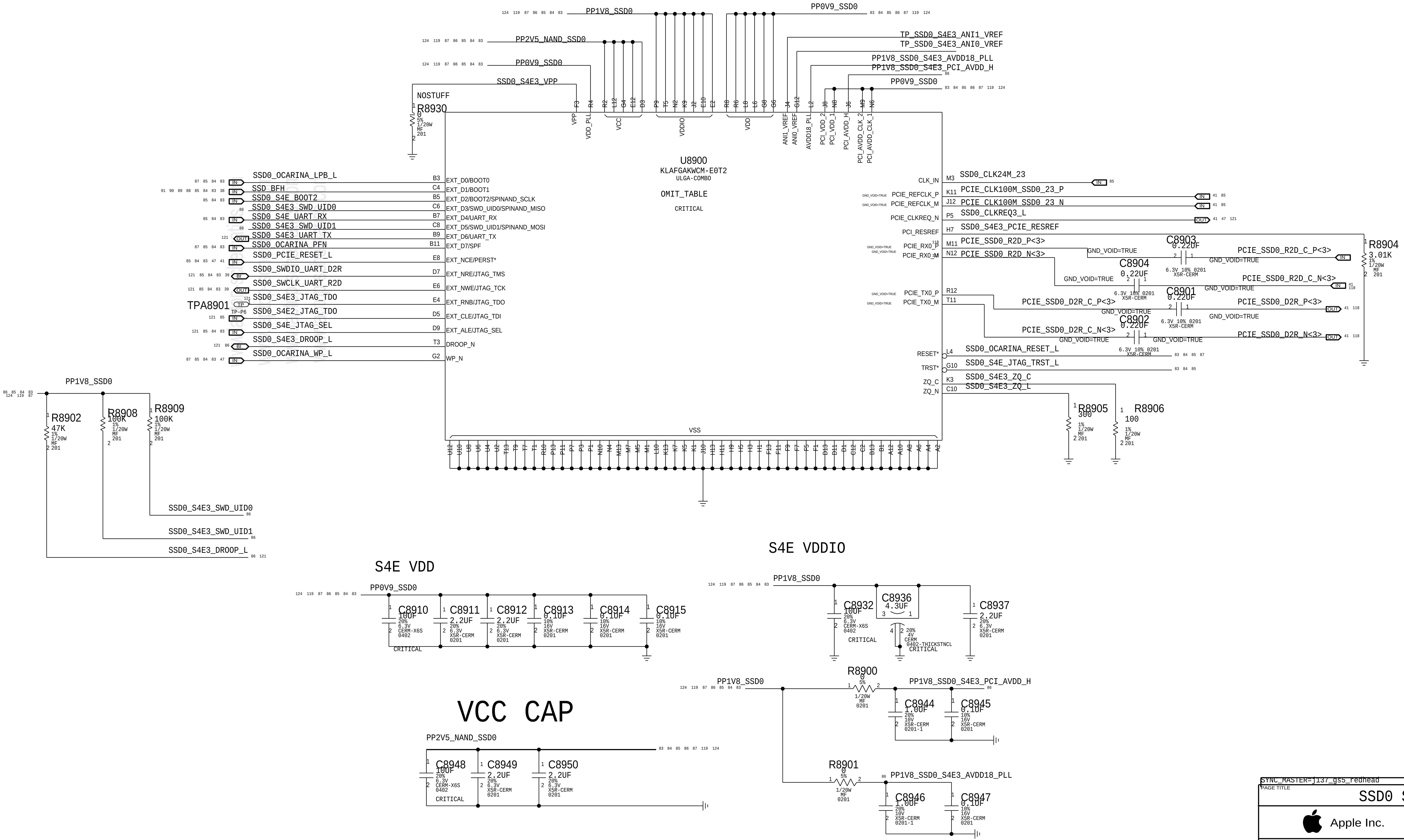
# S4E3




BOM_COST_GROUP=SDD

SYNC_MASTER=j137_gs5_redhead SYNC_DATE=01/26/2017

PAGE TITLE
SSD0 S4E 3

Apple Inc.

| | |
|---|---|
| DRAWING NUMBER | 051-02888 |
| SIZE | D |
| REVISION | 1.0.0 |
| BRANCH | EVT |
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| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 118S0738 | 1 | RES,THICKFILM,200KOHM,0.1%,1/20W,0201 | R9081 | CRITICAL | SSD0_NAND_VCC:2.7V |
| 103S00049 | 1 | RES,THICKFILM,221KOHM,0.1%,1/20W,0201 | R9081 | CRITICAL | SSD0_NAND_VCC:2.5V |

SYNC_MASTER=j137_gs5_redhead SYNC_DATE=01/26/2017

PAGE TITLE
# SSD0 PMIC & VR

Apple Inc.

DRAWING NUMBER 051-02888 SIZE D
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# S4E0

## VCC CAP


BOM_COST_GROUP=SDD

SYNC_MASTER=j137_gs5_redhead SYNC_DATE=01/26/2017

PAGE TITLE

SSD1 S4E 0

Apple Inc.

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REVISION 1.0.0

BRANCH EVT

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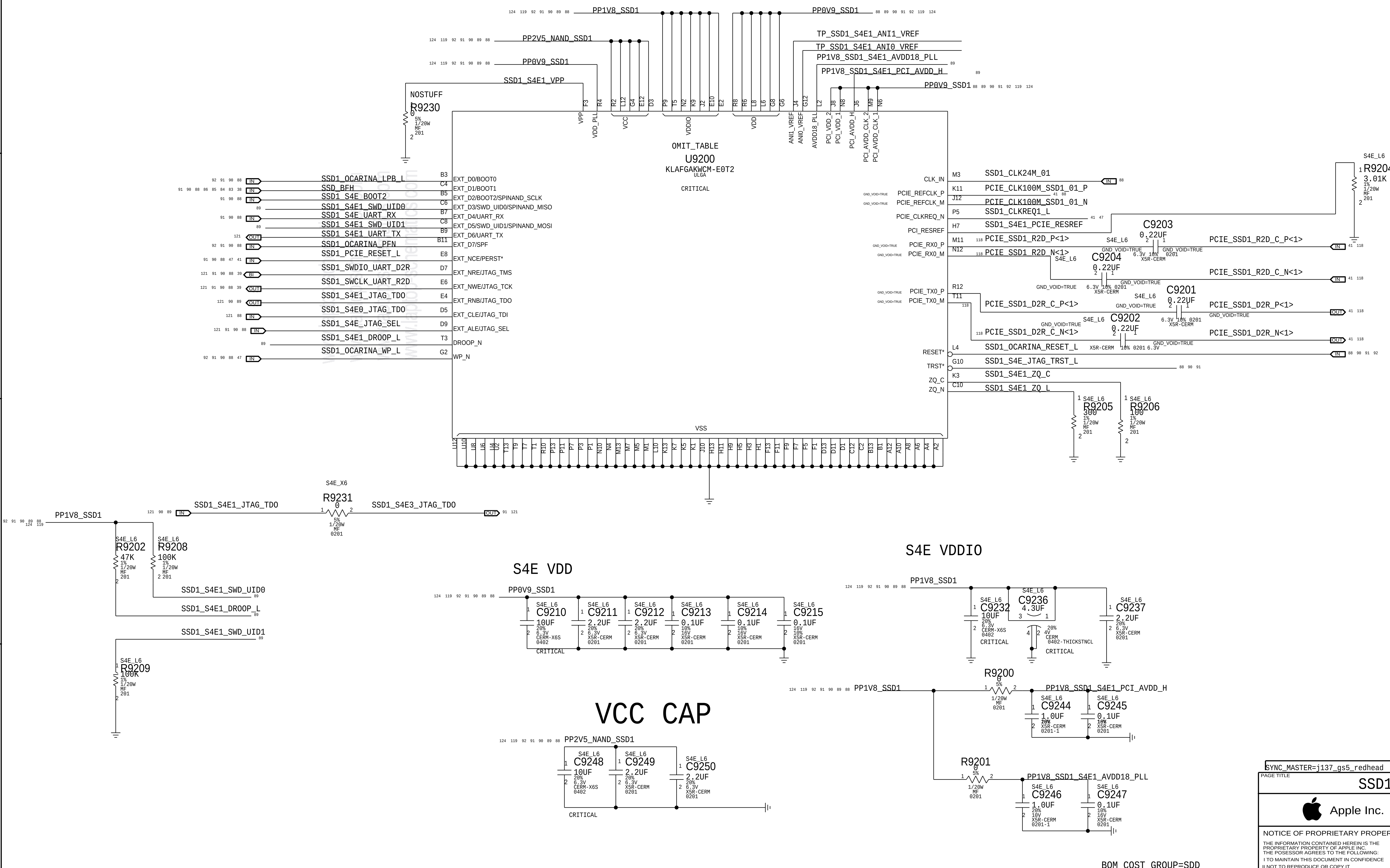
# S4E1




SYNC_MASTER=j137_gs5_redhead SYNC_DATE=01/26/2017

PAGE TITLE: SSD1 S4E 1

Apple Inc.

| | |
|---|---|
| DRAWING NUMBER | 051-02888 |
| SIZE | D |
| REVISION | 1.0.0 |
| BRANCH | EVT |
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# S4E2


SYNC_MASTER=j137_gs5_redhead SYNC_DATE=01/26/2017

PAGE TITLE: SSD1 S4E 2

Apple Inc.

DRAWING NUMBER: 051-02888 SIZE: D

REVISION: 1.0.0

BRANCH: EVT

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# S4E3


PAGE TITLE

## SSD1 S4E 3

Apple Inc.

| DRAWING NUMBER | SIZE |
|---|---|
| 051-02888 | D |

REVISION 1.0.0

BRANCH EVT

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| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 118S0738 | 1 | RES,THICKFILM,200KOHM,0.1%,1/20W,0201 | R9581 | CRITICAL | SSD1_NAND_VCC:2.7V |
| 103S00049 | 1 | RES,THICKFILM,221KOHM,0.1%,1/20W,0201 | R9581 | CRITICAL | SSD1_NAND_VCC:2.5V |

BOM_COST_GROUP=SSD

PAGE TITLE
SSD1 PMIC & VR

Apple Inc.

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BRANCH EVT

GMUX GPIO Expander
DP 2:1 ANALOG MUX
R9802
PP3V3_S5
PP3V3_S5_GPIOX_R
VOLTAGE=3.3V
R9803
PP1V8_S5
PP1V8_S5_GPIOX_R
VOLTAGE=1.8V
C9801 0.1UF
C9802 0.1UF
R9801 NOSTUFF 10K
U9801
PCAL6524
VFBGA
I2C Addr:
Read: 45h
Write: 44h
INT*
ADDR
SCL
SDA
RESET*
VDD(I2C-BUS)
VDD(P)
VSS
DISP_GCON_INT_L
I2C_DISP_SCL
I2C_DISP_SDA
DISP_GCON_RESET_L
R9804 NOSTUFF 10K
GMUX_IOEXP_ADDR_SEL
R9805 0
EG_VR0_PGOOD
P1V8GPU_PGOOD
PVDDCI_PGOOD
GPUVCORE_PGOOD
GPUFB_PGOOD
LAST RAIL PGOOD HAS TO BE VR4_PGOOD
DP_INT_HPD_R
USBC_HPD_DET
TP_BKLT_FAULT_L
DP_INT_EG_HPD
DP_INT_IG_HPD
EDP_MUXSEL_OVR
EDP_IG_PANEL_PWR
EDP_IG_BKLT_EN
EG_LCD_PWR_EN
EG_BKLT_EN
TP_BKLT_BOOST_EN
EG_VR0_EN
EG_VR1_EN
EG_VR2_EN
EG_VR3_EN
EG_VR4_EN
EG_RESET_L_BUFF
EDP_PANEL_PWR_EN
EDP_BKLT_EN
R9806 200
DP_INT_HPD
U9850
CBTL06142EEE
TFBGA
CRITICAL
C9850 0.1UF
C9851 0.1UF
DP_INT_EG_ML_P<0>
DP_INT_EG_ML_N<0>
DP_INT_EG_ML_P<1>
DP_INT_EG_ML_N<1>
DP_INT_EG_ML_P<2>
DP_INT_EG_ML_N<2>
DP_INT_EG_ML_P<3>
DP_INT_EG_ML_N<3>
DP_INT_EG_AUX_P
DP_INT_EG_AUX_N
R9850 4.7K
DP_INT_EG_HPD_R
DP_INT_IG_ML_P<0>
DP_INT_IG_ML_N<0>
DP_INT_IG_ML_P<1>
DP_INT_IG_ML_N<1>
DP_INT_IG_ML_P<2>
DP_INT_IG_ML_N<2>
DP_INT_IG_ML_P<3>
DP_INT_IG_ML_N<3>
DP_INT_IG_AUX_P
DP_INT_IG_AUX_N
R9851 4.7K
DP_INT_IG_HPD_R
R9819 2.26K
LCD_FSS
EDP_MUXSEL_OVR
LCD_MUX_EN
R9807 100K
R9852 100K
EDP_INT_ML_P<0>
EDP_INT_ML_N<0>
EDP_INT_ML_P<1>
EDP_INT_ML_N<1>
EDP_INT_ML_P<2>
EDP_INT_ML_N<2>
EDP_INT_ML_P<3>
EDP_INT_ML_N<3>
EDP_INT_AUX_P
EDP_INT_AUX_N
DP_INT_HPD
Pull down on eDP connector page
MUXSEL is default low.
EG is the default option
C9860 0.1UF
U9860
74LVC1G11GW-S500
SOT363
EG_RESET_L_BUFF
EG_VR0_PGOOD
PLT3V3_RST_L
GPU_RESET_L
R9860 100K
R9810 100K
C9810 0.1UF
P1V8S5_VALID
PMU_SYS_ALIVE
U9810
SOT353
74LVC1G08GW
DISP_GCON_RESET_L_R
R9812 0
DISP_GCON_RESET_L
R9811 0 NOSTUFF
C9840 0.1UF
U9840
SLG4AP41422
STQFN
DP_X_SNK0_HPD
DP_X_SNK1_HPD
DP_T_SNK0_HPD
DP_T_SNK1_HPD
GPUFB_PGOOD
DP_X_SNK0_HPD_EG
DP_X_SNK1_HPD_EG
DP_T_SNK0_HPD_EG
DP_T_SNK1_HPD_EG
USBC_HPD_DET
R9820 47K EG_VR0_EN
R9821 47K EG_VR1_EN
R9822 47K EG_VR2_EN
R9823 47K EG_VR3_EN
R9824 47K EG_VR4_EN
R9825 100K DP_INT_EG_HPD
R9826 100K DP_X_SNK0_HPD_EG
R9827 100K DP_X_SNK1_HPD_EG
R9828 100K DP_T_SNK0_HPD_EG
R9830 100K DP_T_SNK1_HPD_EG
R9832 47K EDP_PANEL_PWR_EN
R9833 47K EDP_BKLT_EN
BOM_COST_GROUP=GRAPHICS
PAGE TITLE
EDP Mux
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REVISION 1.0.0
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SHEET 93 OF 131
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| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 117S0008 | 1 | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | R9961 | | LOADRC:NO |


BOM_COST_GROUP=GRAPHICS



Apple Inc.


PAGE TITLE

# GPU Baffin PCIe

Apple Inc.

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PAGE TITLE

# GPU Baffin Core/FB Power

Apple Inc.

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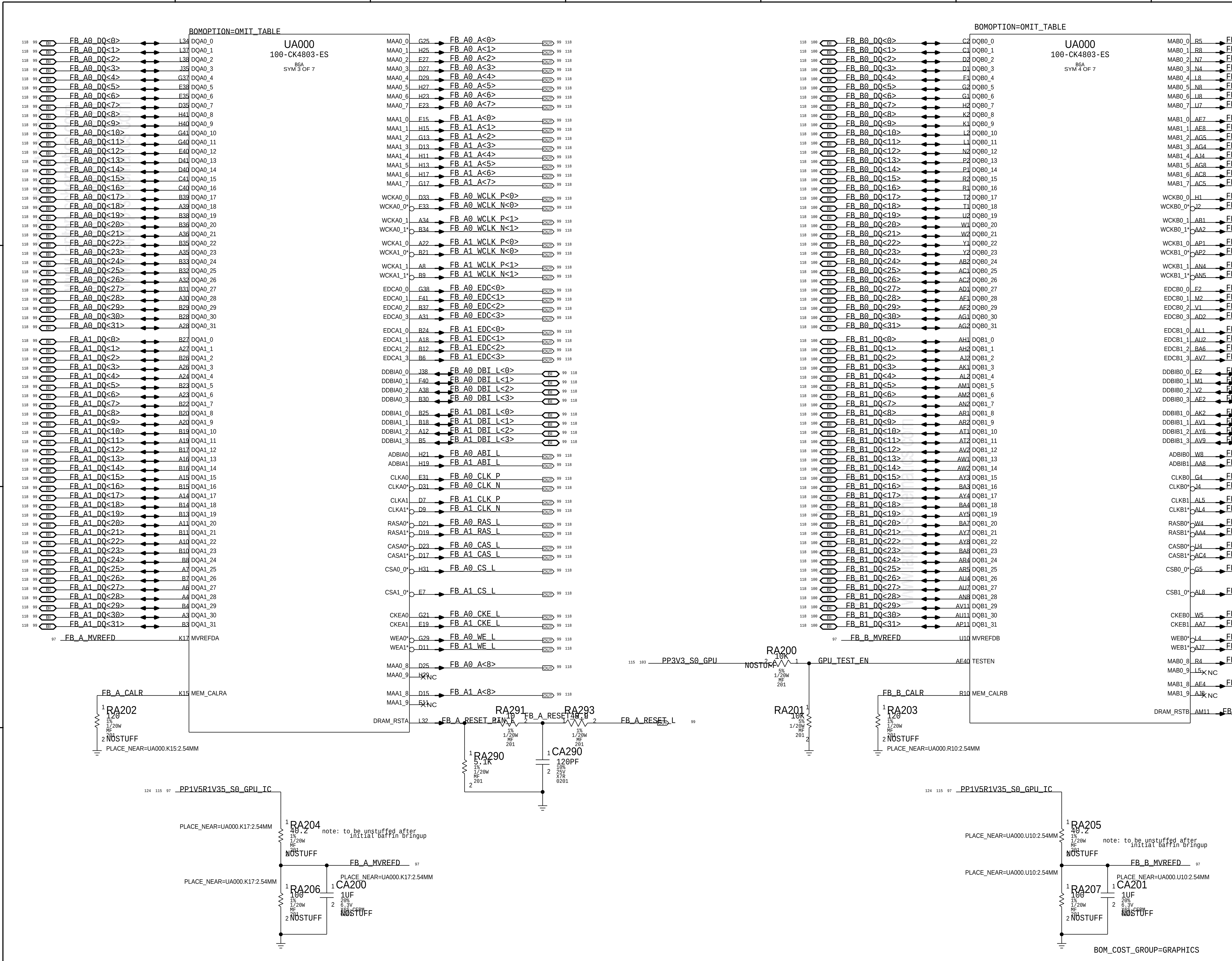


GPU Baffin FB

Apple Inc.

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| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 376S00174 | 1 | MOSFET,CTRL+SYNC,25V,4.1/1.4MO,QFN32,6x6 | QA340 | CRITICAL | |

PAGE TITLE
VR 1.05V GPU & 1.35V FB

Apple Inc.

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PAGE TITLE

GDDR5 VRAM FB 1 [104]

Apple Inc.

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PAGE TITLE

GDDR5 VRAM FB 2

Apple Inc.

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| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 376S00174 | 2 | MOSFET,CTRL+SYNC,25V,4.1/1.4MO,QFN32,6x6 | QA640,QA650 | CRITICAL | |


**TABLE 6. PRE-PWROK METAL VID CODES**

| SVC | SVD | OUTPUT VOLTAGE (V) |
|---|---|---|
| 0 | 0 | 1.1 |
| 0 | 1 | 1.0 |
| 1 | 0 | 0.9 |
| 1 | 1 | 0.8 |

ISL6277A datasheet (December 19,2012)

BOM_COST_GROUP=GRAPHICS

PAGE TITLE
VR GPU Core

Apple Inc.

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PAGE


PAGE TITLE
# GPU Baffin GPIO/CLK/Straps

Apple Inc.

| | |
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| PAGE | |
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BOMOPTION=OMIT_TABLE
UA000
100-CK4803-ES
BGA
SYM 2 OF 7
PP1V8_S0_GPU
RA800
10K
GPU_BP_0
PP1V8_S0_GPU
RA802
10K
GPU_BP_1
PP3V3_S0_GPU
RA805
4.7K
RA806
4.7K
I2C isolation now on CSA 52
level shifting and isolation
done by U5260
GPU_SMB_CLK
GPU_SMB_DAT
EG_BKLT_EN
RA821
10K
GPU_PLLCHARZ_H
GPU_PLLCHARZ_L
NC_EG_BKLT_PWM
GPU_BP_0
GPU_BP_1
GPU_AUD_PORT_CONN<0>
GPU_AUD_PORT_CONN<1>
GPU_AUD_PORT_CONN<2>
GPU_BRD_CFG<0>
GPU_BRD_CFG<1>
GPU_BRD_CFG<2>
GPU_SMBUS_ADDR<0>
GPU_SMBUS_ADDR<1>
TPA802
TPA803
TPA804
TPA805
TPA807
TPA808
TPA809
TPA810
GPU_DBG_8
GPU_DBG_9
GPU_DBG_10
GPU_DBG_11
GPU_MPLS_PS_0
GPU_MPLS_PS_1
GPU_MPLS_PS_2
GPU_MPLS_PS_3
PP1V8_S0_GPU
PP3V3_S0_GPU
RA816
100K
UA802
SLSV1T34AMU-COMBO
UDFN
CRITICAL
GFX_THROTTLE_1V8_L
from H9M
GFX_THROTTLE_L
GPU_GPIO_SVC_R
GPU_GPIO_SVD_R
GPUVCORE_SVT
NC_I2C_GPU_PCC_SCL
NC_I2C_GPU_PCC_SDA
GPU_TX_HALF_SWING
NC_GFX_VDDCI_ALTV_ZERO
GPU_BIF_GEN3_EN_A
GPUVCORE_VR_HOT_L
GPU_ROM_SO
GPU_ROM_SI_R
GPU_ROM_SCLK_R
GPU_ROM_CONFIG<0>
GPU_ROM_CONFIG<1>
GPU_ROM_CONFIG<2>
DP_X_SNK0_HPD_EG
GPU_RESERVED
GFX_VDDCI_ALTV_ONE
GFX_SELF_THROTTLE
DP_X_SNK1_HPD_EG
GPU_GFX_OVERTEMP
GPU_TX_DEEMPH_EN
GPU_VRAM_STRAP
GPU_ROM_CS_L_R
GPU_VGA_DIS
GPUVDDC_PCC_ALERT_L
GFX_SELF_THROTTLE_1V8
to H9M
RA813
1.5K
RA833
1.8K
GPU_JTAG_TRST_L
GPU_JTAG_TDI
GPU_JTAG_TCK
GPU_JTAG_TMS
GPU_JTAG_TDO
DP_T_SNK0_HPD_EG
DP_T_SNK1_HPD_EG
DP_INT_EG_HPD
GPU_VRAM_STRAP
RA823
10K
GPU_CLK_IN
NC_GPU_XTAL_OUT
GPU_TDIODE_P
GPU_TDIODE_N
PLACE_NEAR=LA800.1:2.54MM
LA800
120OHM-25%-1.8A-0.06DCR
0402
VOLTAGE=1.8V
MIN_LINE_WIDTH=0.4000
MIN_NECK_WIDTH=0.2000
PP1V8_S0_GPU
PP1V8_GPU_TSVDD
CA850
0.1UF
PLACE_NEAR=UA000.AM13:2.54MM
CA851
10UF
CA852
1UF
CA853
0.1UF
PLACE_NEAR=UA000.AM13:2.54MM
PLACE_NEAR=UA000.AM13:2.54MM
PLACE_NEAR=UA000.AM13:2.54MM
librarian reordered
pins to follow tx
DP_INT_EG_ML_P<3>
DP_INT_EG_ML_N<3>
DP_INT_EG_ML_P<2>
DP_INT_EG_ML_N<2>
DP_INT_EG_ML_P<1>
DP_INT_EG_ML_N<1>
DP_INT_EG_ML_P<0>
DP_INT_EG_ML_N<0>
DP_X_SNK0_ML_C_P<3>
DP_X_SNK0_ML_C_N<3>
DP_X_SNK0_ML_C_P<2>
DP_X_SNK0_ML_C_N<2>
DP_X_SNK0_ML_C_P<1>
DP_X_SNK0_ML_C_N<1>
DP_X_SNK0_ML_C_P<0>
DP_X_SNK0_ML_C_N<0>
DP_X_SNK1_ML_C_P<3>
DP_X_SNK1_ML_C_N<3>
DP_X_SNK1_ML_C_P<2>
DP_X_SNK1_ML_C_N<2>
DP_X_SNK1_ML_C_P<1>
DP_X_SNK1_ML_C_N<1>
DP_X_SNK1_ML_C_P<0>
DP_X_SNK1_ML_C_N<0>
DP_T_SNK0_ML_C_P<3>
DP_T_SNK0_ML_C_N<3>
DP_T_SNK0_ML_C_P<0>
DP_T_SNK0_ML_C_N<0>
DP_T_SNK0_ML_C_P<1>
DP_T_SNK0_ML_C_N<1>
DP_T_SNK0_ML_C_P<2>
DP_T_SNK0_ML_C_N<2>
DP_T_SNK1_ML_C_P<3>
DP_T_SNK1_ML_C_N<3>
DP_T_SNK1_ML_C_P<2>
DP_T_SNK1_ML_C_N<2>
DP_T_SNK1_ML_C_P<1>
DP_T_SNK1_ML_C_N<1>
DP_T_SNK1_ML_C_P<0>
DP_T_SNK1_ML_C_N<0>
GPU_AUD<1>
GPU_AUD<0>
EG_LCD_PWR_EN
GPU_DIECRACKMON
GPU_TEST_PG_BACON
RA810
1K
PP1V8_S0_GPU
RA811
0
RA815
10K
GPU_CLKREQ_L
DP_INT_EG_AUX_P
DP_INT_EG_AUX_N
DP_X_SNK0_AUXCH_C_P
DP_X_SNK0_AUXCH_C_N
DP_X_SNK1_AUXCH_C_P
DP_X_SNK1_AUXCH_C_N
DP_T_SNK0_AUXCH_C_P
DP_T_SNK0_AUXCH_C_N
DP_T_SNK1_AUXCH_C_P
DP_T_SNK1_AUXCH_C_N
GPU_PLLCHARZ_H
GPU_PLLCHARZ_L
NOSTUFF
CA880
0.1UF
NOSTUFF
CA881
0.1UF
GPU_PLLCHARZ_RC_H
GPU_PLLCHARZ_RC_L
NOSTUFF
RA880
51.1
NOSTUFF
RA881
51.1
BOM_COST_GROUP=GRAPHICS
PAGE TITLE
GPU Baffin DP/GPIO
Apple Inc.
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| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 353S00897 | 1 | IC,TPS62130B-S,3A BUCK CNVTRT,QFN16,3X3M | UA900 | CRITICAL | |

## 3.3V S0 GPU Switch

UA950

| | |
|---|---|
| Part | SLG5AP1453V |
| Type | Load Switch |
| R(on) @ 5.3A | 7.8 mOhm Typ 9.6 mOhm Max |
| Current | 5.3A Max |

## 5V S0 GPU Switch

UA950

| | |
|---|---|
| Part | SLG5AP1453V |
| Type | Load Switch |
| R(on) @ 5.3A | 7.8 mOhm Typ 9.6 mOhm Max |
| Current | 5.3A Max |

PAGE TITLE
# GPU Baffin VSS/Misc

Apple Inc.

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BOM_COST_GROUP=TBT

DRAWING

LAST_MODIFIED=Fri Jan 26 13:26:51 2018

PAGE TITLE

USB-C HIGH SPEED 1

Apple Inc.

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PAGE TITLE

# USB-C HIGH SPEED 2

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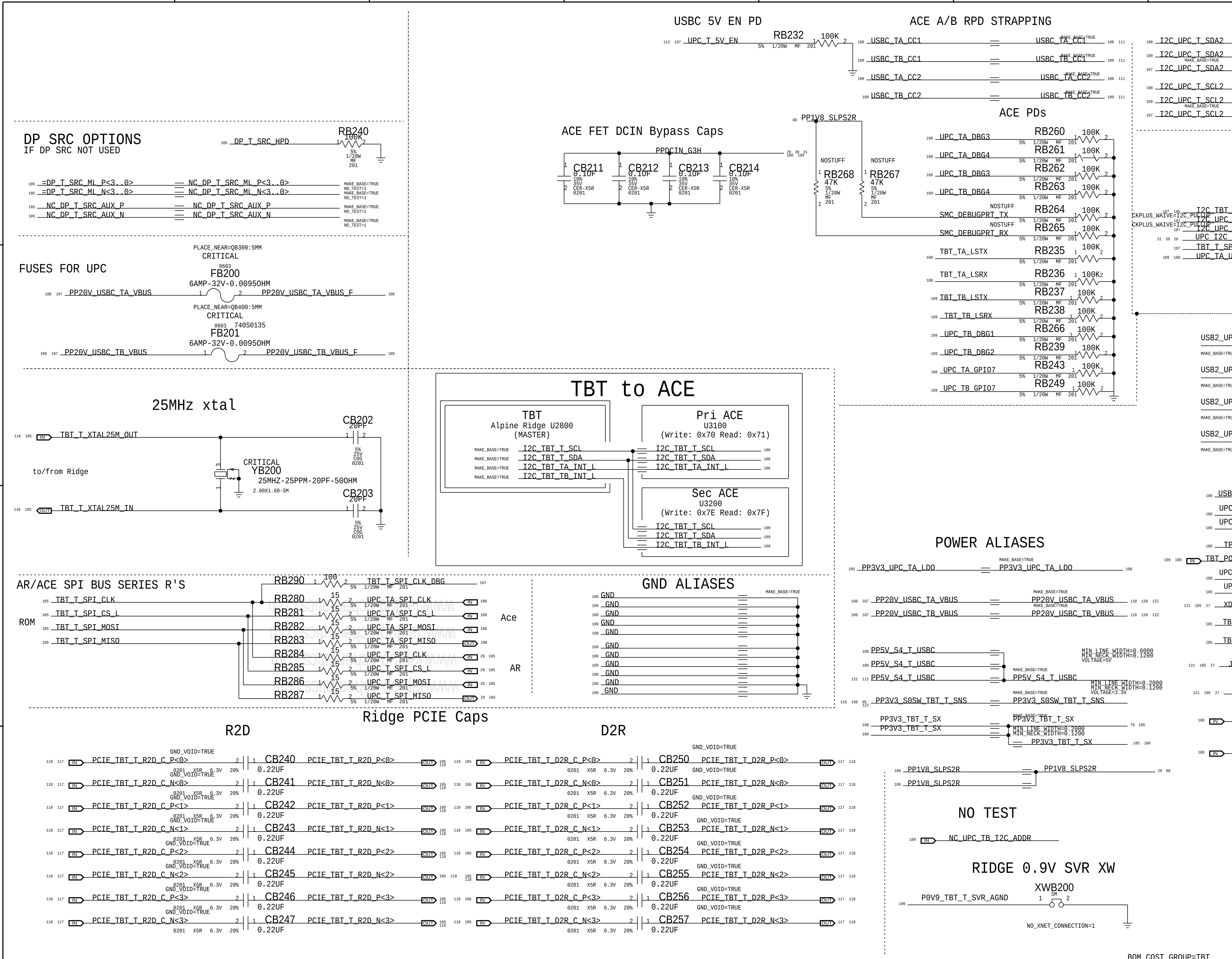


PAGE TITLE

USB-C T Support

Apple Inc.

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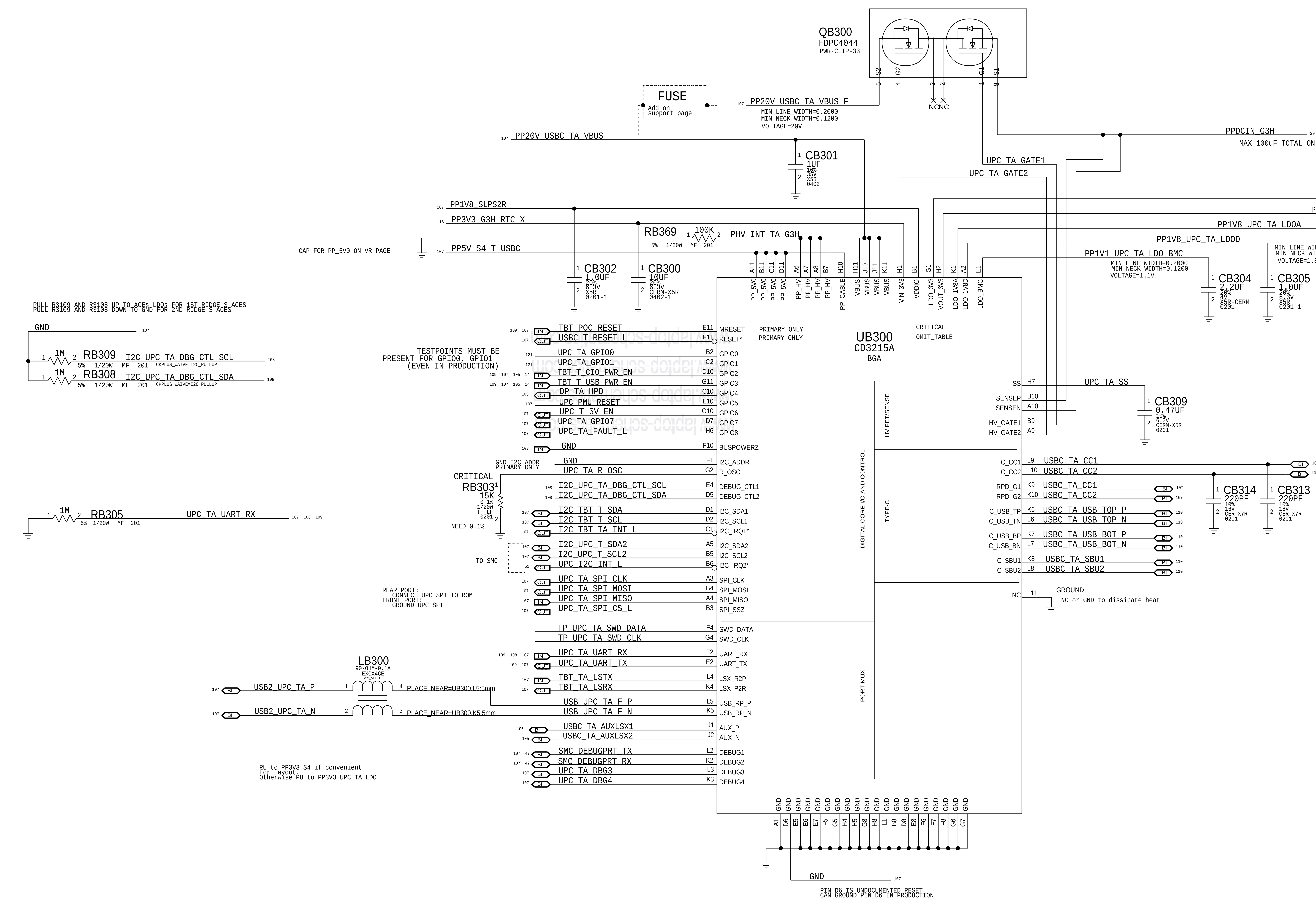
# PRIMARY ACE USB-C PORT CONTROLLER (UPC)




BOM_COST_GROUP=USB-C

PAGE TITLE

USB-C PORT CONTROLLER A

Apple Inc.

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# SECONDARY ACE USB-C PORT CONTROLLER (UPC)


BOM_COST_GROUP=USB-C

| | |
|---|---|
| PAGE TITLE | USB-C PORT CONTROLLER B |
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| REVISION | 1.0.0 |
| BRANCH | EVT |
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| | |
|---|---|
| LAST CHANGE: | Wed Apr 1 22:57:37 2015 |
| PAGE TITLE | USB-C CONNECTOR A |
| DRAWING NUMBER | 051-02888 |
| SIZE | D |
| REVISION | 1.0.0 EVT |
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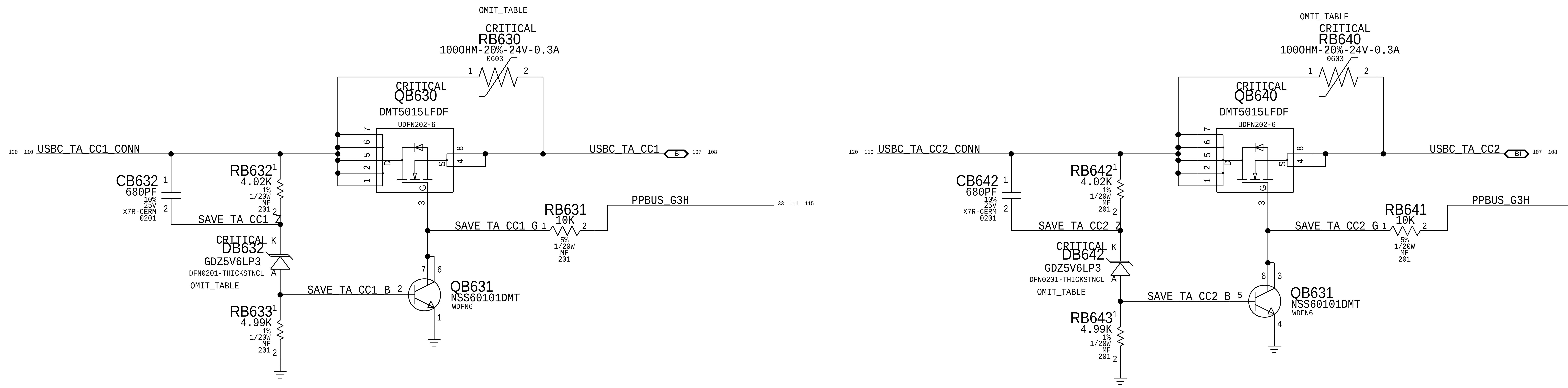
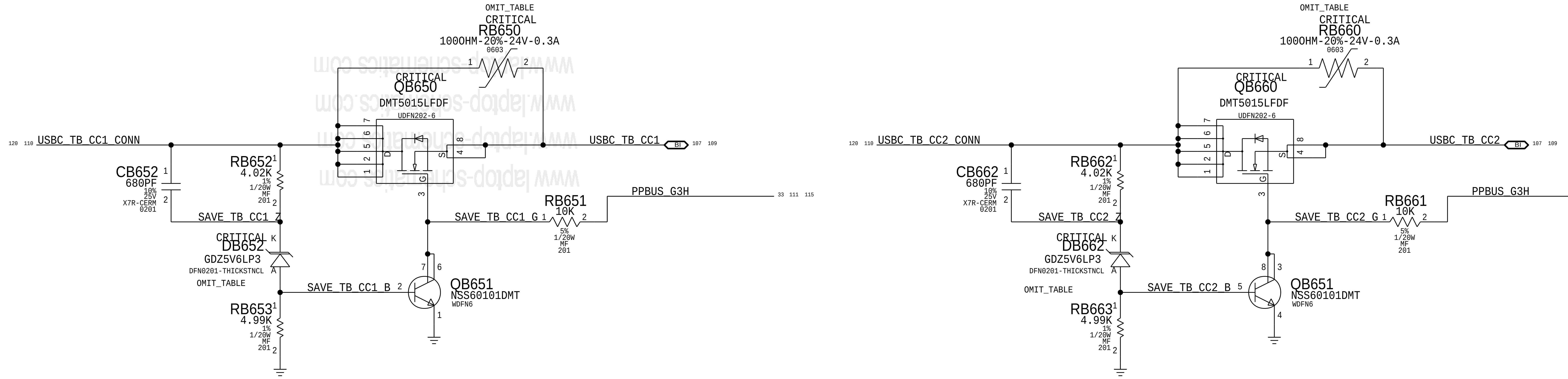
BOM_COST_GROUP=USB-C

# TA CC Protection




# TB CC Protection




| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 371S00181 | 4 | DIODE,ZENER,5.6V,250MW,DFN0201 | DB632,DB642,DB652,DB662 | CRITICAL | |
| 740S00050 | 4 | FUSE,THRMSTR,PTC,470OHM,24V,60MA,0603 | RB630,RB640,RB650,RB660 | CRITICAL | |

BOM_COST_GROUP=USB-C

DESIGN: X1032/MLB_P4BP
LAST CHANGE: Fri Jan 6 16:01:21 2017

PAGE TITLE
USBC T Connector Support

Apple Inc.

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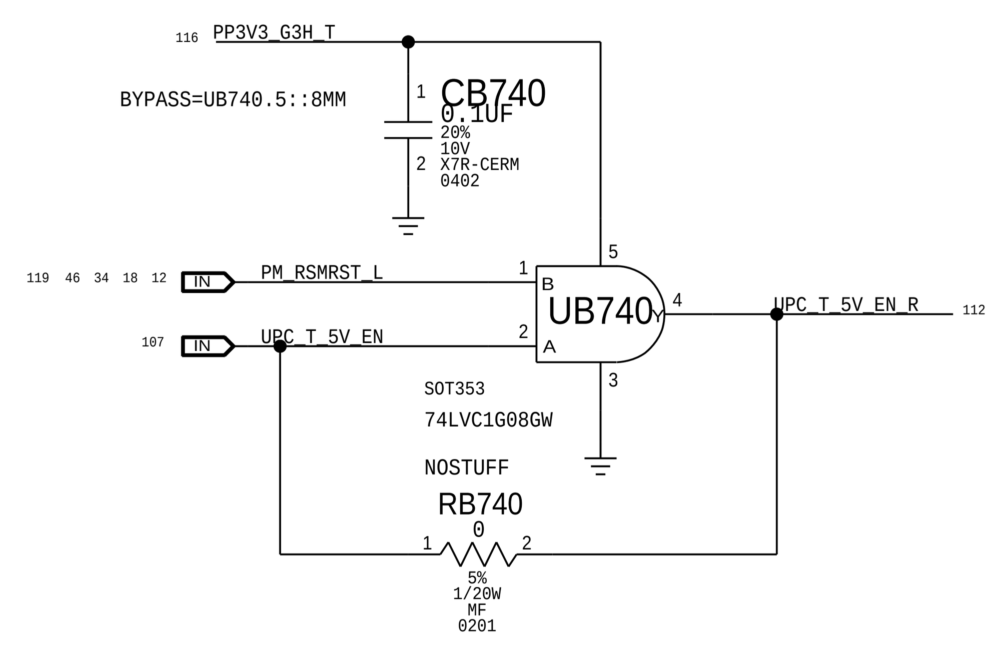


BOM_COST_GROUP=USB-C

PAGE TITLE

# USB-C T 5V VR

Apple Inc.

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SYNC_MASTER=j680_copy SYNC_DATE=02/01/2017

PAGE TITLE

GDDR5 VRAM FB 3

Apple Inc.

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BOM_COST_GROUP=GRAPHICS

BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=j680_copy SYNC_DATE=02/01/2017

PAGE TITLE

GDDR5 VRAM FB 4

Apple Inc.



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PBUS & High Side
PPBUS_G3H
MIN_LINE_WIDTH=0.6000
MIN_NECK_WIDTH=0.1200
VOLTAGE=13.1V
MAKE_BASE=TRUE
PPBUS_HS_CPU
PPBUS_HS_GPU
PPBUS_HS_OTH5V
PPBUS_HS_3V3G3H_T
PPBUS_HS_3V3G3HRTC_X
CPU Rails
PPVCC_S0_CPU
PPVCCSA_S0_CPU
PPVCCGT_S0_CPU
PPVCCIO_S0_CPU
VOLTAGE=0.95V
DDR4 1.2V
PP1V2_S3
VOLTAGE=1.2V
PP1V2_S3_CPUDDR
PP1V2_S0SW
PP0V6_S3_MEM_VREFCA_A
PP0V6_S3_MEM_VREFCA_B
VOLTAGE=0.6V
DDR4 0.6V VTT
PP0V6_S0_DDRVTT
DDR4 2.5V
PP2V5_S3
VOLTAGE=2.5V
PMIC Buck 8 - PCH 1.05V Prime
PP1V05_PRIM
VOLTAGE=1.05V
PMIC Buck8 SW2
PP1V05_S3
PMIC HIO SW
PP1V05_S0SW
GPU Rails
PPVCORE_S0_GPU
PPVDDCI_S0_GPU
PP1V5R1V35_S0_GPU_MEM
VOLTAGE=1.35V
PP1V5R1V35_S0_GPU_IC
PP1V8_GPU
VOLTAGE=1.8V
PP1V8_S0_GPU
PP3V3_S0_GPU
VOLTAGE=3.3V
Digital Ground
GND
MIN_NECK_WIDTH=0.1000
MIN_LINE_WIDTH=0.2000
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Power Alias 1
Apple Inc.
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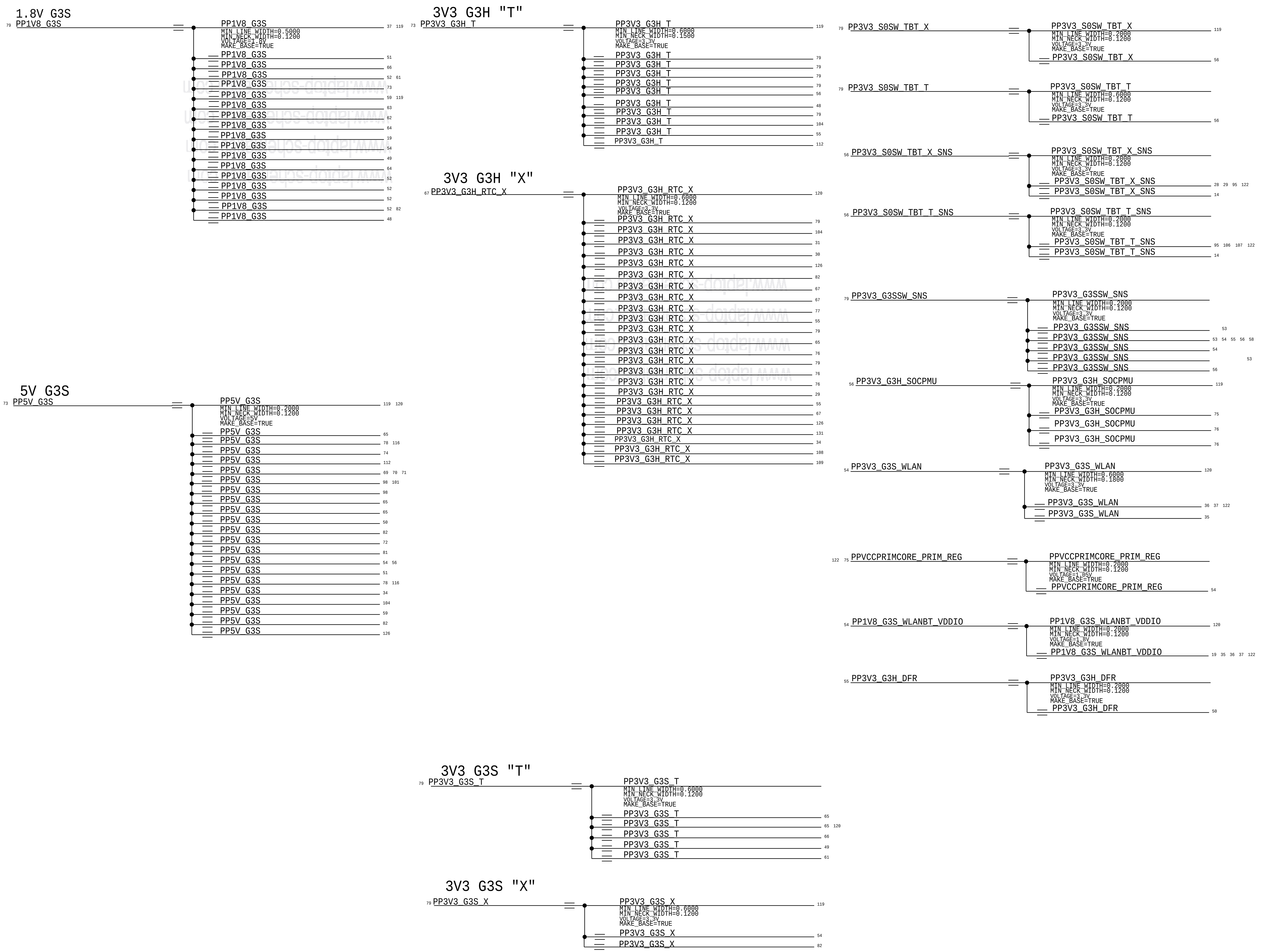


PAGE TITLE

## Power Alias 2

Apple Inc.

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# PEG X16 CPU-GPU

| | |
|---|---|
| PEG_GPU_D2R_P<7..0> (MAKE_BASE=TRUE) | =PEG_D2R_P<7..0> |
| PEG_GPU_D2R_N<7..0> (MAKE_BASE=TRUE) | =PEG_D2R_N<7..0> |
| PEG_GPU_R2D_C_P<7..0> (MAKE_BASE=TRUE) | =PEG_R2D_C_P<7..0> |
| PEG_GPU_R2D_C_N<7..0> (MAKE_BASE=TRUE) | =PEG_R2D_C_N<7..0> |

# GPU Throttle

| H9M Side (After LS) | GPU Side |
|---|---|
| GFX_SELF_THROTTLE_1V8 (MAKE_BASE=TRUE) | GFX_SELF_THROTTLE_1V8 |
| GFX_THROTTLE_1V8_L (MAKE_BASE=TRUE) | GFX_THROTTLE_1V8_L |

# CPU Display Aliases

| | |
|---|---|
| NC_DDI1_ML_C_P<3..0> (MAKE_BASE=TRUE, NO_TEST=1) | NC_DDI1_ML_C_P<3..0> |
| NC_DDI1_ML_C_N<3..0> (MAKE_BASE=TRUE, NO_TEST=1) | NC_DDI1_ML_C_N<3..0> |
| NC_DDI1_AUXCH_C_P (MAKE_BASE=TRUE, NO_TEST=1) | NC_DDI1_AUXCH_C_P |
| NC_DDI1_AUXCH_C_N (MAKE_BASE=TRUE, NO_TEST=1) | NC_DDI1_AUXCH_C_N |
| NC_DDI2_ML_C_P<3..0> (MAKE_BASE=TRUE, NO_TEST=1) | NC_DDI2_ML_C_P<3..0> |
| NC_DDI2_ML_C_N<3..0> (MAKE_BASE=TRUE, NO_TEST=1) | NC_DDI2_ML_C_N<3..0> |
| NC_DDI2_AUXCH_C_P (MAKE_BASE=TRUE, NO_TEST=1) | NC_DDI2_AUXCH_C_P |
| NC_DDI2_AUXCH_C_N (MAKE_BASE=TRUE, NO_TEST=1) | NC_DDI2_AUXCH_C_N |
| NC_DDI3_ML_P<3..0> (MAKE_BASE=TRUE, NO_TEST=1) | NC_DDI3_ML_P<3..0> |
| NC_DDI3_ML_N<3..0> (MAKE_BASE=TRUE, NO_TEST=1) | NC_DDI3_ML_N<3..0> |
| NC_DDI3_AUXCH_P (MAKE_BASE=TRUE, NO_TEST=1) | NC_DDI3_AUXCH_P |
| NC_DDI3_AUXCH_N (MAKE_BASE=TRUE, NO_TEST=1) | NC_DDI3_AUXCH_N |

# Thunderbolt Signals Through PEG

| | |
|---|---|
| PCIE_TBT_X_D2R_P<3..0> (MAKE_BASE=TRUE) | =PEG_D2R_P<11..8> |
| PCIE_TBT_X_D2R_N<3..0> (MAKE_BASE=TRUE) | =PEG_D2R_N<11..8> |
| PCIE_TBT_X_R2D_C_P<3..0> (MAKE_BASE=TRUE) | =PEG_R2D_C_P<11..8> |
| PCIE_TBT_X_R2D_C_N<3..0> (MAKE_BASE=TRUE) | =PEG_R2D_C_N<11..8> |
| PCIE_TBT_T_D2R_P<3..0> (MAKE_BASE=TRUE) | =PEG_D2R_P<15..12> |
| PCIE_TBT_T_D2R_N<3..0> (MAKE_BASE=TRUE) | =PEG_D2R_N<15..12> |
| PCIE_TBT_T_R2D_C_P<3..0> (MAKE_BASE=TRUE) | =PEG_R2D_C_P<15..12> |
| PCIE_TBT_T_R2D_C_N<3..0> (MAKE_BASE=TRUE) | =PEG_R2D_C_N<15..12> |

# GPU ALIAS

| | |
|---|---|
| GPU_RESET_L | GPU_RESET_L (MAKE_BASE=TRUE) |
| EG_PEG_CLK100M_N | EG_PEG_CLK100M_N (MAKE_BASE=TRUE) |
| EG_PEG_CLK100M_P | EG_PEG_CLK100M_P (MAKE_BASE=TRUE) |

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Signal Alias

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# ICT TESTPOINTS , High Speed NO_TEST

## PEG

PEG_GPU_D2R_N<7..0> NO_TEST=1
PEG_GPU_D2R_P<7..0> NO_TEST=1
PEG_GPU_R2D_C_N<7..0> NO_TEST=1
PEG_GPU_R2D_C_P<7..0> NO_TEST=1
PCIE_TBT_X_D2R_P<3..0> NO_TEST=1
PCIE_TBT_X_D2R_N<3..0> NO_TEST=1
PCIE_TBT_X_R2D_C_P<3..0> NO_TEST=1
PCIE_TBT_X_R2D_C_N<3..0> NO_TEST=1
PCIE_TBT_T_D2R_P<3..0> NO_TEST=1
PCIE_TBT_T_D2R_N<3..0> NO_TEST=1
PCIE_TBT_T_D2R_C_P<3..0> NO_TEST=1
PCIE_TBT_T_D2R_C_N<3..0> NO_TEST=1
PCIE_TBT_X_D2R_C_P<3..0> NO_TEST=1
PCIE_TBT_X_D2R_C_N<3..0> NO_TEST=1
PCIE_TBT_X_R2D_P<3..0> NO_TEST=1
PCIE_TBT_X_R2D_N<3..0> NO_TEST=1
PCIE_TBT_T_R2D_P<3..0> NO_TEST=1
PCIE_TBT_T_R2D_N<3..0> NO_TEST=1
PCIE_TBT_T_R2D_C_P<3..0> NO_TEST=1
PCIE_TBT_T_R2D_C_N<3..0> NO_TEST=1
PEG_GPU_D2R_C_N<7..0> NO_TEST=1
PEG_GPU_D2R_C_P<7..0> NO_TEST=1
PEG_GPU_R2D_N<7..0> NO_TEST=1
PEG_GPU_R2D_P<7..0> NO_TEST=1

## BAFFIN FRAME BUFFER

FB_A1_CS_L NO_TEST=1
FB_A0_CKE_L NO_TEST=1
FB_A1_CKE_L NO_TEST=1
FB_A0_WE_L NO_TEST=1
FB_A1_WE_L NO_TEST=1
FB_B1_CS_L NO_TEST=1
FB_B0_CKE_L NO_TEST=1
FB_B1_CKE_L NO_TEST=1
FB_B0_WE_L NO_TEST=1
FB_B0_WE_L NO_TEST=1
FB_B0_CLK_P NO_TEST=1
FB_B1_CLK_N NO_TEST=1
FB_B1_CLK_P NO_TEST=1
FB_B0_RAS_L NO_TEST=1
FB_B1_RAS_L NO_TEST=1
FB_B0_CAS_L NO_TEST=1
FB_B1_CAS_L NO_TEST=1
FB_B0_CS_L NO_TEST=1
FB_A0_DQ<31..0> NO_TEST=1
FB_A1_DQ<31..0> NO_TEST=1
FB_A0_A<8..0> NO_TEST=1
FB_A1_A<8..0> NO_TEST=1
FB_A0_WCLK_N<1..0> NO_TEST=1
FB_A0_WCLK_P<1..0> NO_TEST=1
FB_A1_WCLK_N<1..0> NO_TEST=1
FB_A1_WCLK_P<1..0> NO_TEST=1
FB_A0_EDC<3..0> NO_TEST=1
FB_A1_EDC<3..0> NO_TEST=1
FB_A0_DBI_L<3..0> NO_TEST=1
FB_A1_DBI_L<3..0> NO_TEST=1
FB_A0_ABI_L NO_TEST=1
FB_A1_ABI_L NO_TEST=1
FB_A0_CLK_N NO_TEST=1
FB_A0_CLK_P NO_TEST=1
FB_A1_CLK_N NO_TEST=1
FB_A1_CLK_P NO_TEST=1
FB_A0_RAS_L NO_TEST=1
FB_A1_RAS_L NO_TEST=1
FB_A0_CAS_L NO_TEST=1
FB_A1_CAS_L NO_TEST=1
FB_A0_CS_L NO_TEST=1
FB_B0_CLK_N NO_TEST=1
FB_B0_DQ<31..0> NO_TEST=1
FB_B1_DQ<31..0> NO_TEST=1
FB_B0_A<8..0> NO_TEST=1
FB_B1_A<8..0> NO_TEST=1
FB_B0_WCLK_N<1..0> NO_TEST=1
FB_B0_WCLK_P<1..0> NO_TEST=1
FB_B1_WCLK_N<1..0> NO_TEST=1
FB_B1_WCLK_P<1..0> NO_TEST=1
FB_B0_EDC<3..0> NO_TEST=1
FB_B1_EDC<3..0> NO_TEST=1
FB_B0_DBI_L<3..0> NO_TEST=1
FB_B1_DBI_L<3..0> NO_TEST=1
FB_B0_ABI_L NO_TEST=1
FB_B1_ABI_L NO_TEST=1

## CPU/PCH CLK

CPU_CLK24M_NSSC_CLK_N NO_TEST=1
CPU_CLK24M_NSSC_CLK_P NO_TEST=1
CPU_CLK100M_PCIBCLK_N NO_TEST=1
CPU_CLK100M_PCIBCLK_P NO_TEST=1
CPU_CLK100M_BCLK_N NO_TEST=1
CPU_CLK100M_BCLK_P NO_TEST=1
NC_ITPXDP_CLK100MN
NC_ITPXDP_CLK100MP

## MUX

DP_INT_IG_ML_N<3..0> NO_TEST=1
DP_INT_IG_ML_P<3..0> NO_TEST=1
DP_INT_IG_AUX_N NO_TEST=1
DP_INT_IG_AUX_P NO_TEST=1
DP_INT_EG_ML_N<3..0> NO_TEST=1
DP_INT_EG_ML_P<3..0> NO_TEST=1
DP_INT_EG_AUX_N NO_TEST=1
DP_INT_EG_AUX_P NO_TEST=1

## USB-C X

USBC_XB_D2R_N<2..1> NO_TEST=1
USBC_XB_D2R_P<2..1> NO_TEST=1
USBC_XA_D2R_P<2..1> NO_TEST=1
USBC_XB_R2D_C_P<2..1> NO_TEST=1
USBC_XA_D2R_N<2..1> NO_TEST=1
USBC_XB_R2D_C_N<2..1> NO_TEST=1
USBC_XA_R2D_C_P<2..1> NO_TEST=1
USBC_XA_R2D_C_N<2..1> NO_TEST=1

## DMI

DMI_S2N_P<3..0> NO_TEST=1
DMI_S2N_N<3..0> NO_TEST=1
DMI_N2S_P<3..0> NO_TEST=1
DMI_N2S_N<3..0> NO_TEST=1

## DP - CPU/ACE

DP_X_SNK0_ML_C_N<3..0> NO_TEST=1
DP_X_SNK0_ML_C_P<3..0> NO_TEST=1
DP_X_SNK0_ML_N<3..0> NO_TEST=1
DP_X_SNK0_ML_P<3..0> NO_TEST=1
DP_X_SNK1_ML_C_N<3..0> NO_TEST=1
DP_X_SNK1_ML_C_P<3..0> NO_TEST=1
DP_X_SNK1_ML_N<3..0> NO_TEST=1
DP_X_SNK1_ML_P<3..0> NO_TEST=1
DP_T_SNK0_ML_C_N<3..0> NO_TEST=1
DP_T_SNK0_ML_C_P<3..0> NO_TEST=1
DP_T_SNK0_ML_N<3..0> NO_TEST=1
DP_T_SNK0_ML_P<3..0> NO_TEST=1
DP_T_SNK1_ML_C_N<3..0> NO_TEST=1
DP_T_SNK1_ML_C_P<3..0> NO_TEST=1
DP_T_SNK1_ML_N<3..0> NO_TEST=1
DP_T_SNK1_ML_P<3..0> NO_TEST=1
DP_X_SNK0_AUXCH_C_P NO_TEST=1
DP_X_SNK0_AUXCH_C_N NO_TEST=1
DP_T_SNK0_AUXCH_C_P NO_TEST=1
DP_T_SNK0_AUXCH_C_N NO_TEST=1
DP_X_SNK0_AUXCH_P NO_TEST=1
DP_X_SNK0_AUXCH_N NO_TEST=1
DP_T_SNK0_AUXCH_P
DP_T_SNK0_AUXCH_N
DP_X_SNK1_AUXCH_C_P NO_TEST=1
DP_X_SNK1_AUXCH_C_N NO_TEST=1
DP_T_SNK1_AUXCH_C_P NO_TEST=1
DP_T_SNK1_AUXCH_C_N NO_TEST=1
DP_X_SNK1_AUXCH_P NO_TEST=1
DP_X_SNK1_AUXCH_N NO_TEST=1
DP_T_SNK1_AUXCH_P
DP_T_SNK1_AUXCH_N

## CPU/EDP

## H9M ESPI

ESPI_IO<0> NO_TEST=1
ESPI_IO<1> NO_TEST=1
ESPI_IO<2> NO_TEST=1
ESPI_IO<3> NO_TEST=1
ESPI_IO_R<0> NO_TEST=1
ESPI_IO_R<1> NO_TEST=1
ESPI_IO_R<3> NO_TEST=1
ESPI_IO_R<3> NO_TEST=1
TBT_X_XTAL25M_OUT NO_TEST=1
TBT_X_XTAL25M_IN NO_TEST=1
TBT_T_XTAL25M_OUT NO_TEST=1
TBT_T_XTAL25M_IN NO_TEST=1
GPU_CLK_IN NO_TEST=1

## MEM

MEM_A_CLK_P<1..0> NO_TEST=1
MEM_A_CLK_N<1..0> NO_TEST=1
MEM_A_DQ<63..0> NO_TEST=1
MEM_B_DQ<63..0> NO_TEST=1
MEM_A_DQS_P<7..0> NO_TEST=1
MEM_A_DQS_N<7..0> NO_TEST=1
MEM_B_DQS_P<7..0> NO_TEST=1
MEM_B_DQS_N<7..0> NO_TEST=1
MEM_B_CLK_P<1..0> NO_TEST=1
MEM_B_CLK_N<1..0> NO_TEST=1

## WIFI

50_0_ANT NO_TEST=1
50_1_ANT NO_TEST=1
50_2_ANT NO_TEST=1
50_0_COM NO_TEST=1
50_1_COM NO_TEST=1
50_2_COM NO_TEST=1
50_A_0_DIPLEXER NO_TEST=1
50_A_0_MATCH NO_TEST=1
50_G_0_DIPLEXER NO_TEST=1
50_G_0_MATCH NO_TEST=1
50_A_1_DIPLEXER NO_TEST=1
50_A_1_MATCH NO_TEST=1
50_G_1_DIPLEXER NO_TEST=1
50_G_1_MATCH NO_TEST=1
50_A_2_DIPLEXER NO_TEST=1
50_A_2_MATCH NO_TEST=1
50_G_2_DIPLEXER NO_TEST=1
50_G_2_MATCH NO_TEST=1

## USB-C T

USBC_TA_D2R_P<2..1> NO_TEST=1
USBC_TA_D2R_N<2..1> NO_TEST=1
USBC_TA_R2D_C_P<2..1> NO_TEST=1
USBC_TA_R2D_C_N<2..1> NO_TEST=1
USBC_TB_D2R_P<2..1> NO_TEST=1
USBC_TB_D2R_N<2..1> NO_TEST=1
USBC_TB_R2D_C_P<2..1> NO_TEST=1
USBC_TB_R2D_C_N<2..1> NO_TEST=1

## H9M SSD NAND

PCIE_SSD0_R2D_P<3..0> NO_TEST=1
PCIE_SSD0_R2D_N<3..0> NO_TEST=1
PCIE_SSD0_D2R_P<3..0> NO_TEST=1
PCIE_SSD0_D2R_N<3..0> NO_TEST=1
PCIE_SSD0_R2D_C_P<3..0> NO_TEST=1
PCIE_SSD0_R2D_C_N<3..0> NO_TEST=1
PCIE_SSD0_D2R_C_P<3..0> NO_TEST=1
PCIE_SSD0_D2R_C_N<3..0> NO_TEST=1
PCIE_SSD1_R2D_P<3..0> NO_TEST=1
PCIE_SSD1_R2D_N<3..0> NO_TEST=1
PCIE_SSD1_D2R_P<3..0> NO_TEST=1
PCIE_SSD1_D2R_N<3..0> NO_TEST=1
PCIE_SSD1_R2D_C_P<3..0> NO_TEST=1
PCIE_SSD1_R2D_C_N<3..0> NO_TEST=1
PCIE_SSD1_D2R_C_P<3..0> NO_TEST=1
PCIE_SSD1_D2R_C_N<3..0> NO_TEST=1

## XTAL

TBT_X_XTAL25M_OUT NO_TEST=1
PCH_CLK24M_XTALOUT NO_TEST=1
PMU_XTAL1 NO_TEST=1
PMU_XTAL2 NO_TEST=1

## PCH/AR

PCH_PCIE_WLAN_D2R_N NO_TEST=1
PCH_PCIE_WLAN_D2R_P NO_TEST=1
PCH_PCIE_WLAN_R2D_C_N NO_TEST=1
PCH_PCIE_WLAN_R2D_C_P NO_TEST=1
PCH_PCIE_WLAN_D2R_C_P NO_TEST=1
PCH_PCIE_WLAN_D2R_C_N NO_TEST=1
PCH_PCIE_WLAN_R2D_P NO_TEST=1
PCH_PCIE_WLAN_R2D_N NO_TEST=1

## PCH/SOC

PCIE_SOC_R2D_P<3..0> NO_TEST=1
PCIE_SOC_R2D_N<3..0> NO_TEST=1
PCIE_SOC_D2R_P<3..0> NO_TEST=1
PCIE_SOC_D2R_N<3..0> NO_TEST=1
PCIE_SOC_R2D_C_P<3..0> NO_TEST=1
PCIE_SOC_R2D_C_N<3..0> NO_TEST=1
PCIE_SOC_D2R_C_P<3..0> NO_TEST=1
PCIE_SOC_D2R_C_N<3..0> NO_TEST=1

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High speed No Testpoints
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DFU AND FCT TESTPOINTS placed on top side of MLB, oppsite from CPU.
DFU TEST POINTS <RDAR://31619312>
Possibly useful panda signals
TPC4H8 SOC_DEBUGPRT_RX
TPC4H9 SOC_DEBUGPRT_TX
TPC402 DEBUG_JTAG_SOC_TDO
TPC403 DEBUG_JTAG_SOC_TDI
TPC404 PP1V8_SLPS2R
TPC405 SOC_SOCHOT_L
TPC406 I2C_PWR_SDA
TPC407 I2C_UPC_SCL
TPC4E9 I2C_UPC_SDA
TPC4Z5 I2C_PWR_SCL
PSU Power input
TPC441 PPBUS_HS_GPU
TPC442 PP2V5_S3
TPC443 PP0V6_S0_DDRVTT
TPC444 PP3V3_S5
TPC4Z6 PPVBAT_G3H_CHGR_REG
TPC4Z7 PPDCIN_G3H_CHGR
TPC448
TPC449
TPC450
DEBUG MOJO
TPC451 PCH_UART_DEBUG_R2D
TPC452 PCH_UART_DEBUG_D2R
TPC4F1 SMC_DEBUGPRT_TX
TPC4F2 SMC_DEBUGPRT_RX
TPC453
Soc Access w/o Potassium
TPC454 SOC_USB_VBUS
TPC455 USB_SOC_TYPEC_P
TPC456 USB_SOC_TYPEC_N
TPC457 PMU_ACTIVE_READY
TPC458 SOC_FORCE_DFU
TPC459 SOC_DFU_STATUS
TPC460 UPC_XB_DBG3
TPC461 UPC_XB_DBG4
TPC462 SOC_DOCK_CONNECT
TPC464
DFU NON-CUSTOMER-FACING USB PORTS
PLACE IN ACCESSIBLE BOARD AREA
SWDL
TPC408 USB3_TEST_D2R_N
TPC409 USB3_TEST_D2R_P
TPC410 USB3_TEST_R2D_N
TPC411 USB3_TEST_R2D_P
TPC412
TPC413
TPC414
TPC415
DZC400 DZC401 DZC402 DZC403
5.5V-0.28PF
TPC416 USB2_UPC_TA_N
TPC417 USB2_UPC_TA_P
TPC418
DZC408 DZC409
NON-CUSTOMER-FACING USB PORTS FOR ETHERNET DONGLE(Bristol OS Testing)
PLACE IN ACCESSIBLE BOARD AREA
Ethernet
TPC422 USB3_TEST2_D2R_N
TPC423 USB3_TEST2_D2R_P
TPC424 USB3_TEST2_R2D_N
TPC425 USB3_TEST2_R2D_P
TPC426
TPC427
TPC428
TPC429
DZC410 DZC411 DZC412 DZC413
TPC419 USB2_UPC_TB_N
TPC420 USB2_UPC_TB_P
TPC421
DZC406 DZC407
TPC430 PP5V_G3S
Battery
TPC431 SYS_DETECT_L
TPC432 SMBUS_3V3_BATT_SCL
TPC433 SMBUS_3V3_BATT_SDA
TPC434 TP_BMON_IOUT
TPC435 PPVBAT_G3H_CONN
TPC436 PPVBAT_G3H_CONN
TPC437 PPVBAT_G3H_CONN
TPC4D5 PPVBAT_G3H_CONN
TPC4D6 PPVBAT_G3H_CONN
TPC4D7 PPVBAT_G3H_CONN
TPC4I7 PPVBAT_G3H_CONN
TPC4Z8 SYS_DETECT
TPC4D8
TPC4D9
TPC4E0
TPC4E1
TPC4E2
TPC440
TPC4I6
AUDIO JACK
AUD_CONN_HP_LEFT
AUD_CONN_HP_RIGHT
AUD_CONN_RING2
AUD_CONN_SLEEVE
AUD_CONN_HP_SENSE_L
AUD_CONN_HP_SENSE_R
AUD_CONN_TIP_SENSE
AUD_CONN_SLEEVE_XW
AUD_CONN_RING2_XW
FUNC_TEST=TRUE
GND
TPC4Z9 GND_AUDIO_CODEC
FAN
TPC4N0 SMC_FAN_0_PWM
TPC4N1 SMC_FAN_0_TACH
TPC4N2 SMC_FAN_1_PWM
TPC4N3 SMC_FAN_1_TACH
TPC4N4 PP1V8_G3S
TPC4N5 PP1V8_G3S
TPC4N6
TPC4N7
Power Sequence,status,charging
TPC466 PP3V3_G3H_SOCPMU
TPC468 PPBUS_G3H
TPC469 PP1V2_S3
TPC470 PPVCC_S0_CPU
TPC471 PP5V_G3S
TPC472 PP1V5R1V35_S0_GPU_MEM
TPC473 PPVCORE_S0_GPU
TPC4L6 CPU_PWRGD
TPC474 PMU_ONOFF_R_L
TPC475 PM_SLP_S3_L
TPC476 PM_SLP_S4_L
TPC477 PM_SLP_S5_L
TPC479 PP0V6_S0_DDRVTT
TPC481 PP1V05_S3
TPC482 PM_SYSRST_L
TPC483 PM_PCH_SYS_PWROK
TPC484 PM_PCH_PWROK
TPC485 PLT_RST_L
TPC486 PM_RSMRST_L
TPC487 PM_SLP_S0_L
TPC488 SMC_FIXTURE_MODE_L
TPC489 PP1V8_G3S
TPC490 PP5V_G3S
TPC491 PP3V3_G3H_T
TPC492 PP3V3_G3H_SSD0_SNS
TPC493 PP3V3_S0SW_TBT_X
TPC495 PP3V3_G3H_SOCPMU
TPC4B4 PPVDDCPU_AWAKE
TPC4B6 PVDDQ_EN_R
TPC4B7 PVCCIO_EN
TPC4B8 PP0V9_SLPDDR
TPC4B9 PP1V1_SLPDDR
TPC4C0 PP1V1_SLPS2R
TPC4C1 PP1V8_AWAKE
TPC4J1 SMC_SYSRST_L
TPC4C4 PP1V8_SLPS2R_PMUVDDGPIO
TPC4C5 P1V1_SLPDDR_SOCFET_EN
TPC4C6 PMU_CLK32K_WLANBT
TPC4C7 SOC_WDOG
TPC4C8 PMU_CLK32K_SOC
TPC4C9 PMU_SYS_ALIVE
TPC496
BOM_COST_GROUP=DEBUG
Power Sequence,status,charging
TPC4D0 P5VG3S_EN
TPC4D1 PP0V8_SLPS2R
TPC4D2 PPVDDCPUSRAM_AWAKE
TPC4D3 P5VG3S_PGOOD
TPC4D4 CPUVR_PGOOD
TPC4L7 PPVCCGT_S0_CPU
TPC4F3 PPVCCSA_S0_CPU
TPC4L0 PPVCCIO_S0_CPU
TPC4F4 PP1V05_S0SW
TPC4F5 PP1V2_S0SW
TPC4F6 PP1V05_PRIM
TPC4F7 PP1V8_S5
TPC4F8 PP3V0_G3H_RTC
TPC4F9 ALL_SYS_PWRGD
TPC4G0 PCH_RTC_RESET_L
TPC4G1 PM_THRMTRIP_L
TPC4G2 PM_PWRBTN_L
TPC4G3 CPU_PROCHOT_R_L
TPC4G4 PPVDDCI_S0_GPU
TPC4G5 DISP_GCON_RESET_L
TPC4G6 GPU_RESET_L
TPC4G7 PM_MEMVTT_EN
TPC4G8 ESPI_RESET_L
TPC4G9 GFX_THROTTLE_L
TPC4H0 PMU_COLD_RESET_L
TPC4H1 PP0V82_SLPDDR
TPC4H2 CHGR_EN_MVR
TPC4H3 PMU_PVDDMAIN_EN
TPC4H4 UPC_PMU_RESET
TPC4H5 P3V3G3S_EN
TPC4H6 PP3V3_G3S_X
TPC4H7 P1V8G3S_EN
TPC4L1 SSD_PMU_RESET_L
TPC4L2 PP2V5_NAND_SSD0
TPC4I0 PP2V5_NAND_SSD1
TPC4I1 PP0V9_SSD0
TPC4I2 PP0V9_SSD1
TPC4I3 PP1V8_SSD1
TPC4I4 PP1V8_SSD0
TPC4I5 PP1V2_AWAKE
Soc Access with Potassium
TPC499 USBC_XB_SBU1
TPC4A0 USBC_XB_SBU2
TPC4A1 USBC_XA_CC1_CONN
TPC4A2 PP20V_USBC_XA_VBUS
TPC4A3
Left Side - Top Layer
PPDCIN_G3H TPC4AB
PPDCIN_G3H TPC4AA
PPDCIN_G3H TPC4A5
PPDCIN_G3H TPC4A6
PPDCIN_G3H TPC4A7
PPDCIN_G3H TPC4A8
PPDCIN_G3H TPC4E5
PPDCIN_G3H TPC4E6
TPC4A9
TPC4B0
TPC4B1
TPC4B2
TPC4B3
TPC4E7
TPC4E8
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|---|---|---|---|---|---|---|---|

# FCT TEST POINTS (TOP SIDE OF MLB)

CORE OS DESBUG

## DFR/MESA (FCT)

| Test point | Type | Net |
|---|---|---|
| TPC500 | TP-P5 | |
| TPC501 | TP-P5 | |
| TPC502 | TP-P5 | MIPI_DFR_CLK_CONN_FILT_P |
| TPC503 | TP-P5 | DFR_DISP_VSYNC |
| TPC505 | TP-P5 | DFR_DISP_TE |
| TPC504 | TP-P5 | MIPI_DFR_CLK_CONN_FILT_N |
| TPC506 | TP-P5 | |
| TPC507 | TP-P5 | MIPI_DFR_DATA_CONN_FILT_P |
| TPC524 | TP-P5 | MIPI_DFR_DATA_CONN_FILT_N |
| TPC510 | TP-P5 | DFR_DISP_RESET_L |
| TPC516 | TP-P5 | PP3V3_G3HSW_DFR |
| TPC508 | TP-P5 | DFR_DISP_INT |
| TPC518 | TP-P5 | PP3V3_G3HSW_DFR |
| TPC511 | TP-P5 | |
| TPC512 | TP-P5 | |
| TPC521 | TP-P5 | TP_DFR_TOUCH_PANEL_DETECT |
| TPC522 | TP-P5 | DFR_LID_OPEN_L |
| TPC515 | TP-P5 | |
| TPC520 | TP-P5 | |
| TPC527 | TP-P5 | SPI_DFR_MISO_R |
| TPC528 | TP-P5 | SPI_DFR_CS_L |
| TPC533 | TP-P5 | DFR_TOUCH_INT_L |
| TPC535 | TP-P5 | DFR_TOUCH_CLK32K_RESET_L |
| TPC534 | TP-P5 | I2C_DFR_SCL_R |
| TPC536 | TP-P5 | I2C_DFR_SDA_R |
| TPC538 | TP-P5 | DFR_TOUCH_RESET_L |
| TPC530 | TP-P5 | SPI_DFR_MOSI |
| TPC525 | TP-P5 | |
| TPC526 | TP-P5 | |
| TPC529 | TP-P5 | SPI_DFR_CLK |
| TPC539 | TP-P5 | TP_DFR_TOUCH_ROM_WC |
| TPC540 | TP-P5 | PP1V8_SLPS2RSW_DFR |
| TPC542 | TP-P5 | PP5V_G3S_DFR_FILT |
| TPC541 | TP-P5 | PP1V8_SLPS2RSW_DFR |
| TPC544 | TP-P5 | SPI_MESA_MISO_CONN |
| TPC543 | TP-P5 | PP1V8_MESA_CONN |
| TPC550 | TP-P5 | SPI_MESA_CLK_CONN |
| TPC531 | TP-P5 | |
| TPC532 | TP-P5 | |
| TPC546 | TP-P5 | MESA_INT_CONN |
| TPC547 | TP-P5 | SPI_MESA_MOSI_CONN |
| TPC548 | TP-P5 | MESA_BOOST_EN_CONN |
| TPC552 | TP-P5 | PP16V0_MESA_CONN |
| TPC537 | TP-P5 | |
| TPC554 | TP-P5 | PP3V0_MESA_CONN |
| TPC555 | TP-P5 | PP3V0_MESA_CONN |
| TPC545 | TP-P5 | |
| TPC553 | TP-P5 | |

## TBT/DP

| Test point | Type | Net |
|---|---|---|
| TPC5L7 | TP-P5 | TBT_X_HDMI_DDC_DATA |
| TPC5L8 | TP-P5 | TBT_X_HDMI_DDC_CLK |
| TPC5L9 | TP-P5 | TBT_T_HDMI_DDC_DATA |
| TPC5N0 | TP-P5 | TBT_T_HDMI_DDC_CLK |

## FAN , Keyboard,Trackpad Test Points

| Test point | Type | Net |
|---|---|---|
| TPC556 | TP-P5 | GND_FAN |
| TPC557 | TP-P5 | GND_FAN |
| TPC558 | TP-P5 | PP5V_G3S_FAN_CONN |
| TPC559 | TP-P5 | PP5V_G3S_FAN_CONN |
| TPC560 | TP-P5 | FAN_LT_TACH |
| TPC561 | TP-P5 | FAN_RT_TACH |
| TPC562 | TP-P5 | FAN_LT_PWM |
| TPC563 | TP-P5 | FAN_RT_PWM |
| TPC564 | TP-P5 | |
| TPC565 | TP-P5 | |
| TPC5Z1 | TP-P5 | GND_FAN |
| TPC5Z2 | TP-P5 | GND_FAN |
| TPC5Z9 | TP-P5 | I2C_TPAD3V3_SCL |
| TPC5Y0 | TP-P5 | TPAD_KBD_WAKE_L |
| TPC566 | TP-P5 | I2C_TPAD3V3_SDA |
| TPC567 | TP-P5 | TPAD3V3_ACTUATOR_DISABLE_L |
| TPC568 | TP-P5 | I2C_KBD_SDA |
| TPC569 | TP-P5 | TPAD3V3_SPI_INT_L |
| TPC5Z8 | TP-P5 | IPD_LID_OPEN |
| TPC5Z3 | TP-P5 | PP5V_G3S |
| TPC5Z4 | TP-P5 | PP5V_G3S |
| TPC5Z7 | TP-P5 | PMU_RSLOC_RST_R_L |
| TPC5Z5 | TP-P5 | PP3V3_G3H_RTC_X |
| TPC571 | TP-P5 | |
| TPC588 | TP-P5 | ACT_GND |
| TPC586 | TP-P5 | ACT_GND |
| TPC579 | TP-P5 | KBD_BLC_GSSOUT |
| TPC578 | TP-P5 | TPAD3V3_SPI_EN |
| TPC585 | TP-P5 | PP3V3_G3S_T |
| TPC584 | TP-P5 | PP3V3_G3S_T |
| TPC583 | TP-P5 | KBD_BLC_GSLAT |
| TPC582 | TP-P5 | PP5V_G3S_TPAD_CONN |
| TPC581 | TP-P5 | KBD_BLC_GSSCK |
| TPC580 | TP-P5 | SPI_TPAD3V3_CLK |
| TPC591 | TP-P5 | PPVIN_G3H_TPAD_FUSE |
| TPC590 | TP-P5 | PPVIN_G3H_TPAD_FUSE |
| TPC589 | TP-P5 | PPVIN_G3H_TPAD_FUSE |
| TPC577 | TP-P5 | KBD_BLC_GSSIN |
| TPC576 | TP-P5 | SPI_TPAD3V3_MISO |
| TPC575 | TP-P5 | KBD_BLC_XBLANK |
| TPC574 | TP-P5 | SPI_TPAD3V3_CS_L |
| TPC572 | TP-P5 | SPI_TPAD3V3_MOSI |
| TPC573 | TP-P5 | I2C_KBD_SCL |
| TPC587 | TP-P5 | |
| TPC5M0 | TP-P5 | KBD_INT_L |

## WIRELESS

| Test point | Type | Net |
|---|---|---|
| TPC5H9 | TP-P5 | PP3V3_G3S_WLAN |
| TPC5I0 | TP-P5 | PP1V8_G3S_WLANBT_VDDIO |
| TPC5L0 | TP-P5 | PPVIN_RFLDO_WLANBT |
| TPC5L1 | TP-P5 | PP1V2_WLANBT |
| TPC5L2 | TP-P5 | PP1V5_WLANBT |

## HALL EFFECT

| Test point | Type | Net |
|---|---|---|
| TPC5I1 | TP-P5 | LID_OPEN_LEFT |
| TPC5I2 | TP-P5 | LID_OPEN_RIGHT |

## MEMORY

| Test point | Type | Net |
|---|---|---|
| TPC5I3 | TP-P5 | PVDDQ_PGOOD |
| TPC5I7 | TP-P5 | PP1V2_S3_CPUDDR |

## AUDIO

| Test point | Type | Net |
|---|---|---|
| TPC592 | TP-P5 | AUD_DMIC0_CLK_CONN |
| TPC593 | TP-P5 | AUD_DMIC0_DATA_CONN |
| TPC594 | TP-P5 | PP1V8_DMIC |
| TPC595 | TP-P5 | AUD_DMIC1_CLK_CONN |
| TPC596 | TP-P5 | AUD_DMIC1_DATA_CONN |
| TPC597 | TP-P5 | |
| TPC5A1 | TP-P5 | SPKRCONN_LW_OUTN |
| TPC5B1 | TP-P5 | SPKRCONN_LW_OUTN |
| TPC5B4 | TP-P5 | SPKRCONN_LW_OUTN |
| TPC5A0 | TP-P5 | SPKRCONN_LW_OUTP |
| TPC5G0 | TP-P5 | SPKRCONN_LW_OUTP |
| TPC5G1 | TP-P5 | SPKRCONN_LW_OUTP |
| TPC598 | TP-P5 | SPKRCONN_LT_OUTN |
| TPC5G2 | TP-P5 | SPKRCONN_LT_OUTN |
| TPC5M3 | TP-P5 | SPKRCONN_LT_OUTN |
| TPC599 | TP-P5 | SPKRCONN_LT_OUTP |
| TPC5M4 | TP-P5 | SPKRCONN_LT_OUTP |
| TPC5M5 | TP-P5 | SPKRCONN_LT_OUTP |
| TPC5A6 | TP-P5 | SPKRCONN_RW_OUTP |
| TPC5G6 | TP-P5 | SPKRCONN_RW_OUTP |
| TPC5M7 | TP-P5 | SPKRCONN_RW_OUTP |
| TPC5Y9 | TP-P5 | SPKRCONN_RW_OUTN |
| TPC5G8 | TP-P5 | SPKRCONN_RW_OUTN |
| TPC5M9 | TP-P5 | SPKRCONN_RW_OUTN |
| TPC5A3 | TP-P5 | SPKRCONN_RT_OUTN |
| TPC5L3 | TP-P5 | SPKRCONN_RT_OUTN |
| TPC5L4 | TP-P5 | SPKRCONN_RT_OUTN |
| TPC5A4 | TP-P5 | SPKRCONN_RT_OUTP |
| TPC5L5 | TP-P5 | SPKRCONN_RT_OUTP |
| TPC5L6 | TP-P5 | SPKRCONN_RT_OUTP |
| TPC5B8 | TP-P5 | AUD_CONN_RING_SENSE |
| TPC5B7 | TP-P5 | AUD_CONN_RING2_XW |
| TPC5B6 | TP-P5 | AUD_CONN_TIP_SENSE |
| TPC5B5 | TP-P5 | AUD_CONN_HP_SENSE_R |
| TPC5A7 | TP-P5 | |
| TPC5B3 | TP-P5 | AUD_CONN_HP_SENSE_L |
| TPC5A8 | TP-P5 | AUD_CONN_HP_LEFT |
| TPC5A9 | TP-P5 | AUD_CONN_HP_RIGHT |
| TPC5B0 | TP-P5 | AUD_CONN_RING2 |
| TPC5B2 | TP-P5 | AUD_CONN_SLEEVE |
| TPC5K0 | TP-P5 | |
| TPC5K1 | TP-P5 | AUD_CONN_SLEEVE_XW |
| TPC5A5 | TP-P5 | SPKR_ID0 |
| TPC5A2 | TP-P5 | |

## EDP , CAMERA,ALS

| Test point | Type | Net |
|---|---|---|
| TPC5C0 | TP-P5 | EDP_INT_AUX_N |
| TPC5C1 | TP-P5 | |
| TPC5C2 | TP-P5 | EDP_INT_AUX_P |
| TPC5C3 | TP-P5 | EDP_PANEL_PWR_BUF_EN |
| TPC5C4 | TP-P5 | |
| TPC5C9 | TP-P5 | TP_LCD_IRQ_L |
| TPC5C8 | TP-P5 | EDP_INT_ML_P<0> |
| TPC5C5 | TP-P5 | DP_INT_HPD |
| TPC5C6 | TP-P5 | EDP_INT_ML_N<0> |
| TPC5C7 | TP-P5 | LCD_FSS |
| TPC5D5 | TP-P5 | EDP_INT_ML_P<2> |
| TPC5D3 | TP-P5 | EDP_INT_ML_N<2> |
| TPC5D1 | TP-P5 | EDP_INT_ML_N<1> |
| TPC5D2 | TP-P5 | EDP_INT_ML_P<1> |
| TPC5D0 | TP-P5 | |
| TPC5D7 | TP-P5 | I2C_BKLT_SDA |
| TPC5E0 | TP-P5 | EDP_INT_ML_P<3> |
| TPC5D9 | TP-P5 | I2C_BKLT_SCL |
| TPC5D8 | TP-P5 | EDP_INT_ML_N<3> |
| TPC5D4 | TP-P5 | |
| TPC5E1 | TP-P5 | I2C_TCON_SDA |
| TPC5E4 | TP-P5 | MIPI_FTCAM_DATA_CONN_N<0> |
| TPC5D6 | TP-P5 | |
| TPC5E3 | TP-P5 | I2C_TCON_SCL |
| TPC5F2 | TP-P5 | I2C_CAM_ISOL_SDA |
| TPC5F1 | TP-P5 | MIPI_FTCAM_CLK_CONN_P |
| TPC5F0 | TP-P5 | I2C_CAM_ISOL_SCL |
| TPC5E9 | TP-P5 | MIPI_FTCAM_CLK_CONN_N |
| TPC5E8 | TP-P5 | I2C_ALS_SCL |
| TPC5E2 | TP-P5 | |
| TPC5E6 | TP-P5 | MIPI_FTCAM_DATA_CONN_P<0> |
| TPC5E7 | TP-P5 | I2C_ALS_SDA |
| TPC5F7 | TP-P5 | PP5V_S0_ALSCAM_F |
| TPC5E5 | TP-P5 | |
| TPC5F9 | TP-P6 | PPVIN_S0SW_LCDBKLT_R |
| TPC5F3 | TP-P5 | PPVOUT_S0_LCDBKLT |
| TPC5F4 | TP-P5 | PP3V3_S0SW_LCD |
| TPC5F6 | TP-P5 | PP5V_S0SW_LCD |
| TPC5F5 | TP-P5 | PP3V3_S0SW_LCD |
| TPC5F8 | TP-P5 | |
| TPC5M1 | TP-P5 | BKLT_PWM_MLB2TCON |

## USBC (PLACE NEAR CONNECTOR)

| Test point | Type | Net |
|---|---|---|
| TPC5G3 | TP-P5 | TP_USBC_PP20V_XB |
| TPC5N3 | TP-P5 | TP_USBC_PP20V_XA |
| TPC5G4 | TP-P5 | USBC_XA_SBU1 |
| TPC5G5 | TP-P5 | USBC_XA_SBU2 |
| TPC5G7 | TP-P5 | USBC_XA_CC2_CONN |
| TPC5G9 | TP-P5 | PP20V_USBC_XB_VBUS |
| TPC5H0 | TP-P5 | PP20V_USBC_XA_VBUS |
| TPC5H1 | TP-P5 | TP_USBC_PP20V_TB |
| TPC5H2 | TP-P5 | TP_USBC_PP20V_TA |
| TPC5H3 | TP-P5 | PP20V_USBC_TB_VBUS |
| TPC5H4 | TP-P5 | PP20V_USBC_TA_VBUS |
| TPC5I8 | TP-P5 | PP20V_USBC_XA_VBUS |
| TPC5I9 | TP-P5 | PP20V_USBC_XA_VBUS |
| TPC5J1 | TP-P5 | PP20V_USBC_XB_VBUS |
| TPC5J0 | TP-P5 | PP20V_USBC_XA_VBUS |
| TPC5J2 | TP-P5 | PP20V_USBC_XB_VBUS |
| TPC5J3 | TP-P5 | PP20V_USBC_XB_VBUS |
| TPC5J4 | TP-P5 | PP20V_USBC_TA_VBUS |
| TPC5H5 | TP-P5 | |
| TPC5H6 | TP-P5 | |
| TPC5H7 | TP-P5 | |
| TPC5H8 | TP-P5 | |
| TPC5M2 | TP-P5 | |
| TPC5K2 | TP-P5 | |
| TPC5K3 | TP-P5 | |
| TPC5K4 | TP-P5 | |
| TPC5J5 | TP-P5 | PP20V_USBC_TA_VBUS |
| TPC5J6 | TP-P5 | PP20V_USBC_TA_VBUS |
| TPC5J7 | TP-P5 | PP20V_USBC_TB_VBUS |
| TPC5J8 | TP-P5 | PP20V_USBC_TB_VBUS |
| TPC5J9 | TP-P5 | PP20V_USBC_TB_VBUS |
| TPC5N9 | TP-P5 | USBC_XB_CC1_CONN |
| TPC5N8 | TP-P5 | USBC_XB_CC2_CONN |
| TPC5O1 | TP-P5 | USBC_TA_CC1_CONN |
| TPC5O2 | TP-P5 | USBC_TA_CC2_CONN |
| TPC5O3 | TP-P5 | USBC_TB_CC1_CONN |
| TPC5O4 | TP-P5 | USBC_TB_CC2_CONN |

| | |
|---|---|
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|---|---|---|---|---|---|---|---|

# ICT TEST POINTS, ICT BOUNDARY SCAN TESTPOINTS

## CPU XDP and PCH Test-Points

- XDP_CPU_TCK (TRUE)
- XDP_PCH_TCK (TRUE)
- XDP_CPU_TDI (TRUE)
- XDP_CPU_TDO (TRUE)
- XDP_CPU_TRST_L (TRUE)
- XDP_CPU_TMS (TRUE)
- XDP_PCH_TMS (TRUE)
- XDP_PCH_TDI (TRUE)
- XDP_PCH_TDO (TRUE)
- XDP_CPU_PREQ_L (TRUE)
- XDP_CPU_PRDY_L (TRUE)
- PM_PCH_PWROK (TRUE)
- PM_SYSRST_L (TRUE)
- XDP_PCH_JTAGX (TRUE)
- PCH_ITP_PMODE (FUNC_TEST=TRUE)

H9M BOUNDARY SCAN TESTPOINTS ON FCT TESTPOINT PAGE

## TP Debug ACE Nets (EE tests)

- UPC_XA_GPIO0 — TPC622
- UPC_XB_GPIO0 — TPC624
- UPC_XB_GPIO1 — TPC625
- UPC_TA_GPIO0 — TPC626
- UPC_TA_GPIO1 — TPC627
- UPC_TB_GPIO0 — TPC628
- UPC_TB_GPIO1 — TPC629

## GPU ICT TESTPONTS

ACE

- JTAG_TBT_X_TMS (FUNC_TEST=TRUE)
- JTAG_ISP_TDI (FUNC_TEST=TRUE)
- JTAG_ISP_TCK (FUNC_TEST=TRUE)
- XDP_JTAG_ISP_TDO (FUNC_TEST=TRUE)
- TBT_X_TEST_EN (FUNC_TEST=TRUE)

WLAN

- WLAN_JTAG_TMS (FUNC_TEST=TRUE)
- WLAN_JTAG_TCK (FUNC_TEST=TRUE)
- WLAN_JTAG_TDI (FUNC_TEST=TRUE)
- WLAN_JTAG_TDO (FUNC_TEST=TRUE)
- WLAN_JTAG_TRST_L (FUNC_TEST=TRUE)
- JTAG_TBT_T_TMS (FUNC_TEST=TRUE)
- TP_JTAG_SOC_TRST_L (FUNC_TEST=TRUE)

GPU

- GPU_JTAG_TCK (FUNC_TEST=TRUE)
- GPU_JTAG_TDI (FUNC_TEST=TRUE)
- GPU_JTAG_TDO (FUNC_TEST=TRUE)
- GPU_JTAG_TMS (FUNC_TEST=TRUE)
- GPU_JTAG_TRST_L (FUNC_TEST=TRUE)
- SMC_PCH_SYS_PWROK (FUNC_TEST=TRUE)
- SMC_PCH_PWROK (FUNC_TEST=TRUE)
- SMC_SYSRST_L (FUNC_TEST=TRUE)
- CPU_PWRGD (FUNC_TEST=TRUE)
- PMU_COLD_RESET_L (FUNC_TEST=TRUE)
- SOC_JTAG_SEL (FUNC_TEST=TRUE)
- SOC_TESTMODE (FUNC_TEST=TRUE)
- WLAN_JTAG_SEL (FUNC_TEST=TRUE)
- TBT_T_TEST_EN (FUNC_TEST=TRUE)
- TBT_X_TEST_EN (FUNC_TEST=TRUE)

## SSD BOUNDARY SCAN Test-Points

- SSD0_SWDIO_UART_D2R (TRUE)
- SSD0_SWCLK_UART_R2D (TRUE)
- SSD0_S4E0_JTAG_TDO (TRUE)
- SSD0_S4E0_JTAG_TDI (TRUE)
- SSD0_S4E_JTAG_SEL (TRUE)
- SSD0_S4E1_JTAG_TDO (TRUE)
- SSD0_S4E2_JTAG_TDO (TRUE)
- SSD0_S4E3_JTAG_TDO
- SSD0_S4E2_JTAG_TDO (TRUE)
- SSD1_SWDIO_UART_D2R (TRUE)
- SSD1_SWCLK_UART_R2D (TRUE)
- SSD1_S4E0_JTAG_TDO (TRUE)
- SSD1_S4E0_JTAG_TDI (TRUE)
- SSD1_S4E_JTAG_SEL (TRUE)
- SSD1_S4E1_JTAG_TDO (TRUE)
- SSD1_S4E2_JTAG_TDO (TRUE)
- SSD1_S4E3_JTAG_TDO (TRUE)

## S4E

- PPC603 — PVDDCPUAWAKE_SW0
- PPC604 — PVDDCPUAWAKE_SW1
- PPC605 — PVDDCPUAWAKE_SW2
- PPC606 — PVDDCPUAWAKE_SW3
- PPC607 — PVCCPRIMCORE_SW0
- PPC608 — PVCCPRIMCORE_SW1
- PPC609 — P0V8SLPDDR_SW0
- PPC610 — P0V8SLPDDR_SW1
- PPC611 — P1V8SLPS2R_SW0
- PPC612 — P1V1SLPS2R_SW0
- PPC613 — P1V1SLPS2R_SW1
- PPC614 — P0V9SLPDDR_SW0
- PPC615 — P0V9SLPDDR_SW1
- PPC616 — PVDDCPUSRAMAWAKE_SW0

## EE TESTS SWE

- ISNS_HS_COMPUTING_P — TPC650
- ISNS_HS_COMPUTING_N — TPC651
- ISNS_HS_OTHER5V_P — TPC652
- ISNS_HS_OTHER5V_N — TPC653
- ISNS_HS_3V3_X_P — TPC654
- ISNS_HS_3V3_X_N — TPC655
- ISNS_HS_3V3_T_P — TPC656
- ISNS_HS_3V3_T_N — TPC657
- ISNS_LCDBKLT_P — TPC658
- ISNS_LCDBKLT_N — TPC659
- ISNS_1V0_P — TPC660
- ISNS_1V0_N — TPC661
- ISNS_CPUDDR_P — TPC662
- ISNS_CPUDDR_N — TPC663
- ISNS_CPUVDDQ_P — TPC664
- ISNS_CPUVDDQ_N — TPC665
- CPUVR_ISNS_P — TPC666
- CPUVR_ISNS_N — TPC667
- ISNS_WLAN_N — TPC668
- ISNS_WLAN_P — TPC669
- ISNS_WL1V8_P — TPC670
- ISNS_WL1V8_N — TPC671
- ISNS_LCDPANEL_P — TPC672
- ISNS_LCDPANEL_N — TPC673
- ISNS_CALPE_P — TPC674
- ISNS_CALPE_N — TPC675
- CPUGT_ISNS_R_P — TPC676
- CPUGT_ISNS_R_N — TPC677
- CPUSA_ISNS_P — TPC678
- CPUSA_ISNS_N — TPC679
- ISNS_CPUVCCIO_POS — TPC680
- ISNS_CPUVCCIO_NEG — TPC681
- GFXIMVP_ISNS_R_N — TPC682
- GFXIMVP_ISNS_R_P — TPC683
- VDDCIS0_CS_P — TPC684
- VDDCIS0_CS_N — TPC685
- ISNS_GPU1V8_P — TPC686
- ISNS_GPU1V8_N — TPC687
- GPUFB_CS_P — TPC688
- GPUFB_CS_N — TPC689
- ISNS_GPUFBIC_P — TPC6A0
- ISNS_GPUFBIC_N — TPC6A1
- ISNS_GPU_HS_P — TPC6A2
- ISNS_GPU_HS_N — TPC6A3
- SMC_PBUS_VSENSE — TPC6A4
- PMU_CPU_VSENSE — TPC6A5
- PMU_GPU_CORE_VSENSE — TPC6A6
- EADC1_CPUGT_VSENSE — TPC6A7
- GND_CALPE_AVSS — TPC6A8
- EADC1_CPUSA_VSENSE — TPC6A9
- EADC2_GPU_VDDCI_VSENSE — TPC6B0
- GND_EADC2_COM — TPC6B1
- SMC_DCIN_VSENSE — TPC6B2
- CHGR_CSI_R_P — TPC6B3
- CHGR_CSI_R_N — TPC6B4
- CHGR_CSO_R_P — TPC6B5
- CHGR_CSO_R_N — TPC6B6
- ISNS_PPBUS_MAIN_SSD0_P — TPC6B7
- ISNS_PPBUS_MAIN_SSD0_N — TPC6B8
- ISNS_PPBUS_MAIN_SSD1_P — TPC6B9

## EE TESTS SWE -2

- ISNS_PPBUS_MAIN_SSD1_N — TPC6C0
- ISNS_P3V3_G3W_SSD0_P — TPC6C1
- ISNS_P3V3_G3W_SSD0_N — TPC6C2
- ISNS_P3V3_G3W_SSD1_P — TPC6C3
- ISNS_P3V3_G3W_SSD1_N — TPC6C4
- ISNS_T139_P — TPC6C7
- ISNS_T139_N — TPC6C8

## WP pins of ROMs

- BT_SFLASH_WP_L — TPC6D0 (PLACE_SIDE=TOP)
- TBT_X_ROM_WP_L — TPC6D1 (PLACE_SIDE=TOP)
- TBT_T_ROM_WP_L — TPC6D2 (PLACE_SIDE=TOP)
- GPU_ROM_WP_L — TPC6D3 (PLACE_SIDE=TOP)
- SPI_SOCROM_WP_L — TPC6D4 (PLACE_SIDE=TOP)

## EE TESTS PCH

- PCIE_CLK100M_DEBUG_N — TPC620 (PLACE_NEAR=U1200.AK6:25MM)
- PCIE_CLK100M_DEBUG_P — TPC621 (PLACE_NEAR=U1200.AM6:25MM)
- PCH_PCIE_CLK100M_WLAN_N — TPC631
- PCH_PCIE_CLK100M_WLAN_P — TPC632
- PCIE_CLK100M_SOC_N — PPC633 (PLACE_NEAR=U3900.G12:25MM)
- PCIE_CLK100M_SOC_P — PPC634 (PLACE_NEAR=U3900.G13:25MM)

## S4E

- TPC690 — SSD0_S4E0_UART_TX
- TPC691 — SSD1_S4E0_UART_TX
- TPC692 — SSD0_S4E1_UART_TX
- TPC693 — SSD1_S4E1_UART_TX
- TPC694 — SSD0_S4E2_UART_TX
- TPC695 — SSD0_S4E3_UART_TX
- TPC696 — SSD1_S4E3_UART_TX
- TPC697 — SSD0_S4E1_DROOP_L
- TPC698 — SSD1_S4E2_UART_TX
- TPC699 — SSD0_CLKREQ2_L
- TPC69A — SSD0_S4E3_DROOP_L
- TPC69B — SSD0_S4E2_DROOP_L
- TPC69C — SSD0_CLKREQ3_L
- TPC69D — SSD0_S4E0_DROOP_L

| | |
|---|---|
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BOM_COST_GROUP=WIRELESS

PAGE TITLE
Desense Caps 3

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# Memory Bit/Byte Swizzle

## DDR4 COMMAND/ADDRESS

MAKE_BASE=TRUE

| | | KBL-H DDR4 NET | BIT SWIZZLE | |
|---|---|---|---|---|
| 118 | 22 | MEM_A_DQ<0> | MEM_A_DQ<0> | 7 |
| 118 | 22 | MEM_A_DQ<1> | MEM_A_DQ<1> | 7 |
| 118 | 22 | MEM_A_DQ<2> | MEM_A_DQ<2> | 7 |
| 118 | 22 | MEM_A_DQ<3> | MEM_A_DQ<3> | 7 |
| 118 | 22 | MEM_A_DQ<4> | MEM_A_DQ<4> | 7 |
| 118 | 22 | MEM_A_DQ<5> | MEM_A_DQ<5> | 7 |
| 118 | 22 | MEM_A_DQ<6> | MEM_A_DQ<6> | 7 |
| 118 | 22 | MEM_A_DQ<7> | MEM_A_DQ<7> | 7 |
| 118 | 22 | MEM_A_DQ<8> | MEM_A_DQ<8> | 7 |
| 118 | 22 | MEM_A_DQ<9> | MEM_A_DQ<9> | 7 |
| 118 | 22 | MEM_A_DQ<10> | MEM_A_DQ<10> | 7 |
| 118 | 22 | MEM_A_DQ<11> | MEM_A_DQ<11> | 7 |
| 118 | 22 | MEM_A_DQ<12> | MEM_A_DQ<12> | 7 |
| 118 | 22 | MEM_A_DQ<13> | MEM_A_DQ<13> | 7 |
| 118 | 22 | MEM_A_DQ<14> | MEM_A_DQ<14> | 7 |
| 118 | 22 | MEM_A_DQ<15> | MEM_A_DQ<15> | 7 |
| 118 | 22 | MEM_A_DQ<16> | MEM_A_DQ<16> | 7 |
| 118 | 22 | MEM_A_DQ<17> | MEM_A_DQ<17> | 7 |
| 118 | 22 | MEM_A_DQ<18> | MEM_A_DQ<18> | 7 |
| 118 | 22 | MEM_A_DQ<19> | MEM_A_DQ<19> | 7 |
| 118 | 22 | MEM_A_DQ<20> | MEM_A_DQ<20> | 7 |
| 118 | 22 | MEM_A_DQ<21> | MEM_A_DQ<21> | 7 |
| 118 | 22 | MEM_A_DQ<22> | MEM_A_DQ<22> | 7 |
| 118 | 22 | MEM_A_DQ<23> | MEM_A_DQ<23> | 7 |
| 118 | 22 | MEM_A_DQ<24> | MEM_A_DQ<24> | 7 |
| 118 | 22 | MEM_A_DQ<25> | MEM_A_DQ<25> | 7 |
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| 118 | 22 | MEM_A_DQ<27> | MEM_A_DQ<27> | 7 |
| 118 | 22 | MEM_A_DQ<28> | MEM_A_DQ<28> | 7 |
| 118 | 22 | MEM_A_DQ<29> | MEM_A_DQ<29> | 7 |
| 118 | 22 | MEM_A_DQ<30> | MEM_A_DQ<30> | 7 |
| 118 | 22 | MEM_A_DQ<31> | MEM_A_DQ<31> | 7 |
| 118 | 23 | MEM_A_DQ<32> | MEM_A_DQ<32> | 7 |
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| 118 | 23 | MEM_A_DQ<36> | MEM_A_DQ<36> | 7 |
| 118 | 23 | MEM_A_DQ<37> | MEM_A_DQ<37> | 7 |
| 118 | 23 | MEM_A_DQ<38> | MEM_A_DQ<38> | 7 |
| 118 | 23 | MEM_A_DQ<39> | MEM_A_DQ<39> | 7 |
| 118 | 23 | MEM_A_DQ<40> | MEM_A_DQ<40> | 7 |
| 118 | 23 | MEM_A_DQ<41> | MEM_A_DQ<41> | 7 |
| 118 | 23 | MEM_A_DQ<42> | MEM_A_DQ<42> | 7 |
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| 118 | 23 | MEM_A_DQ<44> | MEM_A_DQ<44> | 7 |
| 118 | 23 | MEM_A_DQ<45> | MEM_A_DQ<45> | 7 |
| 118 | 23 | MEM_A_DQ<46> | MEM_A_DQ<46> | 7 |
| 118 | 23 | MEM_A_DQ<47> | MEM_A_DQ<47> | 7 |
| 118 | 23 | MEM_A_DQ<48> | MEM_A_DQ<48> | 7 |
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| 118 | 23 | MEM_A_DQ<50> | MEM_A_DQ<50> | 7 |
| 118 | 23 | MEM_A_DQ<51> | MEM_A_DQ<51> | 7 |
| 118 | 23 | MEM_A_DQ<52> | MEM_A_DQ<52> | 7 |
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| 118 | 23 | MEM_A_DQ<56> | MEM_A_DQ<56> | 7 |
| 118 | 23 | MEM_A_DQ<57> | MEM_A_DQ<57> | 7 |
| 118 | 23 | MEM_A_DQ<58> | MEM_A_DQ<58> | 7 |
| 118 | 23 | MEM_A_DQ<59> | MEM_A_DQ<59> | 7 |
| 118 | 23 | MEM_A_DQ<60> | MEM_A_DQ<60> | 7 |
| 118 | 23 | MEM_A_DQ<61> | MEM_A_DQ<61> | 7 |
| 118 | 23 | MEM_A_DQ<62> | MEM_A_DQ<62> | 7 |
| 118 | 23 | MEM_A_DQ<63> | MEM_A_DQ<63> | 7 |

MAKE_BASE=TRUE

| | | KBL-H DDR4 NET | BIT SWIZZLE | |
|---|---|---|---|---|
| 118 | 24 | MEM_B_DQ<0> | MEM_B_DQ<0> | 7 |
| 118 | 24 | MEM_B_DQ<1> | MEM_B_DQ<1> | 7 |
| 118 | 24 | MEM_B_DQ<2> | MEM_B_DQ<2> | 7 |
| 118 | 24 | MEM_B_DQ<3> | MEM_B_DQ<3> | 7 |
| 118 | 24 | MEM_B_DQ<4> | MEM_B_DQ<4> | 7 |
| 118 | 24 | MEM_B_DQ<5> | MEM_B_DQ<5> | 7 |
| 118 | 24 | MEM_B_DQ<6> | MEM_B_DQ<6> | 7 |
| 118 | 24 | MEM_B_DQ<7> | MEM_B_DQ<7> | 7 |
| 118 | 24 | MEM_B_DQ<8> | MEM_B_DQ<8> | 7 |
| 118 | 24 | MEM_B_DQ<9> | MEM_B_DQ<9> | 7 |
| 118 | 24 | MEM_B_DQ<10> | MEM_B_DQ<10> | 7 |
| 118 | 24 | MEM_B_DQ<11> | MEM_B_DQ<11> | 7 |
| 118 | 24 | MEM_B_DQ<12> | MEM_B_DQ<12> | 7 |
| 118 | 24 | MEM_B_DQ<13> | MEM_B_DQ<13> | 7 |
| 118 | 24 | MEM_B_DQ<14> | MEM_B_DQ<14> | 7 |
| 118 | 24 | MEM_B_DQ<15> | MEM_B_DQ<15> | 7 |
| 118 | 24 | MEM_B_DQ<16> | MEM_B_DQ<16> | 7 |
| 118 | 24 | MEM_B_DQ<17> | MEM_B_DQ<17> | 7 |
| 118 | 24 | MEM_B_DQ<18> | MEM_B_DQ<18> | 7 |
| 118 | 24 | MEM_B_DQ<19> | MEM_B_DQ<19> | 7 |
| 118 | 24 | MEM_B_DQ<20> | MEM_B_DQ<20> | 7 |
| 118 | 24 | MEM_B_DQ<21> | MEM_B_DQ<21> | 7 |
| 118 | 24 | MEM_B_DQ<22> | MEM_B_DQ<22> | 7 |
| 118 | 24 | MEM_B_DQ<23> | MEM_B_DQ<23> | 7 |
| 118 | 24 | MEM_B_DQ<24> | MEM_B_DQ<24> | 7 |
| 118 | 24 | MEM_B_DQ<25> | MEM_B_DQ<25> | 7 |
| 118 | 24 | MEM_B_DQ<26> | MEM_B_DQ<26> | 7 |
| 118 | 24 | MEM_B_DQ<27> | MEM_B_DQ<27> | 7 |
| 118 | 24 | MEM_B_DQ<28> | MEM_B_DQ<28> | 7 |
| 118 | 24 | MEM_B_DQ<29> | MEM_B_DQ<29> | 7 |
| 118 | 24 | MEM_B_DQ<30> | MEM_B_DQ<30> | 7 |
| 118 | 24 | MEM_B_DQ<31> | MEM_B_DQ<31> | 7 |
| 118 | 25 | MEM_B_DQ<32> | MEM_B_DQ<32> | 7 |
| 118 | 25 | MEM_B_DQ<33> | MEM_B_DQ<33> | 7 |
| 118 | 25 | MEM_B_DQ<34> | MEM_B_DQ<34> | 7 |
| 118 | 25 | MEM_B_DQ<35> | MEM_B_DQ<35> | 7 |
| 118 | 25 | MEM_B_DQ<36> | MEM_B_DQ<36> | 7 |
| 118 | 25 | MEM_B_DQ<37> | MEM_B_DQ<37> | 7 |
| 118 | 25 | MEM_B_DQ<38> | MEM_B_DQ<38> | 7 |
| 118 | 25 | MEM_B_DQ<39> | MEM_B_DQ<39> | 7 |
| 118 | 25 | MEM_B_DQ<40> | MEM_B_DQ<40> | 7 |
| 118 | 25 | MEM_B_DQ<41> | MEM_B_DQ<41> | 7 |
| 118 | 25 | MEM_B_DQ<42> | MEM_B_DQ<42> | 7 |
| 118 | 25 | MEM_B_DQ<43> | MEM_B_DQ<43> | 7 |
| 118 | 25 | MEM_B_DQ<44> | MEM_B_DQ<44> | 7 |
| 118 | 25 | MEM_B_DQ<45> | MEM_B_DQ<45> | 7 |
| 118 | 25 | MEM_B_DQ<46> | MEM_B_DQ<46> | 7 |
| 118 | 25 | MEM_B_DQ<47> | MEM_B_DQ<47> | 7 |
| 118 | 25 | MEM_B_DQ<48> | MEM_B_DQ<48> | 7 |
| 118 | 25 | MEM_B_DQ<49> | MEM_B_DQ<49> | 7 |
| 118 | 25 | MEM_B_DQ<50> | MEM_B_DQ<50> | 7 |
| 118 | 25 | MEM_B_DQ<51> | MEM_B_DQ<51> | 7 |
| 118 | 25 | MEM_B_DQ<52> | MEM_B_DQ<52> | 7 |
| 118 | 25 | MEM_B_DQ<53> | MEM_B_DQ<53> | 7 |
| 118 | 25 | MEM_B_DQ<54> | MEM_B_DQ<54> | 7 |
| 118 | 25 | MEM_B_DQ<55> | MEM_B_DQ<55> | 7 |
| 118 | 25 | MEM_B_DQ<56> | MEM_B_DQ<56> | 7 |
| 118 | 25 | MEM_B_DQ<57> | MEM_B_DQ<57> | 7 |
| 118 | 25 | MEM_B_DQ<58> | MEM_B_DQ<58> | 7 |
| 118 | 25 | MEM_B_DQ<59> | MEM_B_DQ<59> | 7 |
| 118 | 25 | MEM_B_DQ<60> | MEM_B_DQ<60> | 7 |
| 118 | 25 | MEM_B_DQ<61> | MEM_B_DQ<61> | 7 |
| 118 | 25 | MEM_B_DQ<62> | MEM_B_DQ<62> | 7 |
| 118 | 25 | MEM_B_DQ<63> | MEM_B_DQ<63> | 7 |

MAKE_BASE

| | | | | | |
|---|---|---|---|---|---|
| 118 | 22 | TRUE | MEM_A_DQS_P<0> | MEM_A_DQS_P<0> | 7 |
| 118 | 22 | TRUE | MEM_A_DQS_N<0> | MEM_A_DQS_N<0> | 7 |
| 118 | 22 | TRUE | MEM_A_DQS_P<1> | MEM_A_DQS_P<1> | 7 |
| 118 | 22 | TRUE | MEM_A_DQS_N<1> | MEM_A_DQS_N<1> | 7 |
| 118 | 22 | TRUE | MEM_A_DQS_P<2> | MEM_A_DQS_P<2> | 7 |
| 118 | 22 | TRUE | MEM_A_DQS_N<2> | MEM_A_DQS_N<2> | 7 |
| 118 | 22 | TRUE | MEM_A_DQS_P<3> | MEM_A_DQS_P<3> | 7 |
| 118 | 22 | TRUE | MEM_A_DQS_N<3> | MEM_A_DQS_N<3> | 7 |
| 118 | 23 | TRUE | MEM_A_DQS_P<4> | MEM_A_DQS_P<4> | 7 |
| 118 | 23 | TRUE | MEM_A_DQS_N<4> | MEM_A_DQS_N<4> | 7 |
| 118 | 23 | TRUE | MEM_A_DQS_P<5> | MEM_A_DQS_P<5> | 7 |
| 118 | 23 | TRUE | MEM_A_DQS_N<5> | MEM_A_DQS_N<5> | 7 |
| 118 | 23 | TRUE | MEM_A_DQS_P<6> | MEM_A_DQS_P<6> | 7 |
| 118 | 23 | TRUE | MEM_A_DQS_N<6> | MEM_A_DQS_N<6> | 7 |
| 118 | 23 | TRUE | MEM_A_DQS_P<7> | MEM_A_DQS_P<7> | 7 |
| 118 | 23 | TRUE | MEM_A_DQS_N<7> | MEM_A_DQS_N<7> | 7 |

MAKE_BASE

| | | | | | |
|---|---|---|---|---|---|
| 118 | 24 | TRUE | MEM_B_DQS_P<0> | MEM_B_DQS_P<0> | 7 |
| 118 | 24 | TRUE | MEM_B_DQS_N<0> | MEM_B_DQS_N<0> | 7 |
| 118 | 24 | TRUE | MEM_B_DQS_P<1> | MEM_B_DQS_P<1> | 7 |
| 118 | 24 | TRUE | MEM_B_DQS_N<1> | MEM_B_DQS_N<1> | 7 |
| 118 | 24 | TRUE | MEM_B_DQS_P<2> | MEM_B_DQS_P<2> | 7 |
| 118 | 24 | TRUE | MEM_B_DQS_N<2> | MEM_B_DQS_N<2> | 7 |
| 118 | 24 | TRUE | MEM_B_DQS_P<3> | MEM_B_DQS_P<3> | 7 |
| 118 | 24 | TRUE | MEM_B_DQS_N<3> | MEM_B_DQS_N<3> | 7 |
| 118 | 25 | TRUE | MEM_B_DQS_P<4> | MEM_B_DQS_P<4> | 7 |
| 118 | 25 | TRUE | MEM_B_DQS_N<4> | MEM_B_DQS_N<4> | 7 |
| 118 | 25 | TRUE | MEM_B_DQS_P<5> | MEM_B_DQS_P<5> | 7 |
| 118 | 25 | TRUE | MEM_B_DQS_N<5> | MEM_B_DQS_N<5> | 7 |
| 118 | 25 | TRUE | MEM_B_DQS_P<6> | MEM_B_DQS_P<6> | 7 |
| 118 | 25 | TRUE | MEM_B_DQS_N<6> | MEM_B_DQS_N<6> | 7 |
| 118 | 25 | TRUE | MEM_B_DQS_P<7> | MEM_B_DQS_P<7> | 7 |
| 118 | 25 | TRUE | MEM_B_DQS_N<7> | MEM_B_DQS_N<7> | 7 |

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Memory Bite/Byte Swizzle

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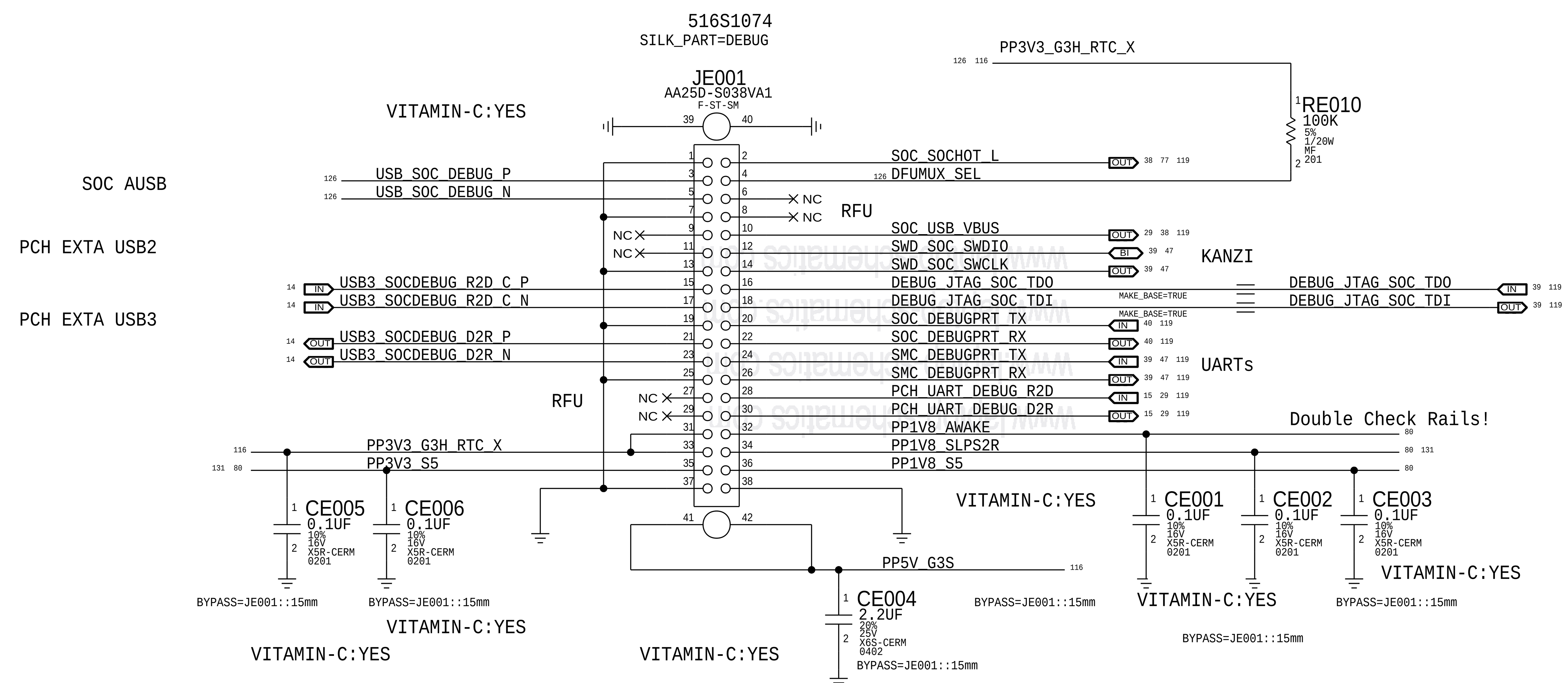
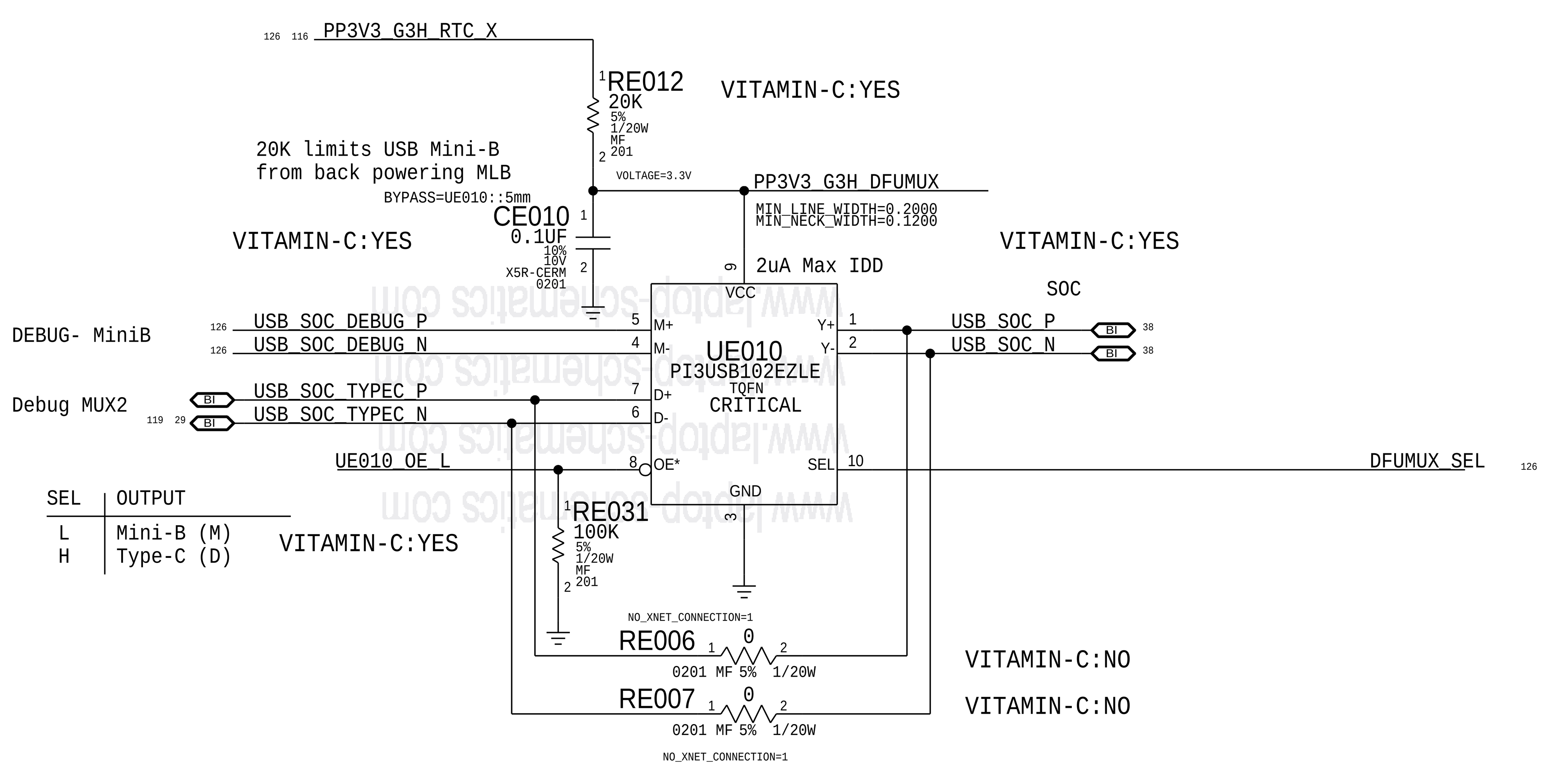
# DEBUG Connector




# SoC USB DFU Mux




| SEL | OUTPUT |
|---|---|
| L | Mini-B (M) |
| H | Type-C (D) |

BOM_COST_GROUP=DEBUG

SYNC_MASTER=Debug SYNC_DATE=05/03/2017

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8 | 7 | 6 | 5 | 4 | 3 | 2 | 1

# Alternate Parts

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 197S00046 | 197S00036 | | ALL | Epson w/ TXC |
| 197S00047 | 197S00036 | | ALL | Kyocera w/ TXC |
| 197S00048 | 197S00036 | | ALL | Murata w/ TXC |
| 376S1080 | 376S0820 | | ALL | |
| 152S00726 | 152S00592 | | ALL | |
| 376S1053 | 376S0604 | | ALL | Diodes alt to Fairchild |
| 353S00636 | 353S4037 | | ALL | |
| 155S00007 | 155S0667 | | ALL | |
| 376S1106 | 376S0678 | | ALL | Fairchild alt to Vishay |
| 371S00127 | 371S00182 | | ALL | |
| 152S00769 | 152S00659 | | ALL | |
| 372S00016 | 372S00015 | | ALL | |
| 107S00041 | 107S00039 | | ALL | |
| 740S00041 | 740S0159 | | ALL | |
| 311S00138 | 311S0436 | | ALL | |
| 152S00725 | 152S00590 | | ALL | |
| 152S00841 | 152S00238 | | ALL | |
| 376S00219 | 376S00079 | | ALL | TI for Fairchild |
| 311S00121 | 311S0398 | | ALL | |
| 311S0596 | 311S0593 | | ALL | |
| 353S3526 | 353S3528 | | ALL | |
| 353S3527 | 353S3528 | | ALL | |
| 197S00131 | 197S00116 | | ALL | |
| 197S00117 | 197S00116 | | ALL | |
| 197S00115 | 197S00116 | | ALL | |
| 152S00398 | 152S00204 | | ALL | |
| 152S00786 | 152S00344 | | ALL | |
| 311S00165 | 311S00164 | | ALL | |
| 128S00050 | 128S0351 | | ALL | |
| 138S00047 | 138S00073 | | ALL | |
| 353S01041 | 353S01042 | | ALL | |
| 155S0665 | 155S00232 | | ALL | |
| 740S00051 | 740S00050 | | ALL | |
| 152S00724 | 152S00311 | | ALL | |
| 155S0694 | 155S0387 | | ALL | |
| 103S0321 | 103S0276 | | ALL | |
| 740S0144 | 740S0118 | | ALL | |
| 740S00028 | 740S0118 | | ALL | |
| 138S00077 | 138S00035 | | ALL | |
| 138S00093 | 138S00035 | | ALL | |
| 138S00056 | 138S1100 | | ALL | |
| 107S00015 | 107S00011 | | ALL | |
| 152S00543 | 152S00484 | | ALL | |
| 155S00018 | 155S0664 | | ALL | Murata w/ Taiyo |
| 152S00359 | 152S00253 | | ALL | Chilisin alt to Cyntec |
| 311S00118 | 311S0489 | | ALL | |
| 155S00365 | 155S00191 | | ALL | |
| 107S00070 | 107S0085 | | ALL | |
| 138S0660 | 138S0684 | | ALL | |
| 138S0846 | 138S0811 | | ALL | |
| 155S0914 | 155S0897 | | ALL | |
| 155S00190 | 155S0897 | | ALL | |
| 138S00117 | 138S00071 | | ALL | |
| 353S01346 | 353S01320 | | ALL | |
| 376S00281 | 376S1147 | | ALL | |
| 376S00230 | 376S00174 | | ALL | |
| 128S00022 | 128S00018 | | ALL | |
| 376S00304 | 376S1187 | | ALL | |

| | | | | |
|---|---|---|---|---|
| 311S0426 | 311S00007 | | ALL | |
| 138S00229 | 138S00107 | | ALL | |
| 353S01614 | 353S4159 | | ALL | |
| 132S00202 | 132S00175 | | ALL | |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 353S00878 | 353S00599 | | ALL | |
| 371S00089 | 371S00085 | | ALL | |
| 107S0248 | 107S0250 | | ALL | |
| 107S00033 | 107S00034 | | ALL | |
| 107S00103 | 107S00058 | | ALL | Cyntec w/ Yageo |
| 311S00013 | 311S0508 | | ALL | Part wi |
| 132S00064 | 132S0409 | | ALL | |
| 128S0364 | 128S0264 | | ALL | Kemet w/ Panasonic |
| 128S00058 | 128S00018 | | ALL | NEC w/ Rohm |
| 107S00071 | 107S00053 | | ALL | |
| 376S00014 | 376S0761 | | ALL | |
| 376S00086 | 376S0761 | | ALL | |
| 107S00044 | 107S00076 | | ALL | |
| 740S00003 | 740S0135 | | ALL | |
| 128S00070 | 128S00007 | | ALL | |
| 740S00019 | 740S00007 | | ALL | Bourns w/ Polytronics |
| 353S00852 | 353S4262 | | ALL | TI w/ OnSemi |
| 138S0775 | 138S0860 | | ALL | |
| 138S0789 | 138S0941 | | ALL | Murata w/ SS |
| 152S00363 | 152S00048 | | ALL | |
| 152S00415 | 152S00140 | | ALL | |
| 155S00204 | 155S0731 | | ALL | |
| 311S00060 | 311S0273 | | ALL | |
| 128S00009 | 128S00007 | | ALL | |
| 138S0738 | 138S1101 | | ALL | Samsung alt to Murata |
| 107S00234 | 107S00235 | | ALL | |
| 311S00156 | 311S00129 | | ALL | |
| 155S0660 | 155S0513 | | ALL | |
| 197S00120 | 197S00118 | | ALL | |
| 152S00368 | 152S00269 | | ALL | |
| 311S0271 | 311S00008 | | ALL | NXP w/ Diodes |
| 311S00104 | 311S00091 | | ALL | |

| | | | | |
|---|---|---|---|---|
| 128S00039 | 128S00038 | | ALL | |
| 128S0302 | 128S00038 | | ALL | |
| 138S1103 | 138S0719 | | ALL | |
| 376S00074 | 376S0855 | | ALL | |
| 311S00180 | 311S00181 | | ALL | |
| 152S00770 | 152S0857 | | ALL | |
| 740S00033 | 740S00002 | | ALL | |
| 155S00166 | 155S0391 | | ALL | |
| 107S0249 | 107S0251 | | ALL | |
| 155S00401 | 155S00067 | | ALL | |
| 152S00369 | 152S00268 | | ALL | Cyntec w/ NEC |
| 371S00074 | 371S0602 | | ALL | |
| 138S0689 | 138S0701 | | ALL | |
| 128S00087 | 128S00011 | | ALL | |
| 128S00031 | 128S00011 | | ALL | |
| 128S00026 | 128S00011 | | ALL | |
| 152S00182 | 152S00703 | | ALL | |
| 152S00705 | 152S00412 | | ALL | |
| 376S00303 | 376S00012 | | ALL | |
| 152S00864 | 152S00851 | | ALL | |
| 311S00179 | 311S00166 | | ALL | |
| 353S01505 | 353S01389 | | ALL | |
| 353S01506 | 353S01404 | | ALL | |
| 311S00057 | 311S0612 | | ALL | |
| 377S0155 | 377S0184 | | ALL | |
| 152S00403 | 152S00322 | | ALL | |

| | | | | |
|---|---|---|---|---|
| 311S00193 | 311S00192 | | ALL | |
| 371S00170 | 371S00125 | | ALL | |
| 376S00302 | 376S1038 | | ALL | |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 103S00248 | 103S00247 | | ALL | 8.06K 0.1% RESISTOR |
| 103S00250 | 103S00249 | | ALL | 18.2K 0.1% RESISTOR |
| 116S00006 | 116S0175 | | ALL | 0OHM RESISTOR |
| 128S00062 | 128S00067 | | ALL | 150UF POLY CAP |
| 128S00069 | 128S00067 | | ALL | 150UF POLY CAP |
| 128S00065 | 128S00067 | | ALL | 150UF POLY CAP |
| 138S00097 | 138S0750 | | ALL | 4.7UF CAP |
| 131S00134 | 131S00041 | | ALL | 12pF Desense:31561601 |
| 107S00101 | 107S00005 | | ALL | 5mOhm :31560708 |
| 107S00087 | 107S00029 | | ALL | 5mOhm :31560843 |
| 107S00086 | 107S00056 | | ALL | 3mOhm : 31560937 |
| 107S00100 | 107S00057 | | ALL | 2mOhm : 31561063 |
| 107S0240 | 107S0255 | | ALL | 1mOhm : 31561401 |
| 107S00139 | 107S0178 | | ALL | 220KOhm :31561213 |
| 128S0296 | 128S0487 | | ALL | 47uF-PT : 31561532 |
| 128S00012 | 128S0487 | | ALL | 47uF-PT : 31561532 |
| 138S0786 | 138S0705 | | ALL | Murata w/ Samsung |
| 353S00832 | 353S00525 | | ALL | IMVP SA |
| 353S00831 | 353S00519 | | ALL | IMVP VCORE |
| 152S00782 | 152SS00772 | | ALL | IMVP CORE INDUCTOR |
| 138S00105 | 138S00037 | | ALL | 10uF CER : 31563978 |
| 138S0714 | 138S0713 | | ALL | 10UF : 31564912 |
| 138S0648 | 138S0703 | | ALL | 4.7uF CER : 31564224 |
| 138S0715 | 138S0732 | | ALL | 1UF CER : 31565015 |
| 152S00845 | 152S00815 | | ALL | |
| 138S00111 | 138S00036 | | ALL | 20uF CER : 31563894 |
| 132S00012 | 132S0401 | | ALL | 0.22uF : 31563715 |
| 138S00084 | 138S00060 | | ALL | 47uF CER : 31564055 |
| 138S0739 | 138S0706 | | ALL | NEC w/ Vishay |
| 138S0945 | 138S0706 | | ALL | NEC w/ Rohm |
| 138S0641 | 138S0700 | | ALL | Murata w/ SS&Taiyo |
| 138S0759 | 138S0762 | | ALL | |
| 138S00049 | 138S0831 | | ALL | |
| 107S00090 | 107S00055 | | ALL | |
| 138S0863 | 138S0853 | | ALL | |
| 138S00006 | 138S0835 | | ALL | |
| 377S0077 | 377S0183 | | ALL | |
| 371S00180 | 371S00077 | | ALL | NXP alt to Diodes |
| 138S00104 | 138S0978 | | ALL | |
| 138S00024 | 138S0986 | | ALL | |
| 152S00434 | 152S1829 | | ALL | |
| 197S00053 | 197S00050 | | ALL | Kyocera w/ TXC |
| 197S00054 | 197S00050 | | ALL | NDK w/ TXC |
| 197S00055 | 197S00050 | | ALL | Murata w/ TXC |
| 335S00270 | 335S00203 | | ALL | Adesto for Micronix |
| 155S00034 | 155S0706 | | ALL | Taiyo w/ Murata |
| 353S00879 | 353S00754 | | ALL | |
| 155S00203 | 155S0894 | | ALL | |
| 152S00461 | 152S00112 | | ALL | |
| 376S00146 | 376S1061 | | ALL | |
| 152S00704 | 152S00413 | | ALL | |
| 152S00199 | 152S00413 | | ALL | |
| 335S00066 | 335S0946 | | ALL | |
| 372S0186 | 372S0185 | | ALL | |
| 311S00177 | 311S00178 | | ALL | |
| 353S00750 | 353S00877 | | ALL | |
| 152S00707 | 152S00689 | | ALL | |
| 376S00282 | 376S1128 | | ALL | |

VRAM ALT SUB-BOMS

| | | | | |
|---|---|---|---|---|
| 685-00222 | 685-00221 | | ALL | |
| 685-00223 | 685-00221 | | ALL | |
| 685-00224 | 685-00221 | | ALL | |
| 685-00225 | 685-00221 | | ALL | |

PAGE TITLE SYNC_MASTER=SBAN SYNC_DATE=04/20/2018

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# Power State Debug LEDs
(For development only)

# System State LEDs

| INPUT | | | | | OUTPUT | | |
|---|---|---|---|---|---|---|---|
| PMU_ACT_RDY | SLP_SCFET_EN | SLP_S0_L | VDDQ_EN | 1V8G3S_EN | R | G | B |
| 0 | 0 | 0 | 0 | 0 | B | 1 | 1 |
| 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 | 1 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 |

# SOC State LEDs


BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE
Dev Support

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